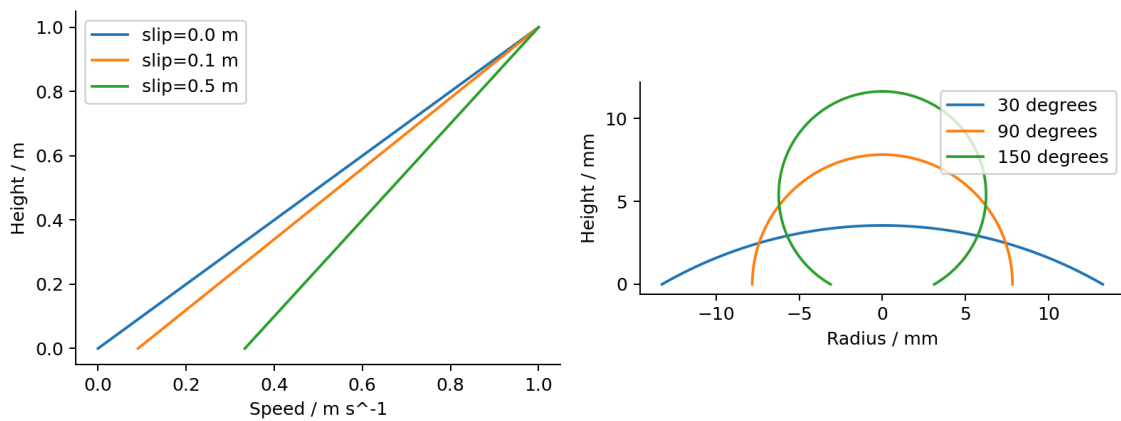


Rheon

Discrete Fluid Simulation

Solids, liquids and contact



Mathematics · Checked contracts · Numerical references · 3D teaching laboratories

4 October 2026

Contents

1. [1 The numerical contract](#)
2. [2 From conservation to a tractable model](#)
3. [3 Staggered geometry and discrete operators](#)
4. [4 Pressure as a constrained correction](#)
5. [5 Solving the pressure system](#)
6. [6 Transport and the price of stability](#)
7. [7 Higher order transport with bounded failure](#)
8. [8 Boundaries and obstacles](#)
9. [9 Liquid interfaces and volume](#)
10. [10 Forces viscosity and material behavior](#)
11. [11 Particle alternatives and hybrid transfers](#)
12. [12 Time stepping and error budgets](#)
13. [13 Memory layout and parallel execution](#)
14. [14 Guidance as a geometric measurement](#)
15. [15 Exact mathematics and finite arithmetic](#)
16. [16 The checked Lean companion](#)
17. [17 A staged Rust implementation](#)
18. [18 Validation and remaining research](#)
19. [19 Forces and moving-boundary work](#)
20. [20 From collision meshes to fluid boundaries](#)
21. [21 Density and real liquid interfaces](#)
22. [22 Viscosity, stress and dissipation](#)
23. [23 Wetting, adhesion and the contact line](#)
24. [24 Slip and wall friction](#)
25. [25 Surface tension and integrated qualification](#)
26. [A Algorithm reference](#)
27. [B Reproduction and evidence](#)
28. [C Primary sources](#)
29. [D Notation and provenance](#)
30. [E A worked matrix laboratory](#)
31. [F Expansion contracts, sources and evidence](#)

1 The numerical contract

Rheon needs to place a moving three-dimensional fluid shape deliberately while leaving memory available for image-generation models. The repository describes a low-resolution, interactive simulation framework and a standalone demonstration application. This book develops the mathematical and engineering foundation for that framework. The recommended first implementation is a bounded Cartesian staggered-grid solver with explicit diagnostics, followed by a deliberately separate free-surface branch. Neither a rendered fluid image nor a successful pressure solve alone establishes the intended result.

The repository at revision 0cb48429789699578b640d30d9d5dc99c6921ce4 contains a README, license and ignore file. It has no existing solver whose behavior can be assumed. Accordingly, algorithms here are specifications and research recommendations until implemented and qualified. The numerical companion executes small reference experiments; the Lean companion checks selected exact discrete contracts. Proposed architecture must not quietly become a claim about software that does not exist.

Spatial accuracy has several meanings

A fluid can be physically plausible while occupying the wrong silhouette, or match a silhouette while violating mass conservation. We distinguish geometric error, numerical error, model error and downstream conditioning error. Geometric error measures interface position against a defined target. Numerical error compares the discretization against the selected mathematical model. Model error compares that model with the real phenomenon. Conditioning error measures how a generated image responds to a guidance representation. These quantities can move in different directions.

Let the simulated state be s , a camera be C , and the deterministic guidance renderer be R_C . An image-generation system applies a map F to that representation and additional inputs z :

$$s \mapsto R_C(s) \mapsto F(R_C(s), z).$$

A bound on the first map says nothing automatic about the sensitivity of the second. ControlNet demonstrates spatial conditioning with several image types, including depth and edges [S14]; it supplies no guarantee that any particular fluid boundary will be preserved exactly. Rheon’s measurable responsibility should end at a documented guidance image and metadata unless the integrated diffusion model is separately evaluated.

For an orthographic camera with world width L_x and image width W , a transverse position error δx produces approximately $W\delta x/L_x$ pixels of displacement. At $L_x = 2$ metres and $W = 1024$, a one-centimetre error is 5.12 pixels. A 64-cell domain of the same width has 31.25 mm cells, or 16 pixels per cell. These arithmetic facts constrain what positionally accurate can mean. Interpolated surfaces vary continuously inside a cell but cannot recover arbitrary unresolved topology.

A staged scope

The first stage should solve incompressible velocity in a stationary box with smoke-like scalar transport. This isolates pressure, advection, boundaries and memory ownership. The second adds solid obstacles and measured geometric error. The third adds liquid interfaces, volume diagnostics and liquid-specific pressure boundaries. Non-Newtonian stress and particle-grid methods are extensions whose cost must be justified by a concrete failure of the baseline.

A source mask, obstacle mask and fluid mask are different fields. A source injects state according to a schedule. An obstacle restricts motion. A liquid mask identifies pressure unknowns and free-surface neighbors. Treating all three as one occupancy array creates contradictory boundary behavior. Visual density is an advected tracer unless a model explicitly equates it with mass density; smoke brightness cannot be used directly as the coefficient in an incompressibility equation.

The first acceptance suite should establish finite state, controlled divergence, stated transport behavior, stable boundary semantics and reproducible guidance. A frame-time target remains a design target until a workload, hardware class, competing model-memory load and variability policy are chosen. Real time is not a theorem and a short reference script is not its benchmark.

Reading the evidence

Each chapter derives discrete contracts before discussing implementation. Worked examples use small systems so sign, scale and conservation failures can be inspected by hand. The experiments extend these checks to a three-dimensional graph and periodic transport. Chapter 16 maps every formal theorem to its assumptions. Chapter 17 defines a Rust implementation sequence; later language ports should inherit verified behavior rather than independently reinterpret the equations.

2 From conservation to a tractable model

Consider a fluid region Ω , velocity $\mathbf{u}$, mass density ρ , pressure p , and body acceleration $\mathbf{f}$. Local mass conservation is

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0.$$

If density is constant along trajectories, then $D\rho/Dt = 0$, and the product rule gives $\nabla \cdot \mathbf{u} = 0$. This is a model assumption about volume preservation, not a statement that pressure is zero. Pressure is the constraint force that makes velocity compatible with incompressibility and boundaries.

For a Newtonian fluid of constant density and viscosity, momentum becomes

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\rho^{-1} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0.$$

Here $\nu = \mu/\rho$ is kinematic viscosity, with units square metres per second. Pressure has units kilograms per metre per second squared. Confusing dynamic viscosity μ with ν introduces a factor of density. Chorin's original projection work provides a foundational velocity-pressure splitting construction [S3]. Rheon's finite operators are specified independently in later chapters.

Nondimensionalization

Choose a length L , speed U , time L/U , and pressure scale ρU^2 . Substitution gives

$$\partial_t \hat{\mathbf{u}} + (\hat{\mathbf{u}} \cdot \hat{\nabla}) \hat{\mathbf{u}} = -\hat{\nabla} \hat{p} + \text{Re}^{-1} \hat{\nabla}^2 \hat{\mathbf{u}} + \hat{\mathbf{f}}, \quad \text{Re} = UL/\nu.$$

Changing scene scale while retaining numerical settings changes the apparent material. A two-metre scene moving at one metre per second with $\nu = 0.01$ has $\text{Re} = 200$. Reducing length by ten at the same speed and viscosity gives 20. Configuration should retain world units or explicitly declare nondimensional parameters, never silently interpret one value as both.

Gravity introduces $\text{Fr} = U/\sqrt{gL}$; surface tension introduces $\text{We} = \rho U^2 L/\sigma$, where σ is force per unit length.

Changing surface tension to obtain a desired shape can be intentional but should not be called calibrated water.

Features with radii comparable to grid spacing have poorly resolved curvature and pressure jumps.

Energy and boundaries

Multiply momentum by $\rho \mathbf{u}$, integrate, and assume sufficient smoothness, incompressibility and vanishing boundary terms, such as periodic boundaries:

$$\frac{d}{dt} \frac{\rho}{2} \int_{\Omega} |\mathbf{u}|^2 dV = -\mu \int_{\Omega} |\nabla \mathbf{u}|^2 dV + \rho \int_{\Omega} \mathbf{f} \cdot \mathbf{u} dV.$$

Advection becomes a boundary flux; pressure vanishes against divergence-free velocity; integration by parts makes viscosity dissipative. Moving walls, inlets and outlets add work and transport terms. Energy gain is not automatically wrong, but it needs an identified source. Vorticity confinement is an added force and belongs in this accounting.

What the model excludes

A constant-density incompressible solver omits acoustic waves, compressibility-driven shocks and phase-transition density changes. A buoyant smoke tracer under a Boussinesq approximation is not combustion. A single-phase liquid with prescribed atmospheric pressure does not model trapped compressible bubbles. An inviscid baseline cannot predict calibrated viscous folding. No finite collection of discrete lemmas proves global regularity of three-dimensional Navier-Stokes flow.

These exclusions guide the API. A material enum should describe implemented constitutive behavior. If the only stress is constant Newtonian viscosity, expose that. If foam means a thresholded visual tracer, say so rather than imply a multiphase physical model.

Splitting errors

Let A denote transport and B another evolution operator. One step $e^{\Delta t B} e^{\Delta t A}$ differs from $e^{\Delta t (A+B)}$ by commutator terms of order Δt^2 locally under suitable smoothness. Over fixed duration, the usual global order is one. Symmetric splitting can improve order under compatible assumptions, but changing fluid regions and boundary handling can invalidate that improvement. Test accuracy on smooth manufactured problems before advertising it for obstacles or free surfaces.

3 Staggered geometry and discrete operators

A Cartesian staggered grid stores scalars at cell centers and velocity components on faces normal to their axes. Harlow and Welch's marker-and-cell formulation is the historical foundation [S2]. Pressure has $n_x n_y n_z$ entries; x-velocity has $(n_x + 1)n_y n_z$, y-velocity has $n_x(n_y + 1)n_z$, and z-velocity has $n_x n_y(n_z + 1)$. These shapes differ even in a cubic domain.

With origin $\mathbf{o}$ and spacing h , a cell center is $\mathbf{o} + h(i + 1/2, j + 1/2, k + 1/2)$. An x-face is at $\mathbf{o} + h(i, j + 1/2, k + 1/2)$. Each component therefore requires its own sampling offset. Using cell-centered offsets for all components creates a half-cell error and can destroy an otherwise correct projection.

Divergence from flux balance

Integrate divergence over a cubic cell, approximate each face flux by its stored normal velocity, and divide by volume:

$$(D\mathbf{u})_{ijk} = \frac{u_{i+1,j,k} - u_{i,j,k}}{h} + \frac{v_{i,j+1,k} - v_{i,j,k}}{h} + \frac{w_{i,j,k+1} - w_{i,j,k}}{h}.$$

The pressure gradient at an interior x-face is $(p_{i,j,k} - p_{i-1,j,k})/h$. The same face appears in two neighboring divergences with opposite signs. This shared-face requirement gives cancellation of internal fluxes.

For the formal model, define a cells-by-faces matrix B . Each oriented internal face column contains -1 at its tail and $+1$ at its head. At unit spacing, physical divergence is $-B\mathbf{u}$, while pressure difference is $B^T p$. The proofs and experiments use this convention. A uniform physical grid inserts factors $1/h$. Cut cells require cell-volume and face-area metrics, not an informal reuse of the unweighted identity.

The adjoint identity

Exchange finite sums:

$$\sum_i p_i \sum_e B_{ie} u_e = \sum_e \left(\sum_i B_{ie} p_i \right) u_e.$$

This exact discrete integration-by-parts identity says $p^T B u = (B^T p)^T u$. Physical divergence and gradient are negative adjoints. The Lean theorem named `adjoint_identity` proves the finite-sum statement for every real matrix of the declared shape. It does not prove that array traversal assembled the right matrix.

Two cells connected by one face give

$$B = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad B^T p = p_1 - p_0, \quad BB^T = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Face velocity 3 gives physical divergence $(3, -3)$, whose sum vanishes. Reversing both orientation and stored velocity sign preserves the physical field; reversing only one is a bug.

Layout and indexing

Use x-fastest flattening:

$$\text{index}(i, j, k) = i + n_x(j + n_y k).$$

The Lean theorem `flatten_in_bounds` proves this lies below $n_x n_y n_z$ for in-range natural coordinates. Machine integers add a separate obligation: shape products and additions must fit before allocation or indexing. Validate shape arithmetic with checked operations when constructing the grid. Retain distinct validated shape types for cells and each face orientation. Repeated handwritten dimension arithmetic spreads the contract across every kernel.

Separate interior stencil loops from boundary updates. Ghost layers simplify sampling but enlarge storage and need one refresh owner. Stale ghost velocities can make transport and pressure use different wall conditions. Tests should include a constant field, linear pressure ramp, one nonzero face and manufactured divergence-free vortex, checking boundaries as well as interiors.

Rectangular cells and physical units

Let cell widths be h_x, h_y, h_z , volume $V = h_x h_y h_z$, and x-face area $S_x = h_y h_z$. Integrating mass flux first, then dividing by volume, gives $S_x/V = 1/h_x$. Thus the rectangular-grid divergence has different denominators per direction. Replacing all three by one average spacing changes both the PDE and the pressure spectrum.

Pressure in pascals divided by x-center separation h_x gives a pressure gradient in pascals per metre. Multiplying by inverse density and time step gives metres per second, as needed for a velocity correction. The x-face correction is

$$u_{i,j,k}^+ = u_{i,j,k}^* - \frac{\Delta t}{\rho_{i,j,k}^x h_x} (p_{i,j,k} - p_{i-1,j,k}).$$

Substituting all three face corrections into divergence gives the familiar seven-point operator. With constant density, an interior row has diagonal $2/(\rho h_x^2) + 2/(\rho h_y^2) + 2/(\rho h_z^2)$, and opposite-direction neighbors receive the corresponding negatives. Every coefficient has units inverse density per square metre. The right-hand side is negative physical divergence divided by time step.

This row-divided form is symmetric when all volumes are equal. For unequal cell volumes, dividing each row by its own volume usually destroys Euclidean symmetry. CG must then use a symmetrized integrated system or a correctly weighted formulation. Calling the nonsymmetric row-divided matrix a Laplacian does not make ordinary CG valid.

Integrated metric construction

Separate topology from geometry. Let B be the unscaled incidence matrix, S the diagonal open-face area matrix, L the diagonal pressure-center distance matrix, and V the diagonal cell-volume matrix. Unknown face speed is u . Let Q be the vector of known outward volume flux through prescribed-velocity boundaries. Then

$$Du = V^{-1}(Q - BSu), \quad Gp = L^{-1}B^T p.$$

The first expression is affine because of known boundary flux. The two-point gradient assumes pressure sample separation is aligned with the face normal and L contains that normal separation. This holds for the selected Cartesian MAC pressure sample locations. Positive areas, distances and volumes alone do not establish consistency for arbitrary cut-cell centroids. Nonorthogonal centroid corrections require a separately derived gradient and matching adjoint operator; they are outside this baseline. An affine-pressure fixture independently checks the normal-gradient consistency and demonstrates failure on a skew connection. With diagonal face density R , correction is $u^+ = u^* - \Delta t R^{-1} G p$. The symmetric integrated pressure system is

$$Kp = b, \quad K = BSR^{-1}L^{-1}B^T, \quad b = (BSu^* - Q)/\Delta t.$$

For positive geometry and density, face coefficient $S_e/(\rho_e \ell_e)$ is nonnegative. K has the same graph energy structure as the Lean model. It is symmetric in the ordinary Euclidean inner product because cell-volume division was left outside the solve. The physical residual relation is now

$$Du^+ = -\Delta t V^{-1}(b - Kp).$$

Consequently a small integrated residual can still produce large divergence in tiny cells. A stopping test should inspect volume-scaled divergence as well as solver norms. The existing Lean theorem is the unweighted algebraic core; this physical metric interpretation is a derived specification.

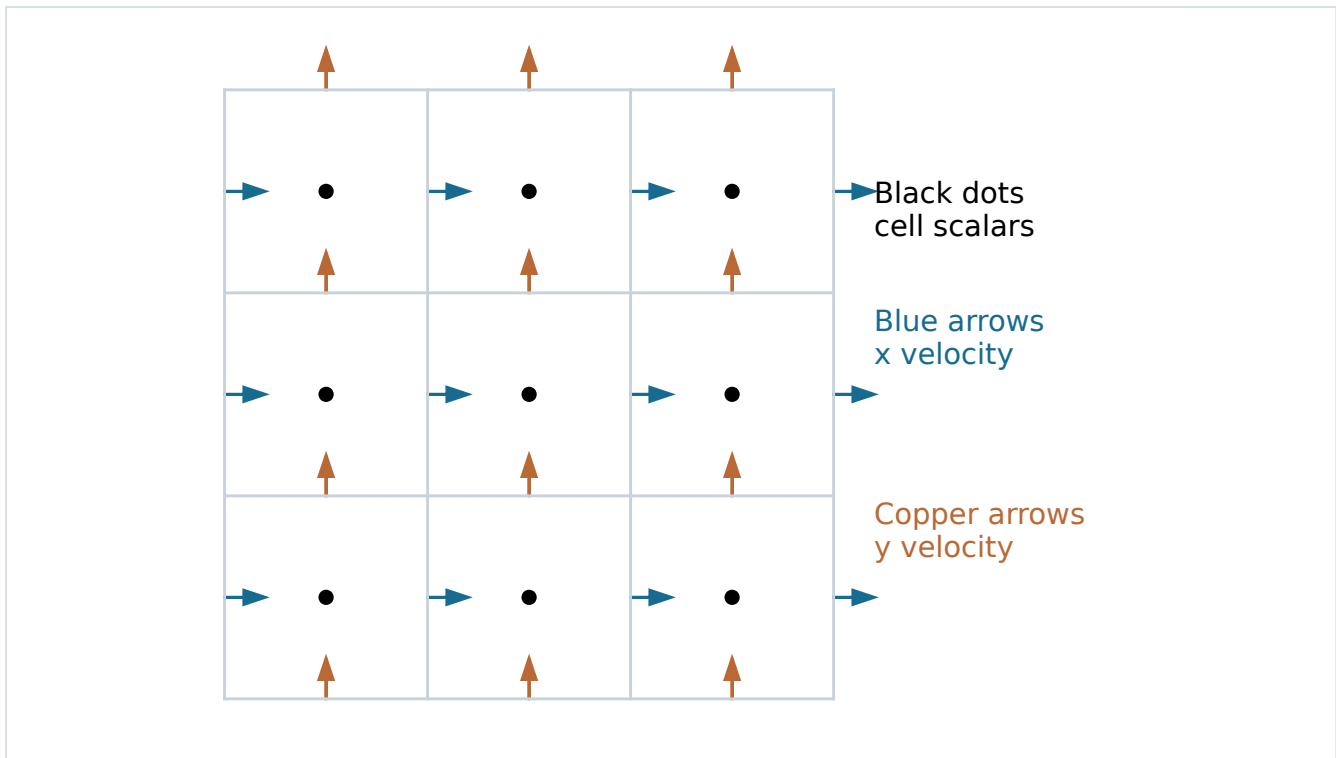
For dimensional checking, $S/(\rho \ell)$ has units metres to the fourth per kilogram. Multiplying by pressure gives cubic metres per second squared, matching volume flux divided by time. V then converts the corrected flux mismatch to inverse seconds. This check catches missing density, distance and volume factors before a simulation is run.

Coefficient storage

For a regular rectangular grid, store dimensions and three reciprocal spacings once. For irregular geometry, each active face needs open area, pressure distance and density information, or a precomputed transmissibility with provenance. One transmissibility value must be shared by both adjacent pressure rows. Independently computing it twice can introduce asymmetry through different clipping or density choices.

At a density discontinuity, derive the coefficient from the selected pressure-gradient model. For two segments with lengths ℓ_a, ℓ_b and densities ρ_a, ρ_b , a series-resistance argument gives transmissibility $S/(\rho_a \ell_a + \rho_b \ell_b)$. This differs from blindly averaging inverse density at the face. The correct choice depends on how pressure and acceleration are integrated; retain the derivation and test a one-dimensional interface fixture.

Cell-volume, face-area and center-distance validity are separate checks. A positive volume does not imply a valid connecting face. A zero area removes a connection; a zero distance is invalid. These distinctions determine graph connectivity and pressure gauges.



Original reference illustration for 3 Staggered geometry and discrete operators

4 Pressure as a constrained correction

Let $\mathbf{u}^*$ be velocity after non-pressure updates and W a diagonal matrix of nonnegative face weights. The graph correction is

$$\mathbf{u}^+ = \mathbf{u}^* - \Delta t W B^T p, \quad A = B W B^T, \quad b = B \mathbf{u}^* / \Delta t.$$

Solving $A p = b$ makes $B \mathbf{u}^+ = 0$, hence physical divergence $D = -B$ vanishes. Density and grid-spacing factors can be absorbed into W and B , but the implementation must state how. A dimensionless graph formula must not be copied into a metre-based solver without scaling.

The pressure quadratic form

$$p^T A p = (B^T p)^T W (B^T p) = \sum_e w_e (p_{\text{head}(e)} - p_{\text{tail}(e)})^2.$$

This is nonnegative when weights are nonnegative; the matrix is symmetric. Theorems `pressure_energy`, `pressure_nonnegative` and `pressure_symmetric` establish these facts. Strictly positive weights make zero energy imply equal pressure across every edge. On a connected closed graph, pressure is constant; on a disconnected graph, each component has its own constant. This connectivity-to-kernel argument is not currently in the Lean inventory.

A closed domain therefore gives a positive semidefinite operator. Pin one pressure per component by eliminating the row and column, or solve in the mean-zero subspace. Replacing only a row with an identity row breaks symmetry unless the column is handled consistently. Prescribed pressure is eliminated and its known contribution moved to the right-hand side.

Compatibility

Each closed component needs a zero-sum pressure right-hand side. Balanced incidence columns imply $\mathbf{1}^T B = 0$. A net inlet into a sealed domain violates compatibility. Subtracting the right-hand-side mean changes the requested problem. It may be an explicitly chosen correction policy but cannot be silent recovery.

The `constant_pressure_nullspace` theorem proves the constant nullspace under balanced columns.

`Internal_flux_conservation` proves matching cancellation. A nonzero total divergence in a fully closed graph, beyond expected rounding, should trigger assembly, boundary timing or unit checks before increasing solver iterations.

Residual as a diagnostic

For approximate pressure p_k , let $r_k = b - A p_k$. Substitution gives, without a convergence assumption,

$$B \mathbf{u}^+ = \Delta t r_k, \quad D \mathbf{u}^+ = -\Delta t r_k.$$

Projection_residual is the central checked theorem. It links stopping to a velocity constraint. A residual norm means different things if the matrix has been multiplied by h^2 , density or time step. Recompute divergence through the production operator and report units. Relative residual alone is insufficient near zero right-hand side; use a documented absolute floor.

Two cells worked completely

Take time step 1, weight 1 and face velocity 3. Then $b = (-3, 3)$. Fix $p_0 = 0$; the reduced equation gives $p_1 = 3$. Corrected velocity is $3 - (3 - 0) = 0$. At $p_1 = 2.9$, residual is $(-0.1, 0.1)$, velocity is 0.1 and physical divergence is $(0.1, -0.1)$. Residual and divergence agree exactly, while acceptable magnitude remains a scene-dependent engineering decision.

For positive weights, minimize velocity change in the W^{-1} metric subject to incompressibility. A Lagrange-multiplier calculation yields the same normal equations. This explains nonincrease of the corresponding kinetic norm for compatible exact projection without boundary work. The current Lean package proves operator identities, not the complete constrained-minimization theorem or its finite-precision version.

The matrix can be algebraically correct and geometrically wrong. The theorem accepts arbitrary B ; it cannot distinguish a neighboring cell from an incorrectly indexed distant one. Independent geometry and stencil tests are essential.

Assembly by boundary case

Every candidate face should be classified before assembling a pressure row. A fluid-fluid face contributes coefficient w to both diagonals and $-w$ to both off-diagonals. A fluid-solid face with prescribed normal speed supplies known boundary flux; it does not add a fictitious pressure neighbor. A fluid-air face with prescribed pressure contributes w to the fluid diagonal and $w p_{\text{air}}$ to its right-hand side. An inactive air-air face contributes no liquid pressure equation.

For example, one liquid cell connected to a prescribed zero-pressure surface at half a cell distance has coefficient $S/(\rho h/2) = 2S/(\rho h)$. Treating the surface as a whole-cell distance halves the coefficient and changes the correction. Conversely, treating a solid as zero-pressure air creates a leaking wall. These two common mistakes can both produce plausible-looking motion while violating the intended boundary.

An isolated liquid cell surrounded entirely by fixed walls has no pressure correction degree of freedom through internal faces. If its prescribed net flux is zero, its pressure is undetermined and velocity is already constrained by boundary data. If net flux is nonzero, the problem is incompatible. It is not repaired by adding a tiny diagonal unless the model explicitly permits compressibility or leakage.

Connected components and gauges

Build a graph using pressure unknowns and strictly positive fluid-fluid transmissibilities. Traverse it with a queue or union-find and record component IDs. A component touching a prescribed-pressure face is anchored by that boundary. A closed component needs a compatibility test and one gauge.

Compute compatibility from integrated flux, using a numerically careful reduction. Its tolerance should account for the magnitudes accumulated and the accepted flux budget. A component with a large inlet and equally large outlet can have cancellation error much larger than an otherwise identical quiescent component. A fixed unscaled epsilon cannot distinguish these cases reliably.

For a closed compatible component, choose a deterministic gauge cell, such as the smallest stable cell index. Remove its row and column; set its known pressure to zero. If using a mean-zero projection instead, apply the projection consistently to residuals, directions and preconditioning. A preconditioner that introduces constant components can corrupt a nominally mean-zero solve.

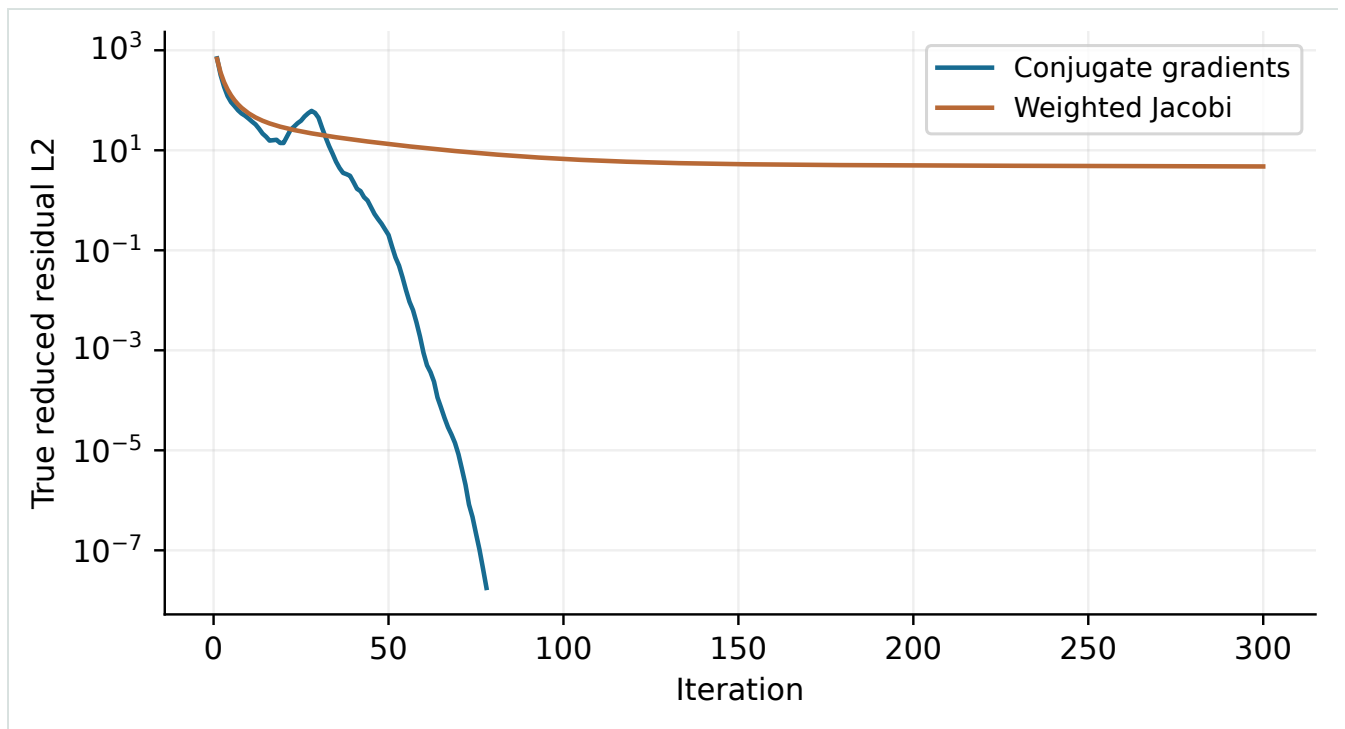
The topology snapshot should retain the mapping between full cell indices and reduced unknown indices. Do not rebuild it nondeterministically inside every matrix application. When geometry changes, invalidate the operator, component map, preconditioner and any pressure warm start whose gauge mapping is no longer meaningful.

Acceptance in physical space

After solving, apply the pressure gradient once using the same transmissibilities or metric factors as assembly. Then recompute physical divergence independently from the corrected face velocities and prescribed boundary data. Compare this with the predicted residual-derived divergence.

Their difference has two sources: floating arithmetic and inconsistent implementation. The companion quantifies a small arithmetic discrepancy on its reference graph. A much larger mismatch is evidence of wrong coefficients, stale geometry, inconsistent density or a boundary overwrite after correction. This paired diagnostic localizes failures better than either pressure residual or divergence alone.

The worked matrix appendix supplies exact chain and cycle fixtures. Add production tests for one closed component, two disconnected components, one prescribed-pressure opening, an isolated cell and incompatible moving-wall flux. These cases test the domain logic that the arbitrary-B theorem intentionally leaves open.



Original reference illustration for 4 Pressure as a constrained correction

5 Solving the pressure system

Pressure often determines both frame cost and volume quality. The solver receives an SPD reduced system or an explicitly handled semidefinite system. It returns pressure, iterations, true residual, termination reason and breakdown diagnostics. A vector alone loses information needed to accept or reject a frame.

Jacobi as a reference

With diagonal D_A , weighted Jacobi is

$$p^{k+1} = p^k + \omega D_A^{-1}(b - Ap^k).$$

Convergence requires spectral radius below one for $I - \omega D_A^{-1}A$ on the solved subspace. Parallel local access is simple, but smooth error can decay slowly as grids grow. A fixed iteration count is not fixed divergence quality. Isolated zero-diagonal cells must be classified, never divided by zero.

Jacobi is useful as an independent reference and multigrid smoother. Compare time to an accepted residual, including reductions and synchronization, not one iteration against an unrelated solver iteration.

Conjugate gradients

For an SPD matrix and SPD preconditioner M , initialize $r_0 = b - Ap_0$, $z_0 = M^{-1}r_0$, $d_0 = z_0$. Then

$$\alpha_k = \frac{r_k^T z_k}{d_k^T A d_k}, \quad p_{k+1} = p_k + \alpha_k d_k, \quad r_{k+1} = r_k - \alpha_k A d_k,$$

$$z_{k+1} = M^{-1}r_{k+1}, \quad \beta_k = \frac{r_{k+1}^T z_{k+1}}{r_k^T z_k}, \quad d_{k+1} = z_{k+1} + \beta_k d_k.$$

Shewchuk develops Krylov methods and conditioning in detail [S5]. Rheon's operational contract additionally detects nonfinite values, nonpositive curvature, zero denominators and exhausted budgets. Check the initial residual before forming any quotient. Zero right-hand side and zero guess should return immediately.

The exact-arithmetic energy-error bound for SPD systems is

$$\|e_k\|_A \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \|e_0\|_A.$$

It is not a runtime prediction. Spectrum, preconditioner cost, precision and memory traffic matter; a singular system requires a specified subspace. The current Lean package does not prove CG convergence.

Preconditioner selection

Diagonal scaling is cheap and parallel. Incomplete Cholesky can reduce iterations but adds factorization, ordering dependence and triangular solves. Modified incomplete Cholesky is discussed in Bridson's notes [S4]. Geometric multigrid attacks smooth error, but coarse levels must retain obstacle and free-surface semantics. A coarse operator that reconnects sealed regions changes the nullspace.

A matrix-free seven-point stencil is attractive for a known regular coefficient structure. Reconstruct neighbor contributions from face weights and diagonals. An explicit sparse matrix remains valuable in tests as an inspectable independent oracle. Comparing two wrappers around one kernel cannot catch shared assembly errors.

Residual and warm starts

Recursive residuals drift from $b - Ap$ in floating point. Recompute the true residual at termination and periodically where needed. Document direction restarts after replacement. Warm starts help only when operator, pressure gauge and topology remain compatible. New disconnected components or grid rescaling require remapping or reset.

The companion uses an 8-cubed closed grid, removes one row and column, and projects deterministic random velocity. It records CG and weighted-Jacobi histories. This compares mechanisms on that small system, not production frame time. Failure leaves the last accepted state intact or publishes a declared degraded preview according to an explicit lifecycle policy.

Residual scales and breakdown policy

Let the integrated solve be $Kp = b$. A relative criterion $\|r\|_2 \leq \tau_{\text{rel}} \|b\|_2$ can be combined with an absolute criterion, but physical acceptance is additionally $\|\Delta t V^{-1} r\|_\infty \leq \tau_{\text{div}}$. The first measures linear-system progress; the second limits maximum cell divergence. They can disagree in cut cells. Physical acceptance must reconstruct the full ungauged residual on every active cell, including eliminated gauge cells. A reduced-system residual alone can hide accumulated imbalance in the removed row. Independently recompute divergence on every active cell after correction. The companion includes a compatible chain where all retained residual entries pass a threshold but the eliminated row fails it.

For a diagonal preconditioner, require positive diagonal entries for every reduced unknown. If the matrix has a zero row, classify the topology before invoking CG. If $r^T z$ or $d^T K d$ is nonpositive in an expected SPD solve, return a breakdown with the iteration and scalar values. Do not take an absolute value or clamp the denominator; that changes the method and conceals the failed assumption.

The source of a breakdown can be an invalid operator, wrong gauge, loss of precision or an inappropriate preconditioner. Retrying in higher precision is useful only after the operator checks pass. A retry should start from a coherent state, recompute true residual and retain the original failure diagnostic.

Why multigrid helps

Jacobi damps different error frequencies at different rates. For a one-dimensional uniform Dirichlet Laplacian, the diagonal-scaled Fourier symbol is $1 - \cos \theta$. Weighted Jacobi's error multiplier is $1 - \omega(1 - \cos \theta)$. Small θ gives a multiplier near one, so smooth error decays slowly. After transferring to a coarser grid, that same physical error becomes less smooth in grid coordinates and easier to correct.

This argument motivates a two-grid method: smooth the current estimate, restrict its residual, solve for a coarse correction, prolong it, and smooth again. Saad provides a primary textbook treatment of multigrid and sparse iterative methods [S22]. The construction below states Rheon's proposed contract rather than asserting an implemented hierarchy.

Transfer operators

Let P map coarse pressure corrections to fine pressure corrections. For uniform cell-centered grids, construct P by trilinear interpolation at fine cell centers expressed in coarse-cell coordinates. The offset is important: a fine center is not generally a coarse node. Near boundaries, interpolation must respect prescribed pressure and component membership.

Choose restriction R consistently with the inner product. In an integrated symmetric system, $R = P^T$ yields the Galerkin operator $K_H = P^T K_h P$. If K_h is SPD and P has full column rank on the solved subspace, then for nonzero coarse x ,

$$x^T K_H x = (Px)^T K_h (Px) > 0.$$

Thus coarse SPD follows from construction, with explicit rank and subspace assumptions. For volume-normalized residuals, volume-weighted restriction is more natural; convert consistently instead of mixing formulas.

On a closed graph, preserving constants requires $P\mathbf{1}_H = \mathbf{1}_h$. Coarse connectivity must not join separate fine components. A coarse cell spanning a thin solid barrier needs multiple component degrees of freedom or a hierarchy policy that preserves separation. Otherwise the coarse correction allows nonphysical communication.

V cycle specification

```
v_cycle(level, rhs, estimate):
    if level is coarsest:
        solve the gauge-fixed coarse system accurately
        return estimate
    apply fixed pre-smoothing sweeps
    residual = rhs - K[level](estimate)
    coarse_rhs = transpose(P[level]) * residual
    coarse_error = zero
    coarse_error = v_cycle(level + 1, coarse_rhs, coarse_error)
    estimate += P[level] * coarse_error
    apply compatible post-smoothing sweeps
    return estimate
```

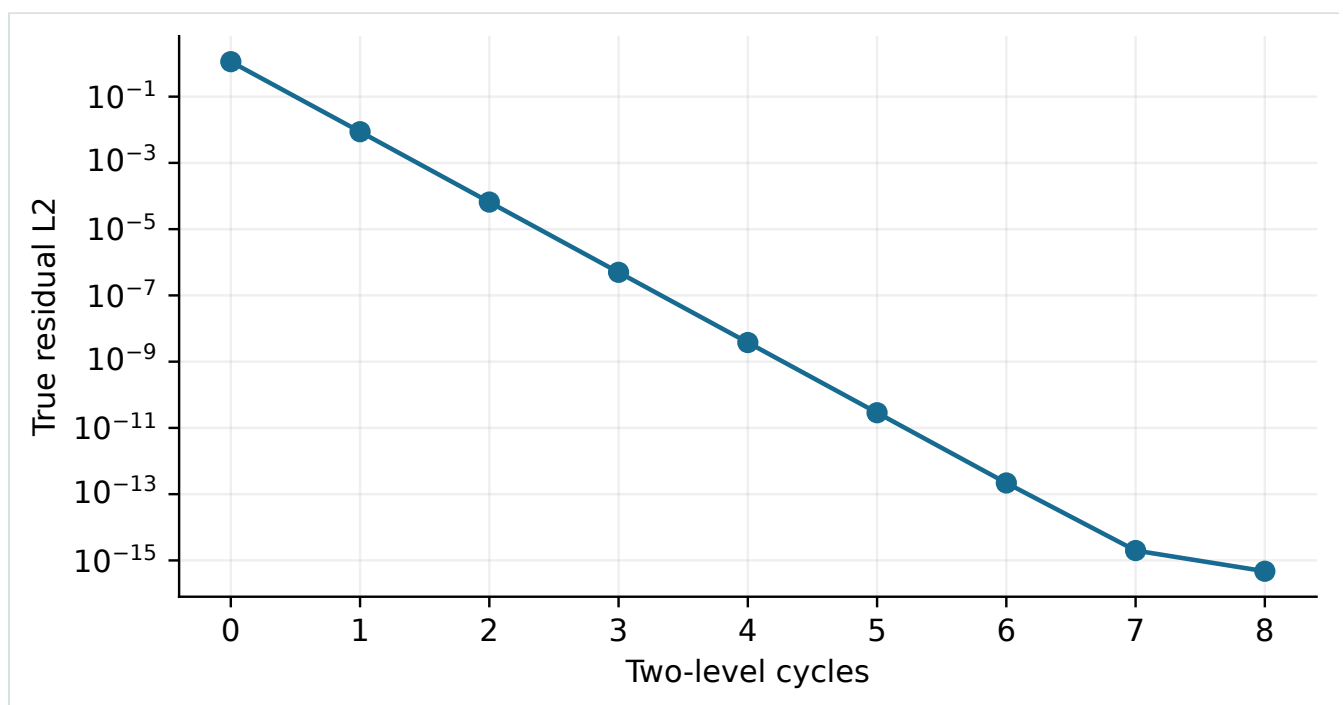
As a standalone solver, monitor the finest-grid true residual after cycles. As a CG preconditioner, the V cycle must act as a fixed linear SPD map on the relevant subspace. Matching adjoint pre/post smoothing and an SPD coarse solve are important. Adaptive iteration counts, nonsymmetric Gauss-Seidel order or changing hierarchy can violate the usual PCG assumptions. A flexible Krylov method is a different algorithmic choice, not a label to add after the fact.

The expanded companion constructs a small Dirichlet one-dimensional hierarchy with interpolation P, Galerkin coarse matrix and weighted Jacobi smoothing. It checks symmetry, positive eigenvalues and measured residual reduction. This is an executable mechanism fixture, not a three-dimensional obstacle-aware multigrid implementation.

Storage and work

A regular three-dimensional geometric hierarchy has cell counts $N, N/8, N/64, \dots$; their sum approaches $8N/7$. This favorable count excludes per-level masks, transfer metadata, coefficients and temporary vectors. An irregular hierarchy can have much more overhead. Count actual simultaneous buffers, especially when used inside PCG.

For low-resolution Rheon scenes, a simpler diagonal-PCG solve may outperform a complex hierarchy once setup and changing geometry are included. Measure complete solve time to a physical tolerance. Multigrid is a promising mechanism, not a required optimization before evidence.



Original reference illustration for 5 Solving the pressure system

6 Transport and the price of stability

A passive scalar satisfies $\partial_t q + \mathbf{u} \cdot \nabla q = 0$ for incompressible source-free flow. Along an exact characteristic it is constant. Semi-Lagrangian transport traces backward from an arrival sample and interpolates the old field. Stam's stable-fluids construction popularized this in interactive graphics [S1]. Stability must name a norm and interpolation rule.

Backtracing

First-order tracing uses $\mathbf{x}_d = \mathbf{x} - \Delta t \mathbf{u}(\mathbf{x})$. A midpoint estimate first samples velocity halfway along that estimate, then traces with the midpoint velocity. Each staggered component uses its own lattice offset; the scalar has another offset.

Trilinear interpolation is a convex combination of eight values when fractional coordinates lie in $[0, 1]$. Nonnegative weights summing to one give

$$\min_i q_i \leq \sum_i w_i q_i \leq \max_i q_i.$$

Convex_sum_bounds proves this finite real statement. Repeated bounded interpolation cannot leave the original global range if boundaries and sources obey it. This does not imply conservation of all sample values.

Row sums and column sums

For $q^+ = Tq$, convex interpolation means nonnegative entries and unit row sums. Conservation of an unweighted total requires unit column sums. For example,

$$T = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \end{bmatrix}$$

maps $(1, 0, 0)$ to $(1/2, 1/2, 1/2)$. Outputs are bounded but total rises from 1 to 1.5. The exact rational arithmetic is checked in Lean. This is a matrix counterexample to an implication, not a claim that this matrix comes from smooth incompressible periodic flow.

Constant translation on a periodic uniform one-dimensional grid happens to give a doubly stochastic interpolation matrix. The companion's box test verifies that special case, not general semi-Lagrangian conservation. Variable velocity and irregular boundaries need separate evidence.

Numerical diffusion

For positive constant speed and Courant number $c = u\Delta t/h$ in $[0, 1]$,

$$q_j^+ = (1 - c)q_j + cq_{j-1}.$$

A Fourier mode has amplification $g = (1 - c) + ce^{-i\theta}$, hence

$$|g|^2 = 1 - 2c(1 - c)(1 - \cos \theta).$$

Amplitude generally decays. Taylor expansion gives leading artificial diffusion $uh(1 - c)/2$ under fixed-Courant refinement. A bounded solver can therefore erase thin jets or smear silhouettes. Arbitrarily large stable steps also degrade trajectory accuracy and obstacle interaction.

Conservative flux updates

Finite volume computes one shared face flux and applies opposite signs:

$$q_j^+ = q_j - \frac{\Delta t}{h}(F_{j+1/2} - F_{j-1/2}).$$

Summation over periodic cells cancels all internal fluxes. Positive-speed upwind flux $F_{j+1/2} = uq_j$ preserves positivity for $0 \leq c \leq 1$, proved by upwind_positive. At $c = 3/2$, $q_j = 1$, $q_{j-1} = 0$, the new value is $-1/2$, also formally checked. Conservation does not imply positivity.

The field determines the contract. A smoke appearance tracer may tolerate measured dissipation. A liquid fraction needs bounded conservative transport and interface geometry. A signed-distance field needs surface-position analysis. One generic quality parameter cannot hide these differences.

Component aware trilinear sampling

For an x-face field, convert world position to lattice coordinates $s = ((x - o_x)/h_x, (y - o_y)/h_y - 1/2, (z - o_z)/h_z - 1/2)$. For a y-face, subtract half from x and z; for a z-face, subtract half from x and y. For cell-centered scalars subtract half from all coordinates.

Let $a = \lfloor s \rfloor$ and $t = s - a$. The eight weights are products of either t_d or $1 - t_d$. For corner $\epsilon \in \{0, 1\}^3$,

$$w_\epsilon = \prod_{d=1}^3 (\epsilon_d t_d + (1 - \epsilon_d)(1 - t_d)).$$

Their sum factors into $\prod_d (t_d + 1 - t_d) = 1$. Nonnegativity follows when each fractional coordinate lies in $[0, 1]$. This derivation explains the exact hypotheses of the interpolation proof. Boundary remapping must preserve or deliberately replace the stencil; simply dropping invalid corners breaks partition of unity.

An affine scalar $q(x, y, z) = a + bx + cy + dz$ is reproduced exactly by trilinear interpolation on a complete stencil in real arithmetic. The expanded companion tests this on all four lattice offsets using independently evaluated world positions. It then deliberately uses the wrong offset for an x-face field to show the expected half-cell bias. This fixture catches a defect that constant-preservation tests miss.

Sampling boundaries

Periodic sampling wraps integer indices modulo the component's periodic extent, accounting for duplicated boundary faces if the storage convention has them. A closed-wall sampler uses its declared ghost extension. A clamped texture sampler repeats edge values; it is not automatically a no-slip boundary condition. An out-of-domain trace at an open boundary needs a prescribed exterior value or a validity failure.

Return a sample together with validity information when the method needs it. Corrected advection needs to know whether forward and reverse stencils are valid and reversible. A scalar value alone cannot communicate that a trace hit a wall or crossed a source discontinuity.

Trace integrators

For an autonomous frozen velocity field, midpoint backtracing is

$$k_1 = \mathbf{u}(\mathbf{x}), \quad k_2 = \mathbf{u}(\mathbf{x} - \tfrac{1}{2}\Delta t k_1), \quad \mathbf{x}_d = \mathbf{x} - \Delta t k_2.$$

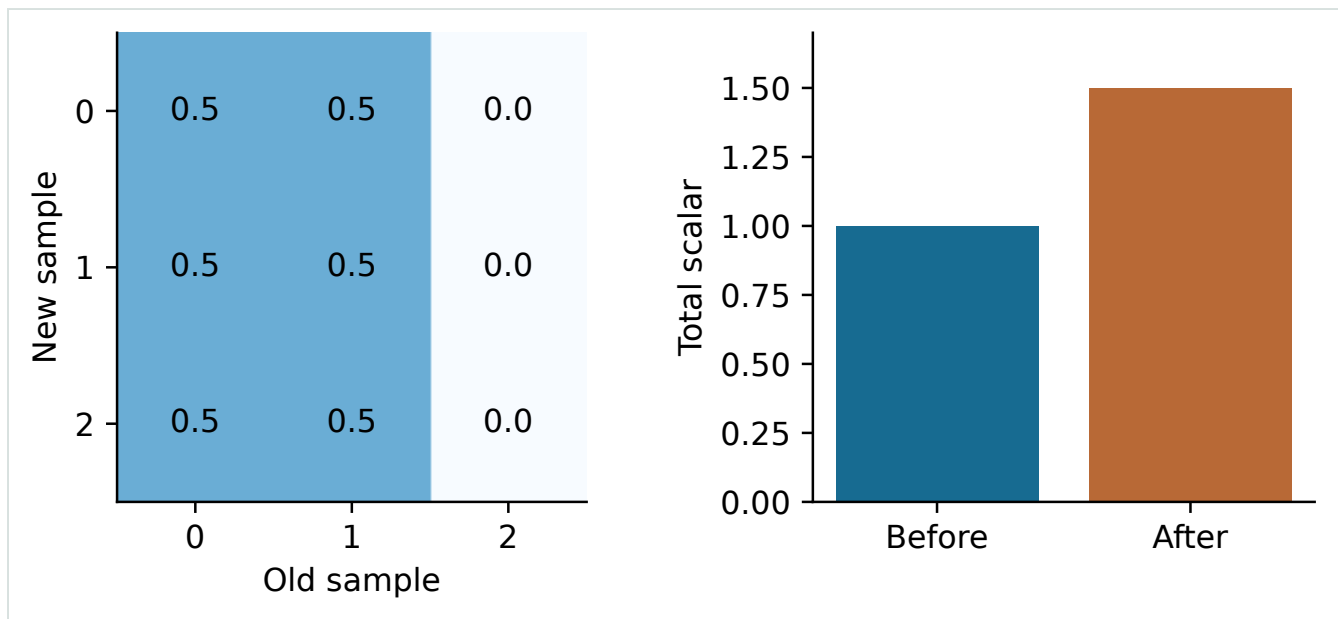
A higher-order trace reduces ODE integration error but cannot improve interpolation order or unresolved velocity detail. For time-varying velocity, temporal interpolation between known levels is another approximation. Name which field drives tracing: old velocity, provisional velocity or projected velocity. These choices change the splitting.

A solid-rotation fixture with $\mathbf{u} = (-\omega y, \omega x, 0)$ has an analytic rotation trajectory. Compare departure points against the exact rotation over one step and over one period. A translation fixture cannot distinguish Euler from midpoint tracing because both are exact for constant velocity. The companion includes the rotation test specifically to expose that difference.

Failure fixtures

Use a constant field to test partition of unity, an affine field to test offsets, a sinusoid to measure dissipation, a top-hat to reveal overshoot and limiter behavior, and a segment crossing a thin wall to test collision handling. Add a trace ending exactly on a lattice boundary, where floor and integer conversion policies matter.

For corrected transport, record the number of reverted samples and the maximum unbounded overshoot before reversion. A zero overshoot on a smooth wave is not evidence for discontinuities. For conservative transport, compare integrated amount separately from min/max bounds. These fixtures should remain distinct because each isolates a different contract.



Original reference illustration for 6 Transport and the price of stability

7 Higher order transport with bounded failure

Linear interpolation is robust but dissipative. Let $S_{\Delta t}$ denote a basic semi-Lagrangian step. A forward-backward correction computes

$$q_f = S_{\Delta t} q, \quad q_b = S_{-\Delta t} q_f, \quad q_c = q_f + \frac{1}{2}(q - q_b).$$

This is the teaching companion's precise formula. Selle and collaborators analyze an unconditionally stable MacCormack construction [S6]. Several related formulas share that informal label; specify the exact correction and limiter rather than relying on the name.

Overshoot and limiting

Forward and reverse interpolation average values. Their composition loses detail; adding half the estimated loss sharpens smooth fields. Subtraction can make effective weights negative, so the convex maximum principle no longer applies. Sharp interfaces can acquire negative concentration or excessive values.

The companion reverts an out-of-range corrected value to the forward value. Bounds come from the old departure interpolation stencil. Clamping is another policy with different truncation behavior. Neither automatically conserves scalar total. The limiter belongs in the method specification and saved configuration.

The inverse trace here reuses one velocity field. It does not invert a time-dependent physical trajectory. Near collisions, emitters and open boundaries, the paths may not represent reversible transport. Mark invalid traces and revert instead of treating arbitrary boundary values as error estimates.

Executed refinement

The companion advects $q(x) = \sin(2\pi x)$ once around a periodic unit interval at Courant number 0.5. At 32, 64, 128 and 256 cells, linear RMS errors are 0.187922, 0.101090, 0.052478 and 0.026743. Limited corrected errors are 0.037630, 0.009022, 0.003211 and 0.001022. These are executed measurements, not assumed order labels.

The corrected scheme retains more amplitude. Refinement ratios vary because the limiter activates near extrema. Do not claim uniform second-order convergence from the unbounded formula alone. Compute $\log_2(E_h/E_{h/2})$, inspect limiter activity and include nonsmooth data. A different step-size sequence asks a different convergence question.

Velocity and conservation

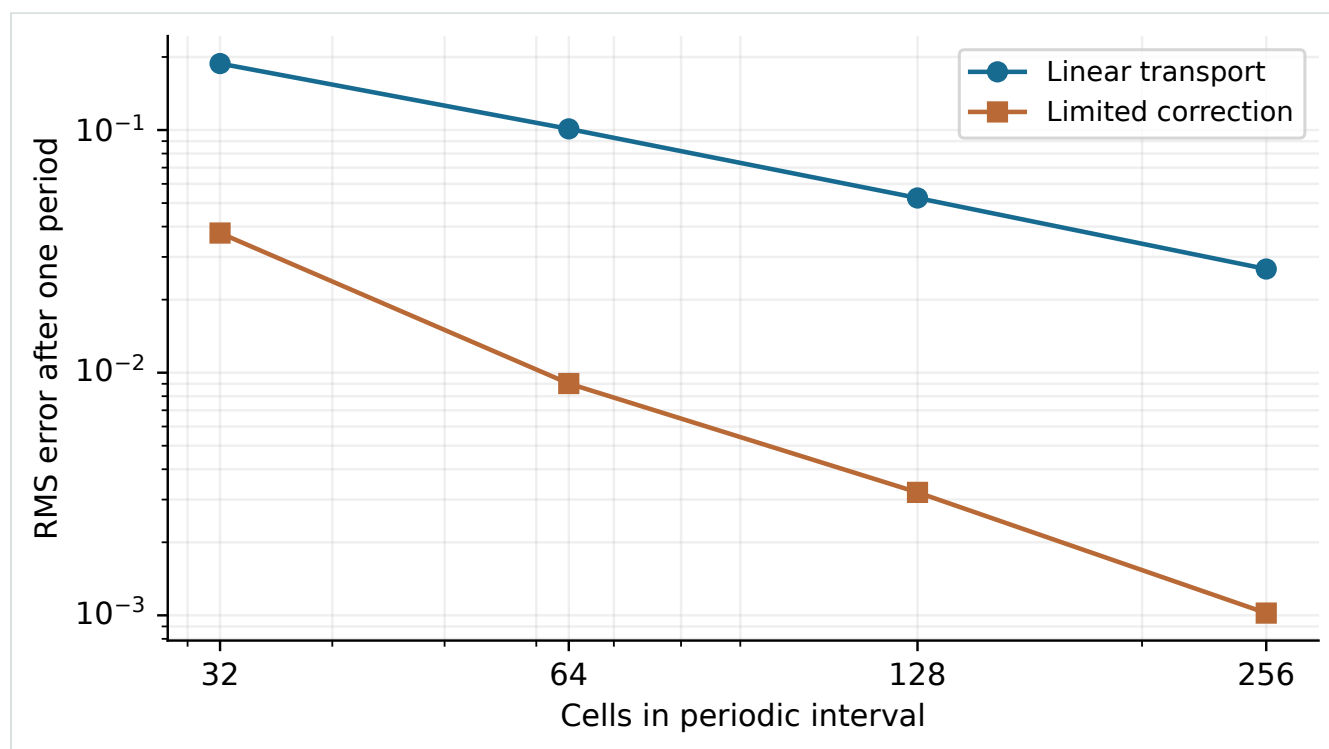
For material volume, use a bounded conservative finite-volume or geometric volume-of-fluid construction. A corrected sampler is an appearance tool unless additional conservation structure is established. Sharp plume images are not mass or momentum evidence.

Velocity components live on different grids and have no scalar positivity requirement. Componentwise limiting may suppress meaningful extrema. Projection removes divergence but cannot undo all momentum or energy errors from interpolation. Measure circulation, energy and vortex displacement as well as divergence. Prescribed-velocity tracer accuracy does not prove self-advected vortex accuracy.

Buffer dependencies

Old data must survive both forward and backward passes; forward data survives correction. Overwriting it early creates an order-dependent in-place scheme. Per-component processing lowers peak memory only if velocity reconstruction does not read an already overwritten component. Immutable input views and separate output slices expose this dependency directly.

The implementation should count fallback samples and invalid backtraces. A method that reverts over most of the domain is functionally close to the first-order baseline, even if its configuration says high order. Report observed behavior rather than only the selected option.



Original reference illustration for 7 Higher order transport with bounded failure

8 Boundaries and obstacles

On a wall moving at $\mathbf{u}_s$, impermeability requires $(\mathbf{u} - \mathbf{u}_s) \cdot \mathbf{n} = 0$. No-slip also matches tangential velocity; free-slip does not. These are distinct models. Prescribed normal wall velocity enters the divergence right-hand side.

Atmospheric-pressure openings instead fix pressure and allow flow. Combining both conditions on one face can overconstrain the system.

Binary geometry

A first obstacle model can classify cells and block faces. The resulting staircase approximates a curved boundary but changes effective volume and creates grid-aligned features. Refinement can move the represented wall discontinuously as classifications change. An accurate linear solve cannot remove geometric error.

A boundary snapshot should have one version consumed by advection, pressure and rendering for a step. If geometry changes during pressure solution, discard or restart the stale candidate rather than publish velocity corrected for a different domain.

Fractional faces

An open area fraction a_e and fluid cell-volume fraction θ_i produce divergence terms proportional to $a_e h^2 / (\theta_i h^3)$. Tiny θ_i creates small-cell stiffness. Clamping coefficients changes the model; removing cells changes topology. Either can be an explicit approximation, not a free fix.

Batty, Bertails and Bridson develop variational solid-fluid coupling on coarse Cartesian grids [S7]. The structural lesson is to choose gradient, divergence and energy together. Rheon's arbitrary-matrix theorem does not establish cut-cell quadrature.

For the linear homogeneous operator D_0 , with cell metric M_c and face metric M_f , require

$$\langle p, D_0 \mathbf{u} \rangle_{M_c} = -\langle Gp, \mathbf{u} \rangle_{M_f}, \quad D_0 = -M_c^{-1} G^T M_f.$$

An attractive divergence formula paired with an unrelated gradient can yield an inconsistent pressure operator. Test this linear weighted adjoint identity with random vectors on every supported geometry class after prescribed boundary flux is separated. For the metric construction in Chapter 3, $M_c = V$, $M_f = SL$, and $D_0 = -V^{-1}BS$. The full physical divergence is affine: $D(u) = D_0 u + V^{-1}Q$. Therefore its identity includes boundary work:

$$\langle p, D(u) \rangle_V = -\langle Gp, u \rangle_{SL} + p^T Q.$$

Omitting the final term is valid only when its value is zero, for example homogeneous prescribed flux. The companion includes nonzero Q to reject that omission. The existing Lean arbitrary-B theorem proves the linear algebraic identity, not moving-boundary assembly.

Trajectory collision

A departure point outside solid material does not prove the path avoided it. A long trace can cross a thin wall and emerge on the far side. Intersect straight traces with geometry and apply the declared rule at the earliest hit. Multi-stage curved traces need compatible approximation. Limiting travel helps but is not collision proof.

Boundary interpolation must define its extension. Constant extrapolation, mirrored normal velocity and no-slip ghost values differ. Do not bury them in a texture sampler mode. Check samples on both sides of walls and at corners.

Diagnostic sequence

Test a sealed box with uniform translation, then stationary fluid under gravity with hydrostatic pressure, then moving obstacles with displaced-volume and wall-work accounting. A wall thinner than one cell must be resolved or explicitly unsupported rather than silently leak.

Report flux by boundary category and closed component. Long-term leaks often originate in one inconsistent face update. More pressure iterations help only if pressure residual is responsible; topology, geometry and buffer timing remain separate suspects.

9 Liquid interfaces and volume

A liquid needs an interface representation beyond velocity. A level set stores ϕ , negative inside, with zero contour at the surface. A signed-distance field has $|\nabla\phi| = 1$ nearby. Osher and Sethian introduced a Hamilton-Jacobi framework for evolving fronts [S8]. Merging and splitting are natural, but interpolation and reinitialization can move the contour.

Reinitialization

Transport satisfies $\partial_t\phi + \mathbf{u} \cdot \nabla\phi = 0$. A common distance-restoring evolution is

$$\partial_\tau\phi + \text{sgn}(\phi_0)(|\nabla\phi| - 1) = 0.$$

Artificial time τ seeks distance-like values with fixed reference sign. Discretization can shift the zero set, especially in thin sheets. Reinitialization frequency, stencil, sign smoothing and termination belong to the method specification. Smooth appearance does not imply correct volume.

A narrow band must cover interpolation support and travel before rebuilding. Outside values need valid classification and reconstruction when activated. Invalid far-field sentinels must not be interpolated into the surface.

Pressure at the free surface

Neglecting air dynamics, surface tension and liquid normal viscous stress, set surface pressure to atmospheric zero. If normal viscous stress is retained, this is only a declared split approximation; a coupled traction condition is required as discussed in Chapter 10 and Batty-Bridson Section 3 [S11]. A pressure connection with interface distance αh uses that shorter distance, increasing its coefficient as α shrinks. A minimum-distance policy changes pressure and interface error and must be evaluated.

Velocity needs extension into a limited air band for transport. The pressure-unknown mask, velocity-validity mask and distance-band mask are distinct. Extending velocity does not mean the whole air region solves the liquid equations.

Volume fractions

A fraction $f_i \in [0, 1]$ gives represented volume

$$V_h = \sum_i f_i V_i.$$

Shared liquid-volume fluxes with opposite signs preserve global volume except for sources and boundaries. Hirt and Nichols established the volume-of-fluid approach [S9]. Geometric reconstruction estimates a local interface to determine swept liquid volume; it is substantially more work than advecting a color scalar.

A cell fraction 0.3 and Courant number 0.5 illustrate the risk: an outgoing flux selected without interface geometry can remove more liquid than exists. Bounding available volume must coordinate faces in multiple dimensions. Clamping final fractions can restore range while destroying total volume. Preserve both by construction or disclose the correction.

Particle correction and extraction

Particle level-set methods use particles carrying side-of-interface information to correct local errors [S10]. Reseeding, radii and collision handling add state and memory. They do not make level-set volume exact.

Select a liquid method from acceptance metrics. Level sets simplify smooth geometry but need drift budgets. Volume fractions expose conservation but need reconstruction. Hybrids increase verification burden. Finish the baseline grid contracts before committing to a liquid branch.

Mesh extraction adds its own error. Edge crossings and ambiguous cases need consistent rules across cells. Normals use physical spacing and safe treatment of small gradients. Validate spheres at several subcell offsets, thin sheets and merging components. Compare volume, surface distance and rasterized silhouette separately. A guidance mesh need not be manufacturing-grade, but its promised topology must be explicit.

Piecewise linear interface reconstruction

A volume-fraction method needs geometry inside partially filled cells. A piecewise linear interface reconstruction chooses a unit normal $\mathbf{n}$ and plane offset d so the liquid region inside a cell is $\{x : \mathbf{n} \cdot x \leq d\}$ and its volume matches $f_i V_i$. For this liquid-side inequality, outward $\mathbf{n}$ is minus the normalized fraction gradient, since f decreases toward air. For a negative-inside level set, outward $\mathbf{n}$ is plus the normalized phi gradient. Use a resolved neighborhood estimate; when its norm is too small, return an unresolved normal or use a documented geometry-derived fallback, never normalize zero. The offset fraction-step fixture checks both signs. Then solve for d by a monotone volume function.

On a unit cube with $\mathbf{n}$ along positive x , the solution is simply $d = f_i$ for fractions in $[0, 1]$. For a general normal, clip the cube polyhedron against the plane and compute its volume, or use a verified analytic formula. Bisection is robust because volume increases monotonically with d . Bracket d using the minimum and maximum plane projections of cube vertices. Stop according to volume and geometric tolerances, not an arbitrary iteration count alone.

Transport then clips the reconstructed liquid region against the swept face volume over the step. In a simple axis-aligned constant-speed case, this is a slab of width $u\Delta t$. In multiple dimensions, directional splitting and unsplit flux constructions handle corner transport differently. A raw sum of independently computed face outflows can exceed available liquid. A production implementation must select one coherent bounded-conservative method and test its multidimensional flux accounting.

The expanded companion only checks the axis-aligned slab case: a cell filled to fraction 0.3 with a positive swept width 0.5 sends the geometrically intersected amount under an explicitly positioned interface. It does not claim a complete arbitrary-normal PLIC implementation. General polyhedral clipping, orientation robustness and multidimensional boundedness remain implementation work.

Curvature from a level set

With outward normal $\mathbf{n} = \nabla\phi/|\nabla\phi|$, mean-curvature sum is $\kappa = \nabla \cdot \mathbf{n}$. In three dimensions a sphere of radius R has $\kappa = 2/R$ under this convention. When air viscous stress is negligible, normal traction balance is $p_{\text{liquid}} - p_{\text{air}} = \sigma\kappa + \mathbf{n}^T \tau_{\text{liquid}} \mathbf{n}$. The simpler Laplace jump $\sigma\kappa$ additionally neglects normal liquid viscous stress. Tangential traction

still requires its own condition. See Chapter 10 and [S11] Section 3; a pressure-only split must declare its approximation. Reversing the sign convention for phi reverses normal and curvature, so the pressure-jump sign must change consistently.

A direct finite-difference curvature formula is

$$\kappa = \frac{|\nabla\phi|^2 \text{tr}(H_\phi) - (\nabla\phi)^T H_\phi \nabla\phi}{|\nabla\phi|^3}.$$

It requires first and second derivatives and becomes ill-conditioned when gradient magnitude is small. Regularizing the denominator changes the curvature estimate. Near underresolved sheets or merged surfaces, a smooth signed-distance assumption may fail even if every value is finite.

The companion evaluates a sphere distance field and central-difference curvature at a known surface point over decreasing h. It compares against $2/R$. This is a derivative-stencil fixture, not a moving-droplet surface-tension validation. Production tests must additionally measure parasitic currents in a stationary droplet, pressure jump and oscillation behavior.

Surface tension time scales

Balance capillary pressure σ/h against inertial acceleration over length h. Acceleration scale is $\sigma/(\rho h^2)$; moving distance h under that acceleration gives time scale $\sqrt{\rho h^3/\sigma}$. This dimensional derivation explains severe restrictions as resolution increases. The stability constant depends on the actual discretization and integration method.

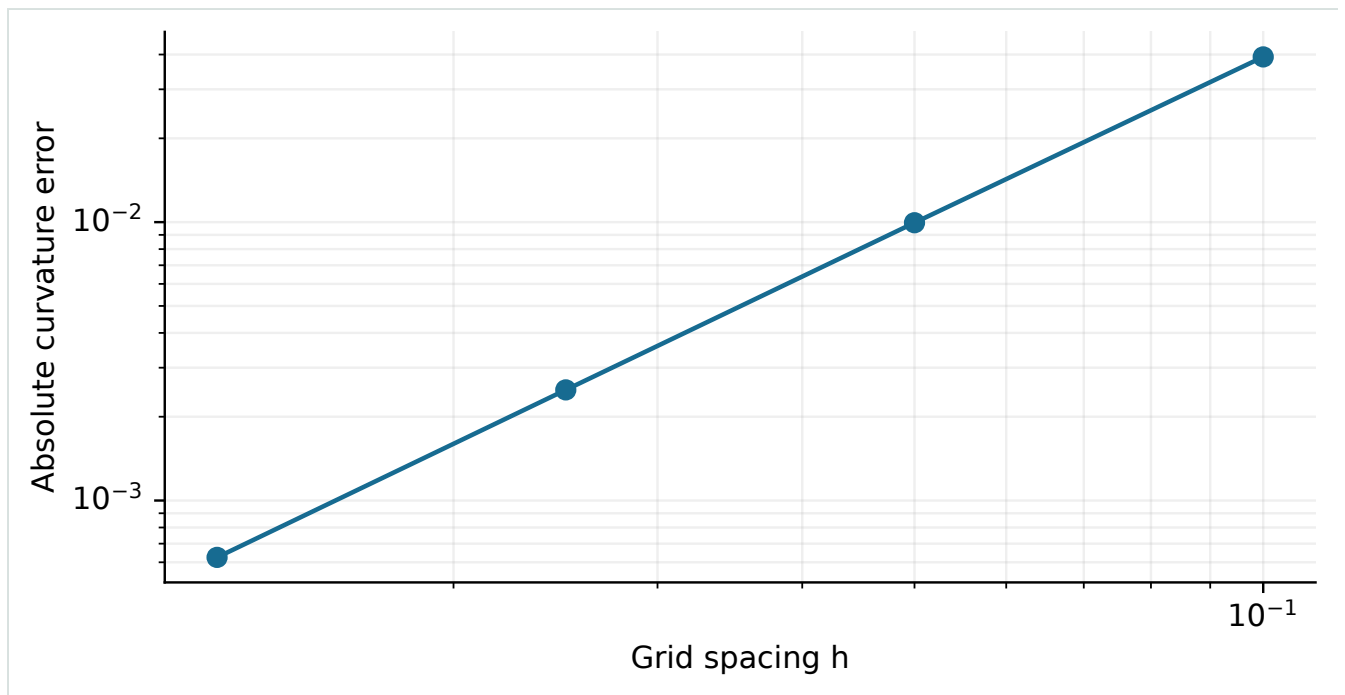
For a 1 cm feature scale, density 1000 kg per cubic metre and surface tension 0.072 newtons per metre, the dimensional time scale is approximately 0.118 seconds. At a 1 mm scale it is about 0.00373 seconds. These are scale estimates, not accepted time-step limits for an unimplemented solver.

Implicit tension may relax a stability restriction but introduces coupled geometry or linearization errors. It does not eliminate the need to resolve curvature or validate the interface. Likewise, increasing surface tension to suppress noise changes the material and can alter the desired guidance shape.

Volume correction as a declared operation

A level-set volume correction sometimes shifts phi by a constant until a target volume is restored. This moves the interface everywhere, including regions already accurate. It can repair a scalar volume metric while damaging local positional accuracy. If used, record shift magnitude, affected volume and resulting silhouette displacement.

A more informative acceptance report retains both volume drift and a spatial surface metric. Two shapes can have equal volume and completely different placement. Rheon's guidance purpose makes that distinction especially important.



Original reference illustration for 9 Liquid interfaces and volume

10 Forces viscosity and material behavior

Gravity is acceleration: $\mathbf{u}^* = \mathbf{u} + \Delta t \mathbf{g}$ needs no density multiplier. Pressure and stress use density according to the selected formulation. Configuration and types should distinguish world lengths, grid coordinates and dimensionless coefficients.

Hydrostatic balance

With upward z and gravity $-g\mathbf{e}_z$, rest requires $\partial_z p = -\rho g$. A linear pressure field with face differences $-\rho g h$ cancels gravity in ideal discrete arithmetic when boundaries agree. Measure speed, divergence and surface drift over many steps instead of judging whether a tank looks still.

Diffusion

One-dimensional explicit diffusion is

$$q_j^+ = q_j + r(q_{j-1} - 2q_j + q_{j+1}), \quad r = \nu \Delta t / h^2.$$

Nonnegative coefficients require $0 \leq r \leq 1/2$, established by `explicit_diffusion_positive`. In isotropic three-dimensional seven-point diffusion, the center coefficient is $1 - 6r$, giving the analogous restriction $r \leq 1/6$. That extension is derived here, not in the current Lean inventory.

Implicit Euler solves

$$(I + \nu \Delta t L) q^+ = q$$

for a positive semidefinite negative Laplacian L . An eigenmode of eigenvalue $\lambda \geq 0$ is multiplied by $1/(1 + \nu \Delta t \lambda)$, between zero and one. Large steps still change decay accuracy; boundaries remain essential.

The companion's 64-cell periodic sine mode uses $\nu = 0.01$, time step 0.001 and 100 steps. Final amplitude 0.9613286500 differs from continuum 0.9612907007 by about 3.79×10^{-5} . This combines spatial and temporal errors in a scalar oracle, not a complete viscous-fluid implementation.

Stress and constitutive choices

Newtonian stress is

$$\tau = 2\mu D(\mathbf{u}), \quad D(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T).$$

Replacing its divergence by $\mu \nabla^2 \mathbf{u}$ uses constant viscosity and incompressibility with compatible boundary terms. Variable viscosity and free surfaces can need coupled stresses. Batty and Bridson's variable-coefficient Stokes work is a primary reference [S11]. State the validity of a simpler componentwise approximation.

Smoke buoyancy may depend on temperature deviation and a density-like tracer. This is not compressible thermodynamics. Vorticity $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ can drive confinement using a normalized gradient of its magnitude crossed with vorticity. Fedkiw, Stam and Jensen use this to compensate for visually lost rotation [S12]. It injects energy; record work and guard near-zero normalization. Resolution changes its effective behavior.

A shear-dependent viscosity may use $\dot{\gamma} = \sqrt{2D : D}$ and regularized power law $\mu = K(\dot{\gamma}^2 + \epsilon^2)^{(n-1)/2}$. Regularization changes zero-shear behavior. Lagged coefficients, Picard iteration and Newton methods have different convergence requirements. Viscoelasticity additionally needs stress history and cannot be modeled by viscosity alone. Delay extensions until their constitutive tests and memory costs are justified.

A discrete variable viscosity energy

Use a discrete strain operator E mapping face velocities to symmetric strain components at appropriate sample locations. Let M contain quadrature volumes and positive viscosity coefficients, with shear factors chosen consistently with tensor contraction. A viscous dissipation quadratic has the form $u^T E^T M E u$. Its nonnegativity follows from positive coefficients, independent of a particular stencil layout.

A backward-Euler viscous update with face mass matrix M_u is

$$(M_u + \Delta t E^T M E) u^* = M_u u^n + \Delta t f.$$

This construction exposes symmetry and dissipation. It also shows why independently smoothing each component can miss cross derivatives when viscosity varies or free-surface traction matters. The pressure constraint can be coupled through a saddle-point system or handled by a specifically justified splitting; those alternatives have different boundary errors.

At an impermeable no-slip solid, prescribed velocity contributes known terms when eliminating boundary degrees of freedom. At a stress-free liquid surface, traction conditions belong to the variational boundary treatment. Reusing a closed-box Laplacian at the free surface silently imposes a different stress model.

The expanded companion assembles a small weighted difference energy $E^T M E$ and checks symmetry, nonnegative spectrum and implicit decay. This validates the algebraic construction only. It does not implement the complete three-dimensional symmetric-gradient stencil or free-surface traction quadrature.

Nonlinear viscosity iteration

For generalized Newtonian viscosity, coefficients depend on strain. A lagged-coefficient iteration takes the current velocity guess, evaluates shear rate and viscosity, builds the linear viscous system, solves it, and repeats until both velocity change and nonlinear residual meet their criteria.

```
u_guess = old_velocity
require max_iterations is a finite nonnegative integer
for iteration in range(max_iterations):
    strain = E(u_guess)
    viscosity = material_law(strain)
    require positive finite viscosity
    solve the frozen-coefficient system for u_candidate
```



```

    evaluate the original nonlinear residual at u_candidate
    if accepted: return u_candidate
    apply declared damping if needed
    u_guess = updated guess
    return NonlinearIterationLimit

```

The declared iteration budget bounds frozen-coefficient solves. Exhausting it (including a zero budget) returns failure without publishing a candidate velocity.

A small iterate change alone can mean stagnation, not convergence. Viscosity clipping can make the linear system manageable but changes the material law. Record the range and active clipping fraction. A yield-stress model needs a stated regularization or complementarity treatment; a high-viscosity threshold is only an approximation.

A simple shear test is the first constitutive oracle. Prescribe a linear velocity profile, calculate constant shear rate analytically, and compare stress against the selected law. Then use a channel-flow case with known or independently computed profile. A visually thick fluid is not sufficient calibration.

Prescribed and dynamic solid coupling

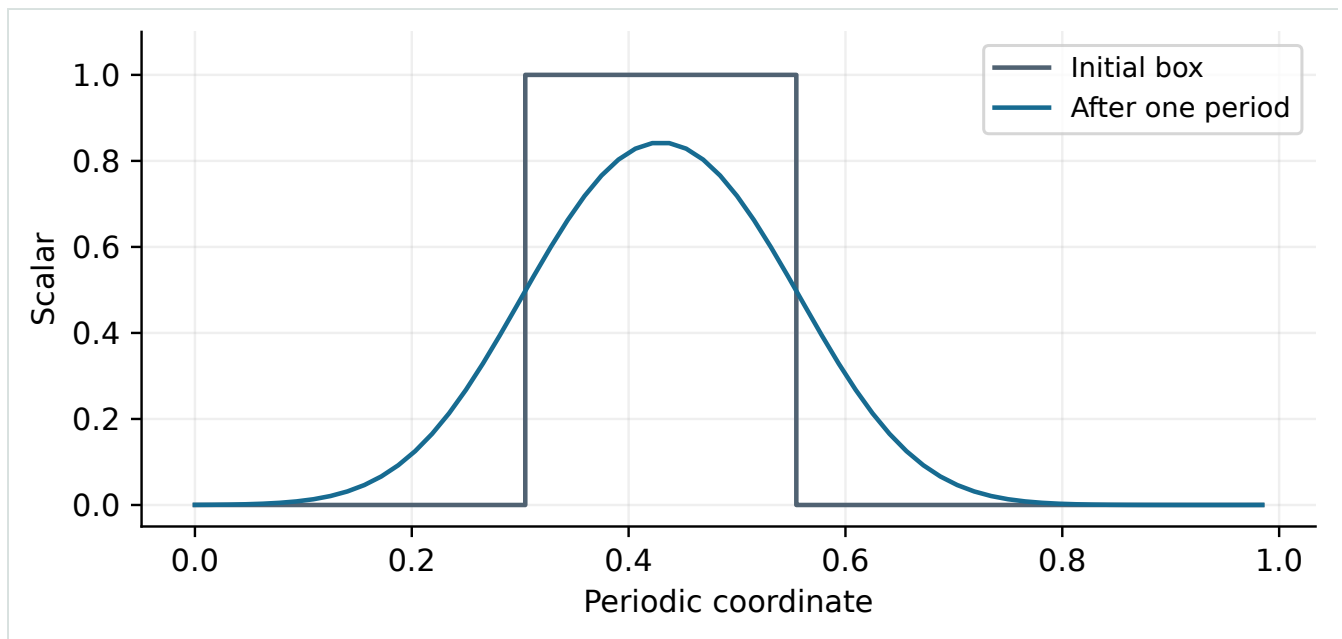
A prescribed obstacle has externally given motion. It can inject work into the fluid without receiving a matching reaction in the simulation. This is a valid one-way model if stated. A dynamic rigid body needs pressure forces and torques integrated over the same geometric interface used by the fluid operator.

For pressure traction $-p\mathbf{n}$, force and torque are

$$\mathbf{F} = - \int_{\partial S} p \mathbf{n} dA, \quad \boldsymbol{\tau} = - \int_{\partial S} (\mathbf{x} - \mathbf{x}_c) \times p \mathbf{n} dA.$$

Discretize these with the same face areas and normals used for flux coupling. Unrelated force quadrature can violate momentum exchange even if fluid divergence is small. The rigid body update also changes wall velocity, so explicit one-way sequencing can have added-mass instability for light bodies. A coupled solve or carefully qualified iteration is a separate capability.

Rheon's initial scope should support prescribed obstacles with explicit wall-work diagnostics. Dynamic two-way coupling should wait for momentum, rest-state and light-body tests. The variational coupling literature [S7] motivates that extension, but the current proof package does not verify it.



Original reference illustration for 10 Forces viscosity and material behavior

11 Particle alternatives and hybrid transfers

Dense grids pay for empty space; particles pay for occupied material and neighborhood organization. Include neighbors, sorting, scratch and surface reconstruction when comparing memory. Particle count alone is not a budget.

SPH

A smoothing-kernel density estimate is

$$\rho_i = \sum_j m_j W_h(\mathbf{x}_i - \mathbf{x}_j).$$

Support defines neighborhoods, often found through uniform bins or sorting. Pairwise force formulations can preserve antisymmetry when consistently evaluated, but free-surface neighbor deficiency and boundary quadrature remain. Müller, Charypar and Gross present an interactive SPH method [S13]; its published performance is not inherited by Rheon.

Weakly compressible pressure laws turn density deviation into force. Higher stiffness reduces compression but introduces fast waves and smaller steps. Visual softness may reflect tolerated compression. Measure density and volume errors before claiming incompressibility.

Position based fluids

Position based fluids iteratively corrects predicted positions using density constraints [S15]. For $C_i = \rho_i/\rho_0 - 1$, a local multiplier has the form

$$\lambda_i = -\frac{C_i}{\sum_k |\nabla_{\mathbf{x}_k} C_i|^2 + \epsilon}.$$

Regularization prevents singular corrections but changes behavior. Lists may become stale as positions change; rebuilding trades work for accuracy. Reconstructed velocity includes position corrections. Finite iterations yield approximate density constraints, not exact incompressibility or calibrated viscosity.

Compare complete frame cost at fixed occupied volume, obstacles, density tolerance, surface metric and memory cap. One density iteration and one pressure iteration are incomparable work units.

PIC and FLIP

PIC transfers particle momentum to a grid, applies grid forces and pressure, then samples velocity back. Repeated averaging dissipates detail. FLIP instead adds grid velocity changes:

$$\mathbf{v}_p^+ = \mathbf{v}_p + \sum_i w_{ip}(\mathbf{u}_i^+ - \mathbf{u}_i^{old}).$$

Blending PIC and FLIP trades damping against retained particle noise. Zero or tiny grid mass needs explicit handling. Parallel deposits need race-free accumulation.

APIC

APIC adds a local affine matrix C_p to particle velocity. A transfer includes

$$m_i \mathbf{u}_i = \sum_p w_{ip} m_p (\mathbf{v}_p + C_p (\mathbf{x}_i - \mathbf{x}_p)).$$

Jiang and collaborators derive conservation properties for specified transfer rules [S16, S17]. They depend on weights, moment matrices and updates. Merely adding nine floats does not inherit angular-momentum conservation. Boundary-truncated support also needs analysis.

Position, velocity, affine matrix and one scalar use 16 single-precision values, or 64 bytes before IDs, padding and scratch. Eight particles per cell in a filled 64-cubed region require 128 MiB for those fields alone; at 128 cubed, 1 GiB. These calculated costs matter in a model-memory-constrained host.

Choose dense grids for bounded low-resolution domains, sparse grids when empty space dominates, and particles when material motion justifies their overhead. Compare identical initial geometry, camera and metrics. No candidate wins every scene.

Complete PIC and FLIP update sequence

At particle positions x_p , evaluate weights w_{ip} on the selected grid lattice. Require partition of unity and, when claiming linear reproduction, $\sum_i w_{ip} x_i = x_p$. Deposit mass and momentum:

$$m_i = \sum_p m_p w_{ip}, \quad g_i = \sum_p m_p w_{ip} v_p, \quad u_i = g_i / m_i \quad (m_i > 0).$$

Store this normalized velocity as the old grid velocity before forces, pressure and collision changes. Apply the grid update to obtain $\tilde{u}_i$. Then

$$v_p^{\text{PIC}} = \sum_i w_{ip} \tilde{u}_i, \quad v_p^{\text{FLIP}} = v_p + \sum_i w_{ip} (\tilde{u}_i - u_i).$$

A blend with parameter $\beta \in [0, 1]$ is $(1 - \beta) v_p^{\text{PIC}} + \beta v_p^{\text{FLIP}}$. Advect positions using a declared grid-velocity interpolant and integrator, rather than assuming noisy FLIP particle velocity is always the best trajectory field. Collisions and reseedling follow a defined stage order.

A face-centered grid performs these deposits separately for each component lattice. Face mass is a transfer weight, not automatically the physical control-volume mass used in a pressure derivation. Keep the transfer normalization and fluid density model distinct. Near empty faces, a threshold must distinguish unavailable velocity from a valid zero.

Constant particle velocity is preserved by PIC transfer when all required mass and weight conditions hold. FLIP can preserve particle modes invisible to the grid because only grid changes are added. A pair of opposite particle velocities depositing to the same grid sample gives zero grid velocity; PIC removes that mode while FLIP retains it under a zero

grid update. This is a concrete noise/dissipation tradeoff, not simply an implementation defect.

Affine transfer moments

For frozen particle positions define offsets $r_{ip} = x_i - x_p$ and moment matrix

$$D_p = \sum_i w_{ip} r_{ip} r_{ip}^T.$$

A local affine particle field is $v_p + C_p r_{ip}$. Deposit

$$g_i = \sum_p m_p w_{ip} (v_p + C_p r_{ip}).$$

After grid evolution, a frozen-position least-squares affine reconstruction is

$$v_p^+ = \sum_i w_{ip} \tilde{u}_i, \quad H_p = \sum_i w_{ip} \tilde{u}_i r_{ip}^T, \quad C_p^+ = H_p D_p^{-1}.$$

These equations assume partition of unity, zero first moment and invertible D. They describe a transfer subproblem. Full time-integrated APIC variants require matching position and affine updates; conservation results from [S16, S17] cannot be transferred to an arbitrary integrator.

To see affine reproduction, let grid velocity be $a + Cx_i$. The particle average becomes $a + Cx_p$ because the weighted offset sum is zero. The cross moment is CD_p , so multiplying by D_p^{-1} returns C. The companion tests this with an independently constructed affine field. It does not test a full APIC liquid simulation.

Moment degeneracy and kernels

Linear hat weights in one dimension give $D = h^2 t(1 - t)$, which vanishes when a particle lies exactly on a grid point. An inverse then becomes undefined. Ignoring the case can generate huge affine coefficients near nodes. A pseudo-inverse, a limiting formula or a higher-order kernel are different choices that require their own consistency tests.

A tensor-product quadratic B-spline uses one-dimensional kernel

$$N(r) = \begin{cases} 3/4 - r^2, & |r| < 1/2, \\ \frac{1}{2}(3/2 - |r|)^2, & 1/2 \leq |r| < 3/2, \\ 0, & \text{otherwise.} \end{cases}$$

With complete regular-grid support, its moment is $D = (h^2/4)I$, giving a simple inverse and 27 collocated nodes per particle in three dimensions. Staggered component lattices require their own shifted supports; three components can mean 81 scalar deposits. The cost is not equivalent to eight-corner trilinear sampling.

Truncating support at a wall breaks moment identities. Renormalizing surviving weights restores their sum but generally not the zero first moment. Recentring changes the effective sample location. Ghost support, constrained reconstruction or boundary-specific transfer can address this, but each must be specified. The companion deliberately

removes one support node and measures the resulting moment defect instead of silently claiming affine preservation.

Momentum bookkeeping

Partition of unity gives total grid mass equal to total particle mass. Zero first moment removes the affine contribution to total linear momentum. Angular momentum has an internal affine contribution:

$$L_p = m_p x_p \times v_p + m_p \sum_i w_{ip} r_{ip} \times (C_p r_{ip}).$$

Comparing only $m_p x_p \times v_p$ would omit represented subparticle rotation. A transfer test should compare the full particle representation with grid angular momentum using the same support. Boundaries can exchange momentum; those impulses must be included rather than blamed on transfer error.

SPH kernels and gradients

For an explicit three-dimensional density fixture, use the compact poly6 kernel

$$W(r) = \frac{315}{64\pi h^9} (h^2 - |r|^2)^3 \quad \text{for } |r| < h,$$

and zero outside. Its gradient is

$$\nabla W(r) = -\frac{945}{32\pi h^9} (h^2 - |r|^2)^2 r.$$

The gradient is zero at coincident positions and at the support boundary under this formula. Other kernels may improve pressure behavior, but replacing the gradient with that of a different kernel changes the constraint derivative. The companion finite-differences the actual density constraint to test this exact gradient relationship.

For equal particle mass m and $C_i = \sum_j m W(x_i - x_j) / \rho_0 - 1$,

$$\nabla_{x_i} C_i = \frac{m}{\rho_0} \sum_{j \neq i} \nabla W(x_i - x_j), \quad \nabla_{x_j} C_i = -\frac{m}{\rho_0} \nabla W(x_i - x_j).$$

These derivatives sum to zero, expressing invariance under uniform translation. Include self density $mW(0)$ while excluding a spurious self-force. A finite-difference test that accidentally omits self density can still have the right gradient but the wrong density target.

PBF correction and neighbor validity

With equal solver weights, set

$$\lambda_i = -\frac{C_i}{\sum_k |\nabla_{x_k} C_i|^2 + \epsilon}.$$

Accumulating all constraint contributions gives

$$\Delta x_i = \frac{m}{\rho_0} \sum_{j \neq i} (\lambda_i + \lambda_j) \nabla W(x_i - x_j).$$

Use Jacobi-style separate correction buffers if all multipliers were computed from the same predicted positions. In-place updates change the iteration into an order-dependent method. Unequal masses require the corresponding inverse-mass-weighted constraint formulation; do not reuse the equal-weight equation unchanged.

A full step predicts positions from forces, builds neighborhoods, iterates density/collision corrections, reconstructs velocity from corrected displacement, then applies any declared viscosity or vorticity postprocess. Recompute density error after the final correction. A fixed iteration count defines a quality budget, not an incompressibility guarantee.

Uniform bins of width h need the 27 neighboring bins in three dimensions for support radius h . Hash collisions must be resolved by cell keys, and every candidate still needs an exact distance test. A cached neighbor list needs a skin radius and a displacement bound before reuse; otherwise particles entering support are missed. Sorting, prefix sums, bin ranges and particle permutations must preserve particle IDs and all state fields consistently.

Complete particle memory accounting

For P particles and K stored directed neighbor entries, a possible PBF layout uses old and predicted positions, velocity and correction vectors: $12P$ floats, or $48P$ bytes. Density and multiplier add $8P$ bytes. Two 32-bit sort/index arrays add $8P$ bytes. CSR neighbor offsets add approximately $4(P+1)$ bytes and neighbor IDs add $4K$ bytes. This totals roughly $68P + 4K$ bytes before boundary data, capacity padding, bin tables, renderer fields and alternative sort scratch.

At $P=100,000$ and mean directed neighbor count 60, these named arrays occupy about 30.8 MB in decimal units. A dense cluster can increase K sharply unless a cap changes the method; dropping neighbors silently changes density and forces. A checked capacity failure is preferable to an unreported truncated neighborhood.

For APIC, the earlier $64P$ -byte state estimate excludes old/new grid velocity, grid mass, sorting, particle output buffers and interface reconstruction. Count peak concurrent state for the actual update schedule. Particle methods save empty space only when these complete costs beat the selected grid alternative for the target scene.

12 Time stepping and error budgets

Physical time and display frames are separate clocks. A frame can contain substeps; a slow frame must not silently enlarge the physical step beyond its accepted regime. Decide whether overload slows simulation, drops presentation frames or changes a declared quality tier. Record the policy in replay metadata.

Restrictions

Directional Courant numbers are $c_x = |u|\Delta t/h_x$ and analogues. Multidimensional positivity depends on the update and can involve their sum. Characteristic transport may remain bounded without that restriction, but travel limits still help trajectory accuracy, collision detection and narrow-band validity.

Explicit diffusion depends on $\nu\Delta t \sum_d h_d^{-2}$. Surface tension introduces a capillary scale proportional to $\sqrt{\rho h^3/\sigma}$, with method-dependent constant. Acceleration and moving walls add constraints. Return a step and its limiting reason, not an unexplained number. If required substeps exceed a cap, return a typed outcome rather than label a partial interval complete.

Error accounting

A conceptual budget is

$$E_{\text{total}} \lesssim E_{\text{model}} + E_{\text{space}} + E_{\text{time}} + E_{\text{solve}} + E_{\text{round}} + E_{\text{surface}} + E_{\text{render}}.$$

This is an accounting guide, not a universal inequality across incompatible norms. Residual controls one constraint error, not advection displacement or mesh extraction. Machine-precision pressure with a coarse staircase obstacle may overspend on a subdominant error.

Refine space at fixed Courant number for smooth transport. Refine time with sufficiently small spatial error for temporal order. Simultaneously changing both mixes effects. For interfaces measure volume, surface distance and silhouette displacement; small average error can hide catastrophic filament loss.

Manufactured solutions

A periodic stream function $\psi = \sin x \sin y$ gives velocity $(\partial_y \psi, -\partial_x \psi)$ with zero continuous divergence. Sampling it does not automatically give zero discrete divergence; a discrete curl construction can ensure cancellation. Distinguish those tests.

Choosing discrete pressure p , computing $b = Ap$ and solving tests the solver for that operator, not the operator's physical accuracy. Add independent analytic and stencil checks. Expected values should not reuse the same indexing routine whose correctness is in question.

Long horizons and outcomes

Track volume or tracer amount, energy, divergence, surface position and rejection count after sources stop. Include rest, translation, rotation and eventually liquid and obstacle cases. Sensitive flows may not match pointwise indefinitely; use bounded time windows and defined geometric or statistical observables. Exact replay remains a separate implementation property.

Outcomes should distinguish converged, iteration limit, invalid input, nonfinite state, incompatible flux and resource limit. A nonconverged preview may be permitted explicitly but must not masquerade as accepted scientific state. Candidate and accepted buffers need clear ownership so cancellation cannot corrupt future steps.

13 Memory layout and parallel execution

A solver leaving no room for diffusion weights fails Rheon’s intended role even if fast alone. The relevant quantity is peak simultaneous memory: persistent state, scratch, render resources, readback buffers and the host application. Persistent simulation arrays alone underestimate it.

An explicit layout

For a cubic grid, face count is $F = 3n^2(n + 1)$. Current and next velocity, eight cell-centered single-precision arrays and a one-byte mask require

$$M(n) = 4(2F + 8n^3) + n^3 \text{ bytes.}$$

This calculated layout excludes allocator overhead, alignment, ghost layers, coefficients, meshes, particles, driver allocations and model weights. The eight scalars might be pressure, right-hand side, residual, search direction, matrix product, preconditioned residual and two tracer buffers. Assign actual lifetimes before reuse.

The formula gives 1.805 MiB at 32 cubed, 14.344 MiB at 64 cubed, 48.305 MiB at 96 cubed and 114.375 MiB at 128 cubed. Doubling every axis multiplies leading volume cost by eight. At 256 cubed this incomplete layout is about 913.5 MiB. Show estimates before allocation.

Ownership and access

Separate velocity arrays naturally match staggered shapes. Particle structures of arrays aid per-field traversal, while tiled layouts may suit transfers reading several fields. Benchmark actual access patterns. Avoid abstractions that conceal shape or introduce unmeasured per-sample dispatch.

A shape owns counts, strides, spacing, origin and checked capacity. State owns persistent fields. Workspace owns scratch and alias rules. A solver borrows them during a step. This exposes reuse without a global pool whose lifecycle is unclear.

For a face joining a and b, the pressure operator contributes $w(p_a - p_b)$ and its negative. Cell gathering writes one output without scatter races. Face scattering emphasizes paired conservation but requires atomics, coloring or local accumulation. Compare reproducibility and bandwidth.

Parallel reductions

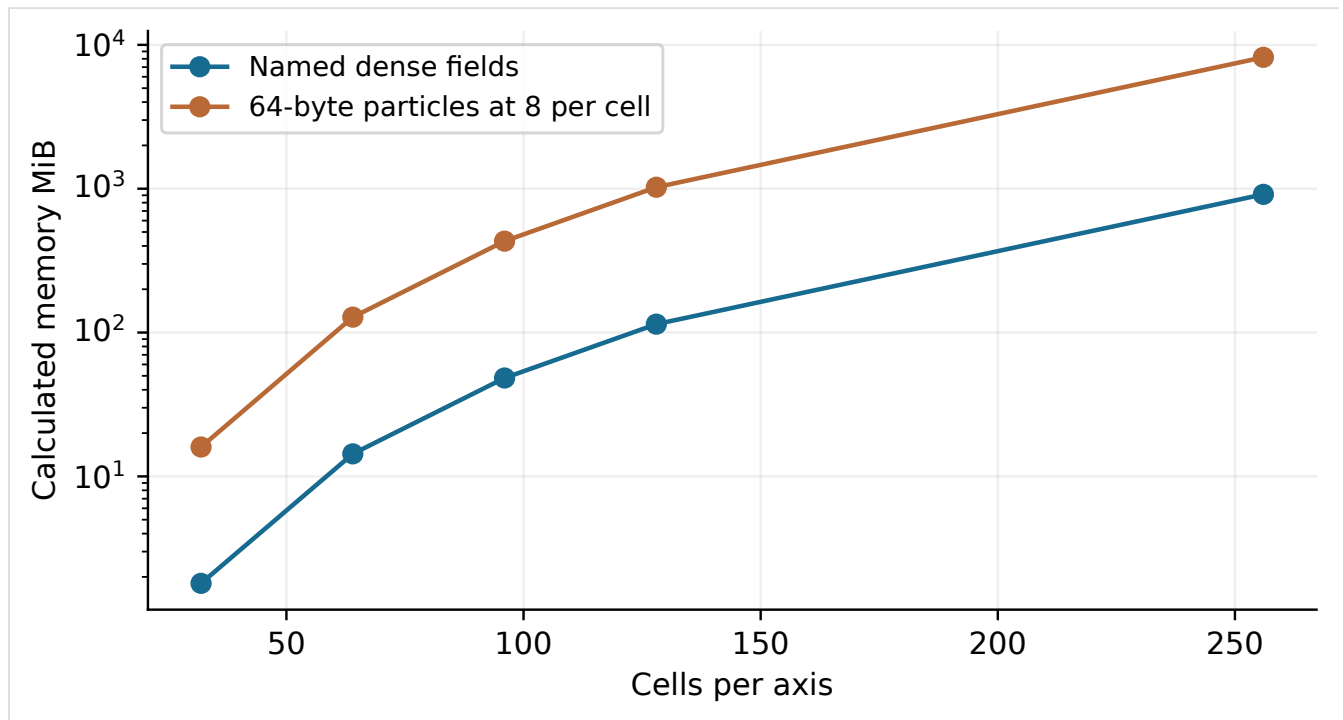
CG dot products are global reductions. Floating addition is nonassociative, so tree shape changes results. Fixed trees support repeatability on a pinned configuration; arbitrary atomics generally do not. Pairwise or compensated accumulation reduces some error at cost. Single-precision fields with double-precision reductions are a candidate to qualify, not a universally available GPU feature.

State whether replay is bitwise on one backend, tolerance-based across backends, or only configuration-reproducible. Save precision, solver tolerance, step policy and algorithm version when they affect the promise.

Sparse grids

Sparse tiles save empty-space cost but add keys, occupancy maps, halos and fragmented work. Benefits depend on active fraction and stencil expansion. A moving plume may activate many tiles at once; reserve or reject before mutating accepted state. Disconnected sparse pressure regions need consistent component gauges.

Begin with dense bounded grids and honest peak reporting. Adopt sparsity when representative scenes show empty-space cost is the limiting resource and topology machinery has a clear owner.



Original reference illustration for 13 Memory layout and parallel execution

14 Guidance as a geometric measurement

Guidance rendering is a measurement layer with a camera model, clipping, sampling and units. It must preserve the spatial information actually simulated. A downstream model may change appearance; Rheon should export reproducible geometric evidence.

Camera coordinates

For camera coordinates (X, Y, Z) with positive Z forward,

$$u = f_x X / Z + c_x, \quad v = f_y Y / Z + c_y.$$

Axial depth Z differs from radial range $\sqrt{X^2 + Y^2 + Z^2}$ off axis. Name the representation, handedness, image origin, clipping and pixel-center convention. Mirroring or shifts may be camera-contract defects rather than fluid errors.

Linearized projection gives

$$\delta u \approx (f_x / Z) \delta X - (f_x X / Z^2) \delta Z.$$

The same world-space error moves more pixels near the camera, and depth errors shift off-axis silhouettes. Evaluate actual camera distributions, not just a world-space norm.

Surface and volume outputs

Liquid can export silhouette, depth and normals in a declared frame. Smoke may integrate extinction. Piecewise constant extinction along a ray gives

$$T = \exp \left(- \sum_i \sigma_i \Delta s_i \right), \quad \alpha = 1 - T.$$

Increasing samples without adjusting segment length changes opacity incorrectly. Concentration-to-extinction is a rendering model, not conservation. A diffuse volume has no unique surface; first-hit depth needs a named threshold or accumulated-opacity rule.

Graphics depth buffers may be nonlinear in distance. Exporting raw values as metre depth is incorrect without conversion. Validate planes at known distances and off-axis spheres. Include validity masks rather than ambiguous zero depth.

Metrics and integration

Silhouette IoU is $|S \cap T| / |S \cup T|$, with an explicit both-empty case. It can hide a displaced thin filament in a large body. Add boundary-distance percentiles and maximum displacement. Compare depth on mutually valid pixels and report missing surfaces separately. Normals use angular error.

Rest scenes under a fixed camera expose flicker from grid crossings, mesh ambiguity and thresholds. Filtering can reduce flicker while shifting edges, which must remain in the error budget.

Export camera calibration, world scale, dimensions, channel semantics, masks, physical time, state ID and renderer version. Resizing requires updated intrinsics or a stated resampling transform. A depth image without its mapping is ambiguous even when attractive.

The first renderer should be simple and deterministic. Its acceptance is geometric agreement with known primitives and state, not photorealism. Integrated diffusion experiments need separately pinned models and seeds.

15 Exact mathematics and finite arithmetic

Lean real numbers are not IEEE floating values. Exact cancellation does not imply identical bits from reordered reductions. Goldberg is a foundational reference for rounding and exceptional values [S18]. The proof-to-runtime gap requires explicit analysis and tests.

Rounding

For suitable finite normal values, rounding assumptions and no exceptional range behavior, a common model is

$$\text{fl}(a \circ b) = (a \circ b)(1 + \delta), \quad |\delta| \leq \varepsilon.$$

Zero results require absolute-error interpretation; underflow, overflow and NaNs need separate treatment. For n-term accumulation, $\gamma_n = n\varepsilon/(1 - n\varepsilon)$ is a standard bound factor when $n\varepsilon < 1$. Cancellation can make final relative error large despite small absolute error.

Pressure subtracts nearly equal velocities near rest. Report absolute divergence and physical energy scales alongside relative residual. Every normalization needs a denominator and zero policy.

An executed arithmetic comparison

The companion's binary64 residual identity discrepancy is approximately 2.49×10^{-15} . Replaying the algebra with binary32 inputs and operations gives 1.70×10^{-6} . These are fixed-problem observations, not universal bounds.

The binary32 replay rounds the binary64 pressure; it does not run a binary32 CG solver. Its purpose is to demonstrate arithmetic discrepancy in a formally exact identity. Production binary32 needs its own convergence, breakdown and residual-drift qualification.

Invalid values and finite range

Validate finiteness explicitly before range checks for spacing, time, density, viscosity, cameras and initial data. NaN ordered comparisons are false. Negative zero may affect serialization or hashes even when harmless numerically; define canonicalization only if the replay contract needs it.

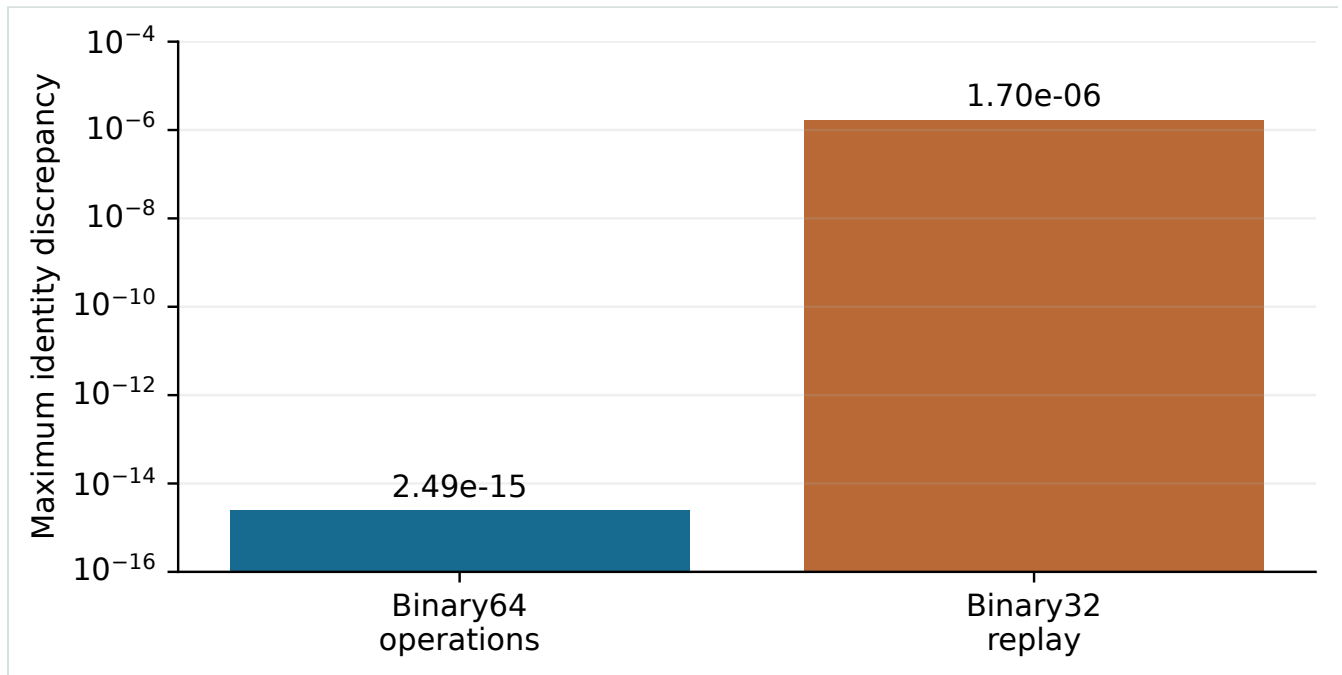
Do not silently replace nonfinite state with zero. Report stage and version, preserve accepted state and use only an explicitly authorized degraded-preview policy. Otherwise the intervention hides the cause.

Natural-number indexing omits machine range. Check dimensions, face counts, byte counts and offsets before allocation, then enforce resource limits. Huge finite world coordinates can overflow integer conversion or lose subcell precision. Clamp only under a declared boundary sampling policy; otherwise reject out-of-domain.

Tolerances

Pressure convergence, geometry classification, interpolation bounds and test comparisons need separate tolerances with units. A global epsilon cannot serve all. A tolerance changes the accepted mathematical set and should be documented.

Use exact rational or higher-precision small oracles. Test both sides of thresholds, zero step, minimum shapes, empty liquid, blocked domains, tiny coefficients and near-contact geometry. Derive tolerances from error models or accepted geometric budgets; never enlarge them merely until a visible defect passes.



Original reference illustration for 15 Exact mathematics and finite arithmetic

16 The checked Lean companion

The foundation has 20 public theorems in three modules; the expanded edition adds thirteen in `Physics.lean` and nine in `BoundedPhysics.lean`, for 42 public theorems in five modules. The bounded module also defines six helpers; its fifteen audited declarations are not fifteen theorems. Lean 4.19.0 and mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned, with locked transitive revisions. Hosted qualification at source head `ecf97d3a943ebe30d20acc1cf01b94dc08ff96a2` compiled all modules and audited 31 declarations including generated equation and proof declarations.

Only `propext`, `Classical.choice` and `Quot.sound` are allowed transitive axioms. Custom assumptions, admitted proofs and native-evaluation assumptions are rejected. Negative tests inject a custom axiom and an admitted theorem and verify rejection. An exact reviewed source inventory requires review when the file surface changes. It is change control; Lean's kernel does the proof check.

Discrete operator inventory

`Adjoint_identity` proves the finite transpose identity for arbitrary real matrices. `Pressure_energy` rewrites the quadratic form as weighted squares. `Pressure_nonnegative` adds nonnegative weights. `Pressure_symmetric` establishes bilinear symmetry.

`Constant_gradient_zero` and `constant_pressure_nullspace` assume balanced columns. They establish constants in the kernel, not that constants are the entire kernel. Disconnected components and zero weights can enlarge it.

`Internal_flux_conservation` proves total incidence cancellation. `Conservative_update` proves unweighted total preservation for its exact update. Unequal physical cell volumes require separate weighting.

`Projection_residual` assumes input incidence equals time step times the declared right-hand side and derives corrected incidence from residual. `Exact_projection` additionally assumes an exact pressure solution. Neither guarantees a solution exists or an algorithm finds it.

Transport inventory

`Blend_lower`, `blend_upper` and `blend_constant` prove two-point bounds and constant preservation.

`Convex_sum_bounds` handles nonnegative finite weights summing to one. `Upwind_positive` proves one-dimensional positivity for Courant number in $[0, 1]$. `Explicit_diffusion_positive` proves one-dimensional three-point positivity for diffusion number in $[0, 1/2]$.

`Cfl_counterexample` checks a negative upwind value outside the restriction. `Interpolation_not_mass_conservative` checks the three-row matrix example of Chapter 6. Both are exact rational counterexamples proved by kernel-checked arithmetic tactics.

Indexing inventory

`Flatten_in_bounds` proves x-fastest natural-number indexing under coordinate bounds. `Staggered_count` proves the cubic face-count identity. They do not verify Rust accesses, allocator behavior or machine overflow.

What remains unproved

No theorem establishes geometry assembly, cut-cell quadrature, positive definiteness of every configured domain, pressure uniqueness, CG convergence, trajectory accuracy, liquid-volume conservation, IEEE error bounds, Rust or GPU equivalence, physical realism, runtime performance or diffusion-image fidelity. No Navier-Stokes existence claim is made.

These boundaries make the proofs useful: exact algebra removes sign and structure ambiguity while implementation and empirical claims retain appropriate evidence. Weighted cut-cell adjointness, component-aware gauges and refinement from array kernels to abstract operators are valuable future work.

Reproduction

From proofs, fetch the pinned mathlib cache, run lake build, then lake env lean AxiomAudit.lean and the negative audit script. The workflow checks unchanged dependency files. Historical success applies to its exact source revision. Changed proof sources require new qualification.

The expansion qualification and all new assumptions are recorded separately in Appendix F and `expansion/proof-qualification.json`. The integrated pinned project compiles all 42 public theorem statements and audits 60 declarations. The current source binding is `expansion/bounded-proof-qualification.json`; the earlier 33-theorem/45-declaration receipt is retained as historical evidence. BoundedPhysics derives planar first contact/clipping and dissipative work from finite-strain backward-Euler equations. Arbitrary-mesh queries, symmetric-gradient assembly and a coupled liquid implementation remain separate research work.

17 A staged Rust implementation

Production should begin in Rust, with Zig and Odin following a stable behavioral contract. This research package is not that production solver. The checked standards snapshot `dcc56f26e884ade260770beceba2501d3746200d` emphasizes owned state, explicit failures and claim-matched evidence [S20].

Core ownership

GridShape validates dimensions, spacing, origin, strides and capacity. BoundarySnapshot owns classification and prescribed data for one version. FluidState owns accepted fields and time. Workspace owns reusable scratch. Stepper produces a candidate and diagnostics. A separate renderer consumes an immutable accepted snapshot.

The numerical core should not depend on GUI, model weights or windowing. A demonstration application is one consumer. Physics configuration and presentation configuration remain separate.

Geometry and pressure first

Implement checked indexing, shapes, coordinates and boundary classification. Test uneven dimensions, minimum grids, overflow, every orientation and linear pressure ramps. Compare independently assembled small matrices with production stencils. Verify adjointness, nullspace and connected-component compatibility.

Then add a typed pressure solver, initially diagonal-preconditioned CG on a gauge-fixed system if selected. Exercise zero residual, incompatible flux, isolated cells, disconnected domains, invalid coefficients and budget exhaustion. Compare direct small solves and hand calculations. Compute residual and divergence through separate paths.

Binary64 simplifies diagnosis; binary32 may be required for memory but needs independent qualification. The real-arithmetic proofs select neither production precision nor hardware.

Transport and forces

Implement component-aware sampling, tracing, collisions and source scheduling. Bounded scalar transport precedes optional correction. Test constants, bounds, translation and limiter behavior. Gravity and constant viscosity follow with hydrostatic and decay tests. Projection and boundary reapplication follow the derived stage order, not buffer convenience.

A step publishes a complete accepted interval or an explicit incomplete outcome. Cancellation preserves accepted state and identifies completed physical time. Renderers must not see half-updated arrays.

Guidance and application

Export calibrated silhouette, depth and normals for primitives and state. Retain camera and state IDs so hosts cannot combine stale images with new metadata. A lagging renderer may be intentional but must consume a coherent snapshot.

Build the demonstration around the core. Acceptance exercises create, advance, pause, reset, parameter changes, cancellation and export. None is proved by the research scripts or Lean algebra.

Liquid and performance

Select an interface method after comparing volume drift, silhouette accuracy and memory. Add its boundary and surface tests before advertising liquid support. Then profile complete frames under intended host model allocation. Record hardware, build, scene, warm-up, variability and peak memory. Optimize measured contributors while retaining diagnostics.

Ports share fixtures, units, serialization and failure meanings. Matching signatures is insufficient. Stabilize Rust first so three languages do not evolve three interpretations of one equation.

Concrete type and function contracts

The following Rust-shaped interfaces are a design specification, not compiled library code. They make coordinate domains and ownership visible:

```
struct CellShape { nx: usize, ny: usize, nz: usize, len: usize }
struct FaceShape { axis: Axis, dims: [usize; 3], len: usize }
struct GridGeometry { cells: CellShape, faces: [FaceShape; 3],
                     origin: [f64; 3], spacing: [f64; 3] }
struct BoundarySnapshot { version: u64, cell_kind: Vec<CellKind>,
                          face_data: [Vec<FaceBoundary>; 3] }
struct FaceField { x: Vec<f32>, y: Vec<f32>, z: Vec<f32> }
struct FluidState { id: u64, time: f64, velocity: FaceField,
                   tracer: Vec<f32> }
struct PressureWorkspace { p: Vec<f64>, rhs: Vec<f64>,
                           residual: Vec<f64>, direction: Vec<f64>,
                           product: Vec<f64>, preconditioned: Vec<f64> }
```

Private constructors validate lengths and finite geometry. A `CellIndex` cannot be constructed from an unchecked integer through the public API. `FaceIndex` includes axis or comes from an axis-specific view. Avoid retaining both an authoritative len and independently mutable dimensions; derived capacities remain immutable after construction.

A proposed `apply_pressure_operator` takes immutable topology, immutable pressure and a mutable output slice of validated length. It must overwrite every active output, including isolated or boundary rows according to classification. A divergence function consumes velocity and the same boundary snapshot but uses an independent direct face-flux traversal for validation. A project function returns a `PressureOutcome` containing diagnostics and never silently reclassifies invalid geometry.

A `sample_velocity` function accepts a world position and boundary policy and returns either a finite vector with validity metadata or a typed sample failure. Advection accepts immutable old fields and separate outputs. A step function accepts a physical interval and returns either an accepted snapshot or a failure carrying the last complete time. It must not expose mutable workspace through callbacks.

Storage and invariants

Use x-fastest contiguous arrays initially. For cells, strides are $1, n_x, n_x n_y$; for x-faces they are $1, n_x + 1, (n_x + 1)n_y$. The y- and z-face strides follow their own dimensions. Validate the multiplication when building shapes, then retain strides for kernel use.

Boundary data should distinguish fluid neighbor, prescribed normal speed, prescribed pressure and inactive face. A floating coefficient of zero is not enough to encode all four meanings. PressureTopology owns reduced indices, component IDs, gauge choices and face transmissibilities. Its version depends on geometry, density policy and active-fluid classification.

State publication increments a version only after diagnostics accept the candidate. A renderer borrows an immutable accepted snapshot. Reset constructs a new coherent state rather than zeroing arrays piecemeal while consumers read them. Parameter changes that alter array shape or topology invalidate dependent workspace and preconditioners explicitly.

Executable acceptance fixtures

The companion provides expected mathematical results for a production test port:

- The three-cell chain in Appendix E must yield pressure $(0, 4, 2)$ and zero corrected internal velocity at time step one half.
- The cycle fixture must retain equal circulation rather than force every face to zero.
- All four lattice samplers must reproduce an affine scalar within precision tolerance; the intentionally wrong offset must fail.
- The pressure operator must match an independently assembled small matrix on rectangular dimensions.
- A closed component with nonzero prescribed net flux must return incompatible flux before iteration.
- A zero right-hand side must terminate without dividing by zero.
- The rational flux fixture must preserve total amount; floating ports compare with an appropriate absolute budget.
- A deliberately truncated particle support must expose moment failure rather than pass the full-support affine assertion.

Add integration fixtures for cancellation after each stage, resource rejection before mutation and stale boundary versions. Those are not mathematical experiments; they require the actual Rust state owner and host boundary.

Dependency and implementation choices

Use established sparse and numerical libraries for reference solves where licensing and deployment allow, while retaining a transparent matrix-free production operator if justified by the memory contract. Pin toolchains and dependencies for qualification, and record the exact compiler and feature set used in numerical comparisons.

No unsafe code is required for the first bounded CPU implementation. Introduce unsafe indexing or SIMD only after profiling identifies a meaningful contributor and the validated shape contract supports a reviewed safety argument. Bounds-check elimination by the compiler is preferable to a manually duplicated unchecked path when performance is equivalent.

GPU kernels require a separate buffer-layout and dispatch contract, including alignment, workgroup limits, synchronization and readback. A CPU test cannot establish a shader's binding layout or race freedom. Keep backend-independent fixtures but qualify each backend through its real execution path before advertising support.

18 Validation and remaining research

Formal algebra, numerical fixtures, integration tests and visual review prove different things. A proof build cannot replace camera-export validation; a beautiful render cannot replace divergence or volume checks.

Executed evidence

The companion builds a closed 8-cubed graph with 512 cells and 1344 faces. It gauge-fixes pressure and projects deterministic random velocity. CG uses 78 iterations at the chosen tolerance. Divergence L2 decreases from 49.74366159 to 4.4429×10^{-10} . Unweighted kinetic energy decreases from 642.9265376 to 401.9458778. The scalar update retains total 512 in the executed binary64 computation.

These are sparse-reference results, not Rust, cut-cell or laboratory evidence. Energy decrease is an observation consistent with projection, not a general proof. An exact rational three-cell flux example updates to 199/630, 96/175 and 61/450, summing exactly to one.

Transport and diffusion experiments are described in Chapters 7 and 10. Scripts assert stated properties and preserve data used by figures. No script timing is claimed as production performance.

Production evidence matrix

Geometry tests cover shape arithmetic, transforms and orientation. Operator tests cover nullspaces, adjointness, symmetry and compatibility. Solver tests cover residual/divergence, termination and singularity. Transport tests cover smooth and discontinuous data, variable velocity and boundaries.

Forces need rest and decay tests. Liquids need sphere transport, volume drift, merging, thin sheets, rest and contact. Rendering needs calibration, invalid depth, normals and temporal rest stability. Integration needs coherent state versions, resource rejection, cancellation and replay. Each class requires an owner and diagnosis.

Convergence and performance

Report grid, time step, norm, duration, boundaries and reference with each error plot. Several refinements reveal whether slopes stabilize. Coarse points can be pre-asymptotic and fine points can reach solver or rounding floors; neither should be discarded without explanation. Limiters and topology changes can lower observed order.

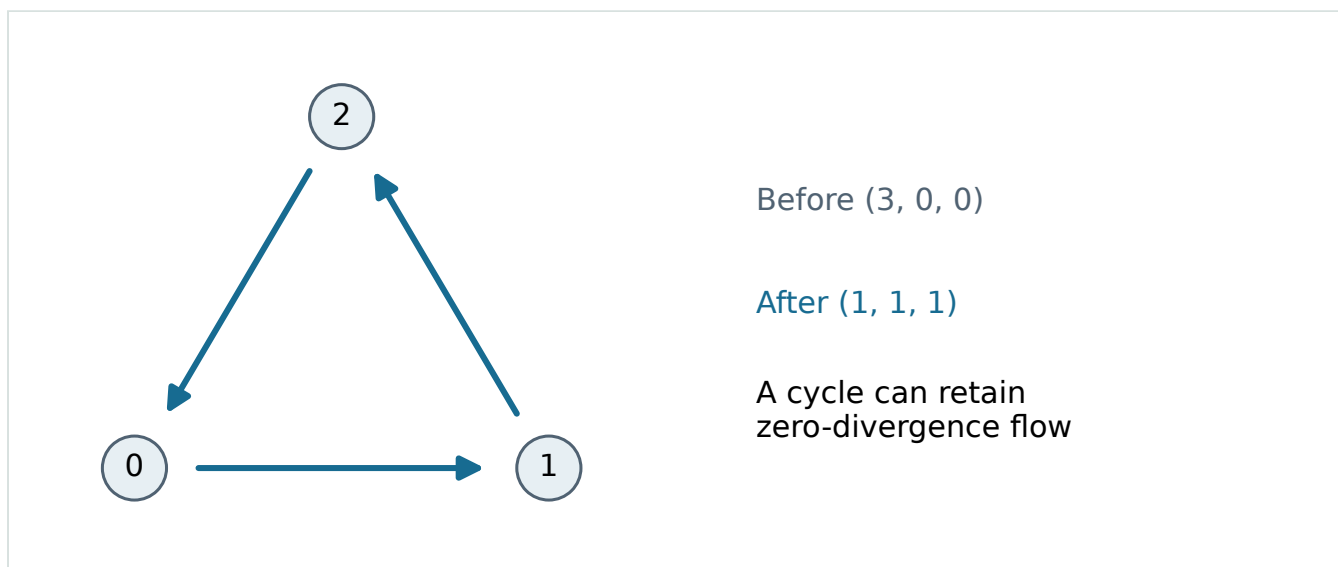
Choose representative sparse and dense scenes, obstacles, sources and output resolution. Measure simulation, pressure, rendering, readback and full-frame distributions, plus peak memory with host model allocations. The README supplies no numeric budget or target hardware. Until chosen, memory arithmetic is an estimate and real time remains an objective.

Next research

The highest-value integrated study links coarse simulation error to spatial conditioning quality. Measure how displaced interfaces, missing thin features and flicker affect a pinned diffusion pipeline. More simulation detail may consume memory without improving final guidance.

Numerical priorities are component-aware pressure, obstacle sampling and bounded liquid volume. Formal priority is connecting array assembly to the abstract model. Non-Newtonian stress, APIC and sparsity follow needed capability rather than the project's name.

Every result should retain its level of certainty: derived identity, checked theorem, executed fixture, measured implementation or proposed design. That distinction makes a small reusable solver dependable.



Original reference illustration for 18 Validation and remaining research

19 Forces and moving-boundary work

The accepted smoke demonstration supplies one localized acceleration in a fixed box. The expansion studies externally forced liquids and prescribed moving solids; it does not add those capabilities to the Rust core. The first decision is whether motion is prescribed or dynamically coupled. A scripted paddle can supply arbitrarily large work without losing momentum. A dynamic paddle requires a body mass, inertia, reaction force and compatible time integration.

Units before parameters

Use metres, seconds and kilograms internally. Density ρ is kg/m³, dynamic viscosity μ is Pa s, and kinematic viscosity $\nu = \mu/\rho$ is m²/s. An acceleration a changes velocity by Δta ; a force density f , in N/m³, changes it by $\Delta tf/\rho$. A source of liquid volume is neither of these. Its addition must appear in the volume ledger and pressure compatibility condition.

For a face mass $m > 0$, an acceleration-only update satisfies

$$\frac{1}{2}m(u + \Delta ta)^2 - \frac{1}{2}mu^2 = m\Delta tau + \frac{1}{2}m\Delta t^2 a^2.$$

The second term explains why logging old-velocity power alone does not reproduce a finite explicit step's energy change. `force_work_identity` checks this exact scalar identity. The reference fixture verifies it for a signed acceleration; it makes no general stability claim.

One geometry interval

A rigid prescribed boundary has world velocity $u_s(x) = v + \omega \times (x - x_c)$. Retain pose, translation, rotation and version for the interval consumed by transport, pressure, viscosity and output. Sampling only the final mesh pose can miss a swept collision. Sampling wall velocity from one pose and cut volumes from another breaks displaced-volume accounting even if pressure residual is tiny.

For an outward fluid normal, sealed moving-wall flux is positive when fluid-domain volume grows. Reynolds transport gives $\sum_e Q_e = \dot{V}$, with $Q_e = A_e u_s \cdot n_e$. Incompressibility on a moving control volume uses this geometric volume change, rather than imposing a zero sum of all wall fluxes blindly. A translating obstacle inside a fixed sealed container preserves total fluid-domain volume; a piston moving inward does not unless a compensating outlet or source exists.

For fixed-volume cell constraints choose an outward-positive internal incidence B (the negative of Chapter 16's head-positive pressure incidence) and write $Bu + Q = s$, where internal incidence columns sum to zero and s is an integrated volume source. Summing yields the necessary component condition $\sum Q = \sum s$. `sealed_flux_compatibility` proves this implication under explicit balance and solved-constraint hypotheses. It does not prove a solution exists. Moving-cell schemes must additionally supply their geometric conservation law.

Pressure and external work

Prescribed boundary data make divergence affine. The weighted adjoint relation includes pressure work $p^T Q$, as Chapter 8 derives. Separate known boundary terms before testing a homogeneous transpose identity. For two-way coupling use the same normals, areas and pressure interpolation for wall flux, force and torque. Otherwise an apparently correct pressure solution can exchange inconsistent momentum. The variational solid-fluid paper provides the primary coupled formulation [E1].

The original three-cell amount ledger stores balanced and incompatible prescribed flux vectors. Their sums illustrate the necessary component condition; the separate 27-cell projection reference executes a dense pressure solve. No moving-wall pressure solve is claimed. The 3D mesh lab shows prescribed translation and the velocity at a selected collision; it is not a moving-solid fluid solve.

Implementation contract and experiments

Proposed ForceField distinguishes acceleration from force density; BoundaryInterval owns immutable start/end geometry and motion. Reject nonfinite input, stale geometry, unclassified boundary and incompatible component source before publication. An accepted report retains force work, wall work, source volume, physical time and geometry version. Failure leaves the previously accepted state coherent.

Begin with gravity and a flat tank. Then use a translating closed obstacle, a rotating paddle and a piston with a specified outlet. Compare volume change against integrated boundary flux, include both signs of wall motion, and refine temporal sampling. Dynamic light-body added-mass tests belong to a later two-way implementation; prescribed-body examples do not establish them. Mesh query accuracy and liquid volume transport remain separate obligations even when the exact algebra passes.

Work in a frozen-geometry operator

Let M be a positive mass matrix, K a nonnegative strain operator, C a tangential sample map, and Λ a nonnegative wall-friction matrix. With wall samples w , generalized force f and equation residual e , suppose

$$M(v - u) + \Delta t \{ K v + C^T \Lambda (C v - w) - f \} = e.$$

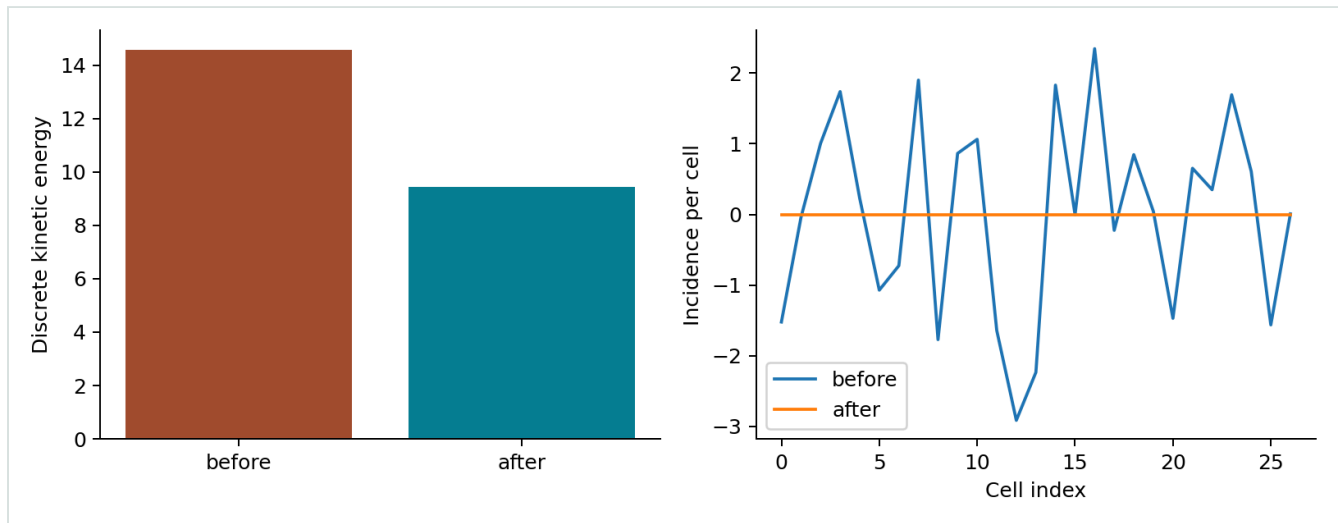
For $r = C v - w$ and kinetic energy E , the finite work budget is

$$E(v) - E(u) + \frac{1}{2} \|v - u\|_M^2 + \Delta t v^T K v + \Delta t r^T \Lambda r = \Delta t v^T f - \Delta t w^T \Lambda r + v^T e.$$

The minus sign in actuator work follows from wall force $-\Lambda r$ on the fluid. Laboratory-frame energy can increase while relative friction dissipates. The residual must be taken from this unscaled equation, with its norm and normalization declared. This richer matrix statement is a next proof target, numerically checked in small synthetic systems; it is not one of Physics.lean's thirteen compiled contracts. Prescribed normal-wall pressure work requires its own affine constraint term.

An affine projection $M(v-u) + A^T \lambda = 0$, $A v = b$ gives $E(v) - E(u) + \frac{1}{2} \|v - u\|_M^2 = -\lambda^T b$. Here λ is impulse-scaled. A homogeneous stationary constraint has zero right side; a moving constraint supplies work. Feasibility is separate from pressure gauge uniqueness. For two-way solids, combine fluid and body masses and use one adjoint constraint J .

Common translation conserves the corresponding generalized momentum only if J annihilates that mode; angular momentum additionally needs rotation modes and consistent interaction points. These are frozen-geometry specifications, not completed moving-solid proofs.



Original reference illustration for 19 Forces and moving-boundary work

20 From collision meshes to fluid boundaries

A rendered triangle soup is not automatically a closed collision solid. Rheon's proposed boundary pipeline accepts an oriented watertight mesh in a declared world transform, or a deliberately separate thin-shell model. It does not invent an inside for open edges. Static and prescribed rigid motion are the first research targets; deforming meshes require vertex velocities and swept geometry.

Admission and representation

Validate finite vertices, valid triangle indices, nonzero triangle area at the declared tolerance, edge incidence, coherent orientation and supported topology. Self-intersections invalidate a simple signed-volume interpretation. Nonuniform scale affects distances and normals: transform a normal with the inverse transpose and normalize after checking its length. Negative determinant transforms require orientation handling. Record original topology and transformed geometry hashes.

An unsigned nearest distance comes from the closest point on triangles. Its sign requires an inside/outside policy, such as consistently oriented winding or parity with robust edge/vertex rules. A signed distance used for classification is a derived numerical field, not an authoritative replacement for the mesh. Thin features can disappear when sampled at cell centers; face openings and cell volumes need their own geometry queries.

Bridson's author-hosted mesh-query and geometric-predicate implementations provide relevant original implementation references [E2]. Their availability does not prove Rheon's independently written code. The present companion implements segment/triangle intersection and closed-cube tests directly, with explicit finite tolerances. It does not advertise a production robust-predicate package.

Earliest path collision

For a segment $x(t) = x_0 + t(x_1 - x_0)$, seek the smallest hit $t \in [0, 1]$. A triangle test solves the barycentric system with two in-plane coordinates and segment parameter; parallel or nearly degenerate configurations need a specified classification. Selecting the nearest endpoint distance is insufficient. An endpoint can be outside the solid while the segment crosses it twice.

The original fixture uses a triangulated closed cube, traces an outside-to-outside segment and finds its first hit at $t = 0.3125$. It also checks a miss and a trace beginning inside. The browser shows the complete segment, earliest hit and triangle mesh, so the evidence remains visible rather than hidden behind a texture effect. A boundary control translates the geometry and uses the corresponding regenerated hit records.

At a hit, collision response depends on what is transported. A passive tracer trace may stop at the surface; a liquid interface needs its own boundary/contact-angle extension; a velocity sample needs prescribed normal motion plus the chosen tangential law. Repeatedly clamping a point can pin a contact line unintentionally. No-slide and no-slip mean different operations.

Checked planar first contact and clipping

The mesh-query fixture and the exact collision contract have different domains. The new `BoundedPhysics.lean` contract uses an infinite stationary plane in three dimensions. Its oriented plane value is

$$g(x) = \sum_{i=0}^2 n_i x_i - c, \quad x(t) = (1-t)a + tb.$$

The permitted half-space is $g(x) \geq 0$. The normal points into that half-space and need not have unit length; g is a sign/plane-value function, not necessarily a signed distance. Assume strictly $g(a) > 0$ and $g(b) < 0$. These hypotheses exclude a zero denominator and imply

$$t_* = \frac{g(a)}{g(a) - g(b)}, \quad 0 < t_* < 1.$$

`wall_segment_affine` derives $g(x(t)) = (1-t)g(a) + tg(b)$ from the actual finite-vector definitions. `wall_hit_range` proves the range of t . `wall_hit_on_surface` proves $g(x(t)) = 0$. `wall_first_hit` proves every $0 \leq t < t_*$ remains strictly in the permitted side. Finally, `clipped_segment_in_halfspace` proves every point on the segment from a to $x(t_*)$, parameterized by $0 \leq s \leq 1$, has nonnegative plane value, including contact. The hit and clipping conclusions are derived, not assumed.

The exact-rational witness uses plane values 1 and -1 , giving $t = 1/2$. *The unclipped continuation at $t = 3/4$ has value $-1/2$. If the proposed end instead has value $1/2$, the formula gives $t = 2$ outside the segment; if the start/end values are -1 and -2 , it gives -1 .* The endpoint sign assumptions are therefore essential. Starting on the plane, grazing, same-side traces and moving planes require separate policies.

This proves a bounded planar geometric contract. It does not prove triangle containment, ordering among mesh facets, watertightness, cut-cell construction or floating-point robustness. A finite-facet query must establish both that its intersection lies inside the facet and that no earlier relevant facet was missed. The closed-cube browser fixture remains an independently executed numerical query, not an implementation refinement proved by these planar lemmas.

Cut geometry and pressure

For a face opening A_e , fluid control volume V_i and sample separation ℓ_e , a candidate pressure weight is $w_e = A_e / (\rho_e \ell_e)$. Positive weights preserve the finite energy structure. Shared openings must be computed once for both adjacent cells. Independent face estimates can violate cancellation. The scalar volume and face metrics must agree with the chosen divergence/gradient pair, not merely look plausible.

Tiny volumes create stiffness and amplify residual-to-divergence scaling. A minimum fraction, cell merging or topology removal is a model policy requiring comparison against unclipped geometry. A binary staircase is a valid educational baseline with a disclosed geometry error; it is not continuous collision-mesh support. Basilisk's original embedded-boundary source exposes how geometry enters gradients, interpolation and small-cell handling [E3].

Qualification ladder

Compare cube intersections against hand arithmetic, then oblique planes and spheres against analytic distances. Add grazing rays, edge hits, reversed winding, repeated triangles, very small and large scales, and subcell wall offsets. Use refinement to distinguish geometry error from linear-solver error. Validate moving traces in relative coordinates for rigid translation, then rotation against an independent swept reference.

The research admission contract returns `UnsupportedOpenMesh`, `DegenerateTriangle`, `AmbiguousSign` or `UnresolvedThinFeature` where appropriate. It does not silently substitute a box. The finite Lean conservation contracts assume a valid assembled incidence and shared fluxes; they do not verify these geometry predicates. A complete mesh-to-cut-cell implementation remains future production work.

Mesh validity is part of the numerical method

A render mesh is not automatically a valid collision domain. The conversion should produce an explicit report containing:

- Unit scale, coordinate transform, winding convention, and sign convention.
- Watertightness and manifold checks for volumetric obstacles.
- Degenerate triangles, self-intersections, duplicate faces, and open boundaries.
- Minimum feature/gap scale relative to Δx , particle spacing, kernel support, and collision thickness.
- A declaration that a surface is a finite-volume solid, a zero-thickness shell, or a intentionally thickened approximation.

For a closed solid, a signed distance function can describe inside and outside if its sign construction is valid. A zero-thickness open sheet has no enclosed solid volume. A sub-grid sheet can disappear entirely if an algorithm samples only whether cell centers lie inside it. Inflating it avoids that disappearance but changes displaced volume, passage width, wetted area, and pressure forces.

Guendelman, Selle, Losasso, and Fedkiw's thin-shell method uses ray-based visibility information to prevent interpolation and finite-difference stencils from crossing an infinitesimally thin triangulated barrier. It also treats incompressibility near that barrier. This is the important conceptual precedent: detect which samples may communicate across the shell, rather than treating every nearby sample as a neighbor. [R18](#)

For cut cells, store fluid cell volumes and open face areas. Also represent connectivity: an internal baffle can split one Cartesian cell into disconnected fluid pieces that cannot share a single pressure unknown without introducing a false connection. Scalar volume fractions alone are not enough for all sub-cell topologies. The correct fallback may be refinement, multi-region cut cells, or rejecting the feature as unresolved.

Small cut cells introduce their own explicit transport restriction. When fluid volume tends to zero faster than an open face area, the local transit time tends to zero. Basilisk's embedded-boundary code documents this issue and implements a redistribution approach for tracer overflow. Any adopted merge/redistribution rule needs separate conservation, positivity, accuracy, and barrier-connectivity checks. Clamping a tiny volume to a larger value is a model change unless the resulting discrepancy is accounted. [R19](#)

Moving and deforming walls

For a rigid wall, evaluate local velocity at the actual interaction point:

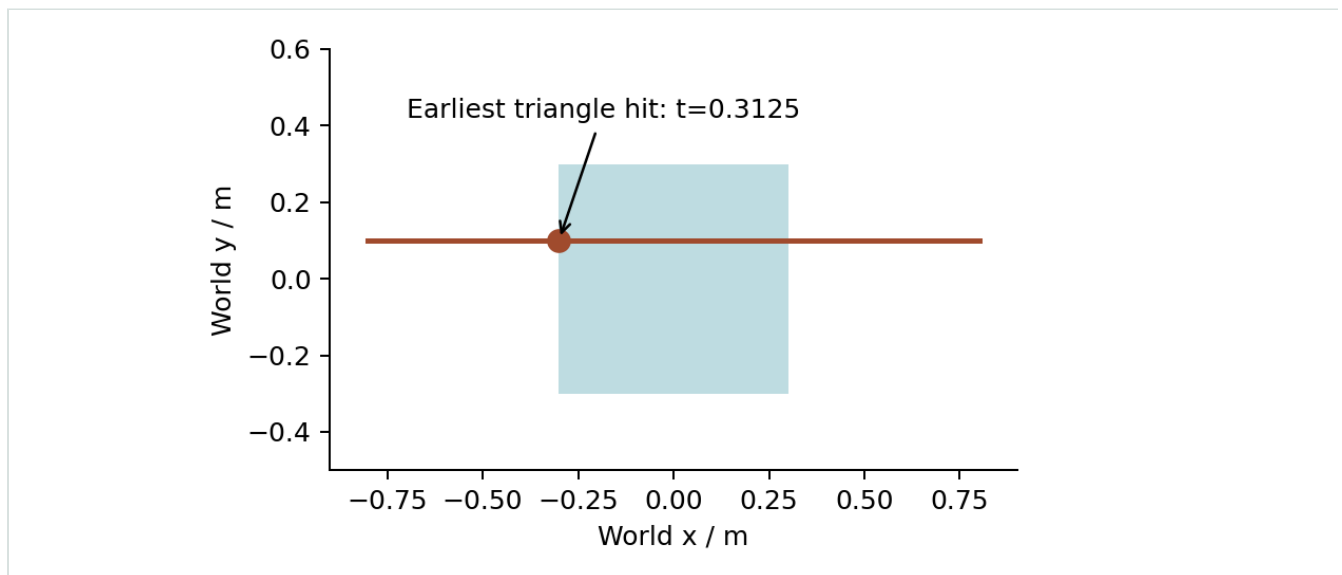
$$U_w(x) = V + \Omega \times (x - c).$$

A single object translation velocity is incorrect for a rotating paddle. Use the same point and frame for wall traction, solid force, and solid torque. For a two-way solid, accumulate equal-and-opposite impulses and the moment arm. For a prescribed wall, accumulate the actuator work; do not report fluid energy growth as a solver defect without including it.

When geometry moves, a cut cell can be uncovered or swallowed. Update swept volumes and associated mass/momentum consistently. For a space-time control volume, a discrete geometric conservation law is the prerequisite for transporting a uniform field without generating a fictitious disturbance. Reinitializing newly uncovered cells by arbitrary extrapolation may be robust, but it is not automatically conservative or work-consistent.

Collision detection for advected particles should consider the segment or swept trajectory, not only its endpoint, whenever a particle can traverse a thin wall in one substep. Surface-only pushout can repair position while leaving pressure fluxes, density estimates, and wall work wrong. Do not use a smoothed rendering normal as the geometric contact normal without analyzing the resulting gap and traction error. Nonuniform scaling also requires correct normal transformation and a recomputed distance metric.

Precomputed rigid-boundary maps may be transformed with a rigid body; arbitrary deformation changes the precomputed geometry. A deforming mesh needs an updated representation and checks for new self-intersections or inverted elements. A narrow band must cover the actual query reach, including kernel support and swept motion, not merely the visible surface.



Original reference illustration for 20 From collision meshes to fluid boundaries

21 Density and real liquid interfaces

A liquid state needs more than a smoke concentration. Its interface locates a liquid domain, its density supplies inertial mass, and its volume ledger records transport and sources. Changing the grayscale opacity coefficient does not produce a liquid. This chapter specifies an incompressible, immiscible Newtonian research model. Compressible flow, phase change and dissolved materials are separate extensions.

Constant material and multiple phases

A single liquid with negligible air dynamics can use constant positive ρ_l in active liquid cells and a pressure boundary at the interface. A two-phase model solves momentum in both phases with densities $\rho_l, \rho_a > 0$. These choices have different pressure masks and inertia. The choice must remain visible in state and output metadata.

For variable density, the pressure coefficient uses inverse density at the pressure connection. Across a one-dimensional layered connection of lengths ℓ_1, ℓ_2 , integrating $\partial_x p = \rho a$ gives effective $1/\rho_e = (\ell_1 + \ell_2)/(\rho_1 \ell_1 + \rho_2 \ell_2)$. This is the harmonic transmissibility of the inverse-density coefficient, not a universal average independent of geometry. Volume-fraction mixing is another declared approximation. Use the same coefficient in assembly and velocity correction.

`positive_transmissibility` proves positivity from positive area, density and distance.

`weighted_pressure_energy` proves the corresponding weighted-square sum is nonnegative; Chapter 16's existing matrix identity explains its operator interpretation. Neither identifies the physically correct density interpolation at an unresolved interface.

Hydrostatic oracle

Take upward y , gravity $-g$, and free surface at H . For piecewise layers,

$$p(y) = p_a + g \int_y^H \rho(s) ds.$$

A discrete face increment $p_{j+1} - p_j = -\rho_{j+1/2} g \Delta y$ cancels a gravity step when the same face density is used. The reference tank has a lower layer of density 1000 and an upper layer of 800 kg/m³, each 0.5 m deep. With $g=9.81$ m/s² its bottom gauge pressure is 8829 Pa. These are illustrative parameters, not a measured material calibration or a stable stratified-flow simulation.

The educational tank displays the stored layer density and pressure samples. Increasing density changes pressure and inertia; it does not change equilibrium cap geometry at zero gravity. That distinction prepares the later capillary lab. `hydrostatic_increment` checks the exact cancellation identity for a prescribed face balance; it does not prove long-time rest after transport, interface reconstruction or rounding.

Conservative represented volume

Let $m_i = f_i V_i$, with $0 \leq f_i \leq 1$. Update $m_i^+ = m_i - \Delta t (BF)_i + \Delta t s_i$ using one signed shared face flux. In this amount ledger B is outward-positive (positive at the face tail, negative at its head), the negative of Chapter 16's pressure incidence. The conservation result is unchanged by this global sign choice. Balanced incidence columns imply

$$\sum_i m_i^+ = \sum_i m_i + \Delta t \sum_i s_i.$$

`volume_source_balance` proves this identity in exact arithmetic for arbitrary finite cell volumes. The proof operates on represented amounts; it does not establish fraction bounds or determine geometrically correct fluxes. Post-update clamping can violate the identity. The original three-cell volume fixture checks unequal volumes and a nonzero source using rational arithmetic before exporting floating values.

Chapter 9's axis-aligned PLIC example remains a local geometric oracle. Full three-dimensional bounded conservative transport needs one coherent multidimensional outflow scheme, consistent reconstruction normals and source treatment. A level-set branch instead needs measured volume drift and an explicit correction policy. Neither is selected merely because it yields a smoother surface image.

State and failure contract

Proposed `LiquidState` retains interface representation, positive material parameters, volume/source totals, pressure-unknown mask and velocity-validity mask. Those masks are distinct. Air velocity extension supports transport without making extended air cells physical liquid. Rebuild topology and preconditioners when classification or density policy changes. A failed interface reconstruction must not publish pressure and geometry from different versions.

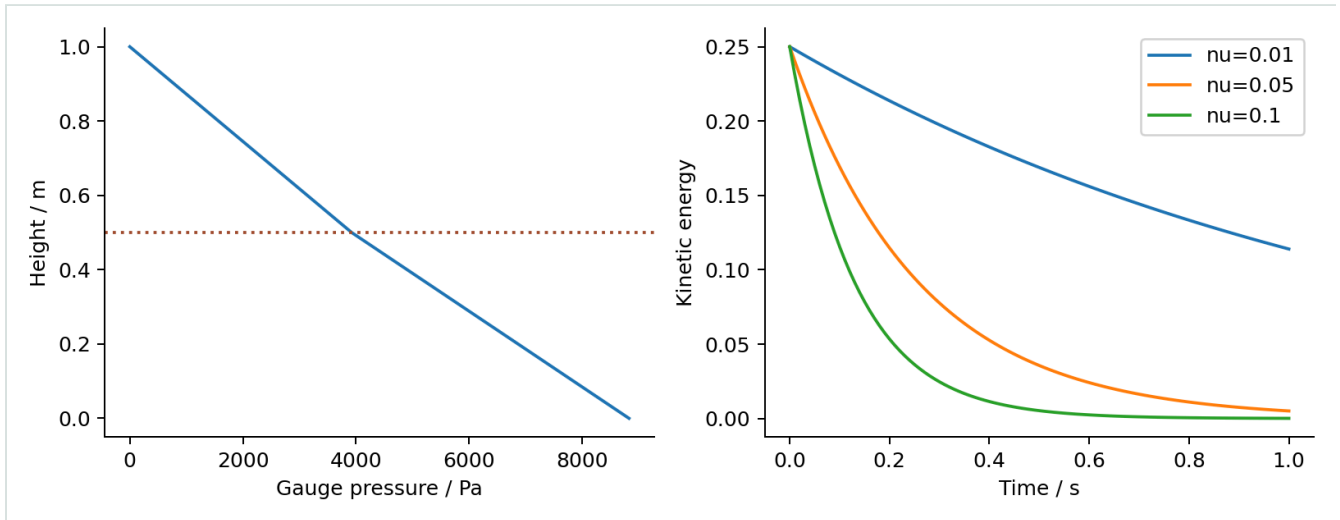
Acceptance progresses through hydrostatics, translating sphere, oblique interface transport, merging volumes and thin films. Report both represented volume error and spatial interface error; equal volume does not establish equal shape. Two-phase density-ratio tests additionally require momentum-transfer and pressure-jump checks. The original VOF and level-set works remain primary method references [S8–S10], while the present fixtures are independently constructed.

Holding one material quantity fixed

Increasing density at fixed dynamic viscosity lowers $v = \mu/\rho$; increasing density at fixed v raises μ . A single-phase prescribed gravity acceleration stays the same, while hydrostatic pressure and inertia change. A fluid material API must record which viscosity quantity is supplied. The inspected original `SPlisHSPlasH` Weiler2018 code multiplies its configured viscosity by rest density to obtain μ [R14](#). Copying a GUI value between solvers without converting its definition is therefore unsafe as a material comparison.

Represented particle mass, reference density, estimated density and particle count also differ. For uniform reference sampling $m = \rho_0 V_0$. Removing particles removes represented mass unless explicitly accounted as boundary/source flux. A planar 2D formulation must declare an out-of-plane thickness or use per-unit-depth quantities; a 3D coefficient cannot be transferred by a screenshot comparison.

For pressure-driven planar Poiseuille flow with $G = -dp/dx > 0$, $u(y) = Gy(H-y)/(2\mu)$ and flow per width $Q' = GH^3/(12\mu)$. The equivalent acceleration is G/ρ , with units m/s^2 . This is a useful future force/density/viscosity cross-check: at fixed G the steady speed depends on μ ; at fixed acceleration the effective pressure gradient changes with ρ . No such channel solver has been run in this edition.



Original reference illustration for 21 Density and real liquid interfaces

22 Viscosity, stress and dissipation

Density and viscosity have different roles. Density weights inertia; dynamic viscosity relates deformation rate to stress; kinematic viscosity controls a simple diffusion time scale. Treating a viscosity slider as an arbitrary velocity damping factor hides both the material units and boundary work. The first liquid research model is Newtonian, with positive density and nonnegative viscosity.

Symmetric strain

For $D = (\nabla u + \nabla u^T)/2$, Newtonian stress is $\tau = 2\mu D$. Its power dissipation is $2\mu D : D \geq 0$. The cross derivatives matter: rigid rotation has zero symmetric strain even though velocity gradients are nonzero. A scalar component Laplacian is equivalent to the incompressible constant-viscosity interior equation only under the matching assumptions and boundary terms.

Batty and Bridson's original viscous-free-surface paper separates normal and tangential traction and builds a compatible variational discretization [E4]. Their later variable-coefficient Stokes formulation supports irregular moving boundaries [S11]. Basilisk's original viscosity implementation is another useful code reference for coupled stress components [E5]. These sources motivate a choice; they do not qualify Rheon's future stencil.

Finite dissipative update

Let face masses form positive diagonal M , let E map face velocity to strain samples, and let nonnegative quadrature coefficients form W . Put $K = E^T W E$. Homogeneous stationary boundaries and zero external force give backward Euler

$$M(v - u) + \Delta t K v = 0.$$

Taking the inner product with v and completing a square yields

$$\|u\|_M^2 - \|v\|_M^2 = \|u - v\|_M^2 + 2\Delta t v^T K v \geq 0.$$

`implicit_energy_identity` checks the finite identity from an explicit scalar work equation.

`implicit_energy_nonincrease` adds nonnegative time and dissipation to obtain the inequality. Its finite sums use arbitrary positive masses, so variable density is represented without assuming uniform kinetic weights. It proves neither that an iterative solve reaches the work equation nor that E is the actual symmetric-gradient stencil.

Moving walls and body forces add work on the right. Energy need not decrease then. A diagnostic should compare measured energy change against viscous dissipation and declared external work, with residual and rounding budgets, rather than reject every energy increase.

Checked bridge from strain to the implicit equations

The earlier Physics contracts require a scalar work identity. The new BoundedPhysics contracts derive that identity for a defined fixed finite-strain operator. For arbitrary finite dimensions n and m , let E map velocity degrees of freedom to strain samples, and let μ_e be fixed material/quadrature weights. Define

$$(Ev)_e = \sum_i E_{ei} v_i, \quad (Kv)_i = \sum_e E_{ei} \mu_e (Ev)_e, \quad \mathcal{D}(v) = \sum_e \mu_e (Ev)_e^2.$$

Thus $K=E^T \text{diag}(\mu)E$. `viscous_work` derives $\sum_i (Kv)_i v_i = \mathcal{D}(v)$. `dissipation_nonnegative` derives $\mathcal{D}(v) \geq 0$ when every weight is nonnegative. The exact coordinate backward-Euler hypothesis is

$$m_i v_i + \Delta t (Kv)_i = m_i u_i \quad \text{for every } i.$$

`backward_euler_work` derives $\sum_i m_i (u_i - v_i) v_i = \Delta t \mathcal{D}(v)$ directly from these equations.

`backward_euler_energy_nonincrease` then applies the existing finite energy theorem: with $m_i \geq 0$, $\Delta t \geq 0$ and $\mu_e \geq 0$, $\sum_i m_i v_i^2 \leq \sum_i m_i u_i^2$. The coordinate update and positive-weight hypotheses replace an assumed dissipative-work conclusion.

The theorem permits zero masses and weights, so it does not prove invertibility, existence or uniqueness. E is arbitrary and fixed during the step; no theorem here identifies it with an assembled symmetric-gradient stencil or proves a rigid-rotation null mode. This is an unforced homogeneous step. Moving-wall work, free-surface traction, nonlinear coefficients, approximate solves and IEEE rounding remain outside its statement.

For the exact witness $E=[-1,1]$, $\mu=2$, unit masses, $\Delta t=1/4$ and $u=(1,-1)$, the computed update is $v=(1/2,-1/2)$. The weighted squared norm falls from 2 to 1/2; kinetic energy with its 1/2 factor falls from 1 to 1/4. The derived work is $1/2=\Delta t \cdot 2$. A negative weight $\mu=-1/2$ instead gives squared norm 32/9, while an arbitrary candidate not satisfying the update can also increase it. These are exact rational witnesses, not liquid simulations or material calibration.

Shared shear-mode oracle

The reference code constructs a 32-cell periodic difference matrix with spacing $1/32$ m. A sine mode has discrete eigenvalue $\lambda_h = 4 \sin^2(\pi/32)/h^2$. With $\Delta t = 0.01$ s and ν chosen from 0.01, 0.05 or 0.1 m²/s, backward Euler damps amplitude by $(1 + \nu \Delta t \lambda_h)^{-n}$.

An independently assembled dense solve verifies the modal formula at each step. The browser displays the resulting velocity arrows in a three-dimensional slab and reports mass-weighted kinetic energy. The same profiles and energy records generate the book figure. A large implicit step remains damped but does not reproduce exact exponential decay; the error is shown rather than treated as a performance improvement.

Boundary and material tests

Use affine shear to check the stress law and rigid rotation to check the strain nullspace. Add periodic decay, no-slip channel flow, prescribed-wall Couette flow and free-surface traction. The first two are local operator tests; a complete free surface additionally needs pressure and stress coupling. Explicitly record whether a split assumes zero normal viscous stress.

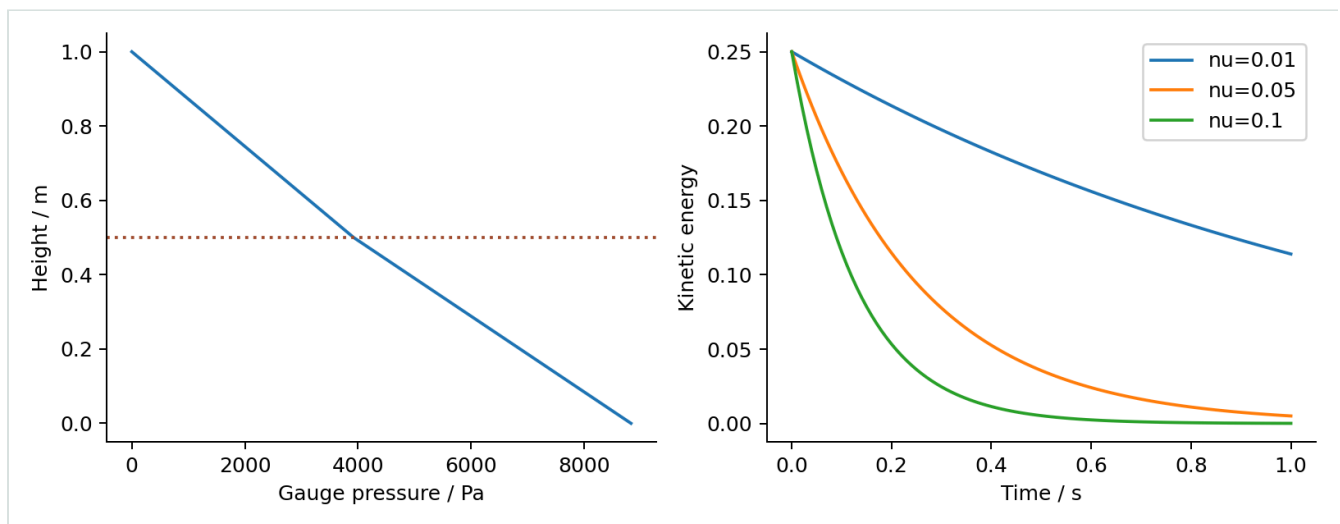
Generalized Newtonian and viscoelastic materials remain later research. A shear-dependent viscosity needs an admissible range and nonlinear residual; viscoelasticity needs history and its own stress evolution. The existing finite-budget nonlinear specification in Chapter 10 remains valid. This expansion does not quietly turn it into an implemented material law.

A stable operator can damp the wrong mode

Positive semidefiniteness only says that a quadratic energy decreases. It does not imply that the stencil preserves rigid rotation. For Ω about the z axis, points at $(-1,0,0)$ and $(1,0,0)$ have velocities $(0,-\Omega,0)$ and $(0,\Omega,0)$. Pairwise vector-difference smoothing sees a difference of magnitude $2|\Omega|$ and penalizes it, although the continuum symmetric strain of the rigid field vanishes. A future assembly theorem must show $\text{Br}=0$ for sampled rigid translation and rotation under explicit geometry assumptions. Only then does $\mathbf{K}=\mathbf{B}^T\mathbf{W}\mathbf{B}$ annihilate those modes.

Isolate viscosity for that fixture: a fully rotating drop still needs centripetal pressure, advection and capillary shape dynamics. Check an affine shear field separately, so a stencil that simply applies zero force everywhere cannot pass. At a free surface, normal and tangential traction belong to the stress condition; independent scalar diffusion with convenient component boundary conditions can violate it even when the interior formula matches $\mu\Delta u$.

For comparison with particle methods, the original Weiler et al. viscosity work evaluates conservation, spatial/temporal refinement and spurious viscous forces [R12](#). Those criteria transfer as questions to ask of a grid method; its published performance and accuracy do not transfer as Rheon results. PBF positional density constraints, artificial pressure and postprocessed smoothing offer a different real-time model [R15](#). Label their coefficients by their actual equations instead of silently assigning μ or σ units.



Original reference illustration for 22 Viscosity, stress and dissipation

23 Wetting, adhesion and the contact line

Wetting describes a liquid's preference for solid contact in relation to surrounding phases. Adhesion names interfacial energetic attraction; slip describes tangential motion; viscosity describes bulk stress. These are not interchangeable controls. A liquid can have a small equilibrium contact angle and substantial slip, or a large angle with strong viscous drag. A wall that numerically pins every trace can look sticky without implementing an adhesion model.

Equilibrium surface energy

For a smooth homogeneous ideal solid, write interfacial energies γ_{sv} , γ_{sl} and liquid-vapor tension $\sigma > 0$, all in N/m. The equilibrium Young relation is

$$\gamma_{sv} - \gamma_{sl} = \sigma \cos \theta_e.$$

The reversible work of adhesion is $W_A = \gamma_{sv} + \sigma - \gamma_{sl}$, giving $W_A = \sigma(1 + \cos \theta_e)$. `young_adhesion_identity` proves this implication from a stated algebraic Young balance. `adhesion_bounds` assumes a cosine value in $[-1,1]$ and proves $0 \leq W_A \leq 2\sigma$. These are energetic identities, not a microscopic derivation, an existence theorem for a droplet, or a proof of dynamic contact angle.

Only the difference of solid interfacial energies is inferred by an angle. Their individual values remain unidentified. Nor does W_A specify a bulk attractive body force. Adding an arbitrary wall attraction while also imposing the same contact angle can count energetic effects twice. A chosen disjoining-pressure or precursor-film model requires its own potential, length scale and calibration.

Constant-volume cap geometry

In negligible gravity, an axisymmetric spherical cap with curvature radius R has height $H = R(1 - \cos \theta)$, contact radius $a = R \sin \theta$, and volume

$$V = \frac{\pi}{3} R^3 (2 - 3 \cos \theta + \cos^3 \theta).$$

For fixed V , solve this equation for R before changing angle. Drawing every cap with the same sphere radius changes liquid volume and confounds the demonstration. The original fixture holds V at 10^{-6} m^3 , samples angles from 30 to 150 degrees, and independently integrates circular cross sections to check volume. All cap meshes, pressure jumps and adhesion readouts use these same stored records.

Basilisk's original 3D sessile-drop test provides a primary implementation and analytic comparison [E6]. The present cap lab evaluates analytic equilibrium geometry; it does not simulate relaxation. Density and viscosity therefore do not affect its equilibrium shape. Gravity and finite Bond number invalidate this spherical-cap oracle.

Contact-angle reconstruction

A negative-inside level set gives an outward liquid normal. Let the solid normal point into the accessible fluid. At a flat horizontal solid, the cap's liquid normal obeys $n_l \cdot n_s = \cos \theta$. Curved or angled surfaces require a local wall frame and a contact-line tangent. A two-dimensional ghost slope copied directly into three dimensions can impose the wrong angle.

Afkhami and Bussmann's original height-function method accounts for the orientation of the contact-line projection in 3D [E7]. Its geometry and refinement results are method-specific. Basilisk's documented contact implementation includes restrictions at shallow angles and unresolved height support [E8]. Rheon's proposed admission rule reports unresolved normals rather than advertising all-angle accuracy.

Dynamics and hysteresis

A sharp moving contact line with strict no-slip has a classical stress-singularity difficulty. Cox's original analysis addresses a small-capillary-number regime with a microscopic slip or other regularization scale [E9]. A static angle alone is not a complete dynamic law. A proposed model must identify microscopic regularization, apparent-angle observation scale and contact-line mobility.

Advancing and receding angles can define a pinned interval with motion outside it. That is an additional hysteresis model, not an automatic consequence of Young's relation. Heterogeneous solids, roughness and contamination can change observed behavior. Start with a declared smooth equilibrium model, then qualify contact-line speed, angle and volume under refinement before adding hysteresis. No such dynamic contact-line solver is implemented or formally verified by this edition.

Moving contact lines

A static angle closes an equilibrium geometry problem. It does not determine how a contact line moves. Huh and Scriven's original creeping-flow analysis shows the divergent stress and dissipation associated with a sharp moving contact line under the classical adherence condition. A finite grid may conceal the singularity by supplying numerical slip; refining that grid changes the concealed length scale. [R04](#)

Choose and state one regularization:

- Resolve a physical or effective Navier slip length and a compatible dynamic contact-angle closure.
- Use a diffuse interface with specified interface thickness, mobility, wall free energy, and wall relaxation law.
- Use a precursor-film/disjoining-pressure model with its own microscopic length scale.
- Pair an explicitly labeled engineering angle-speed law with effective slip, diffusion or another declared microscopic cutoff. Advancing/receding limits and pinning are optional additional models; the angle-speed law alone does not remove the sharp no-slip singularity.

For a low-capillary-number hydrodynamic approximation, the familiar small-angle form is

$$\theta_{app}^3 - \theta_{micro}^3 \simeq 9 Ca \log(L/\ell_{micro}), \quad Ca = \mu U_{cl}/\gamma.$$

Here U_{cl} is signed, the angle is in radians, and L is the stated observation scale. This expression is not a universal all-angle, high-speed law. The general Cox construction concerns small Ca , scale separation, and a two-fluid viscosity ratio; at finite gas viscosity the full angle function replaces the cubic shorthand. Use the expression as a labeled asymptotic comparison only within those assumptions. [R05](#)

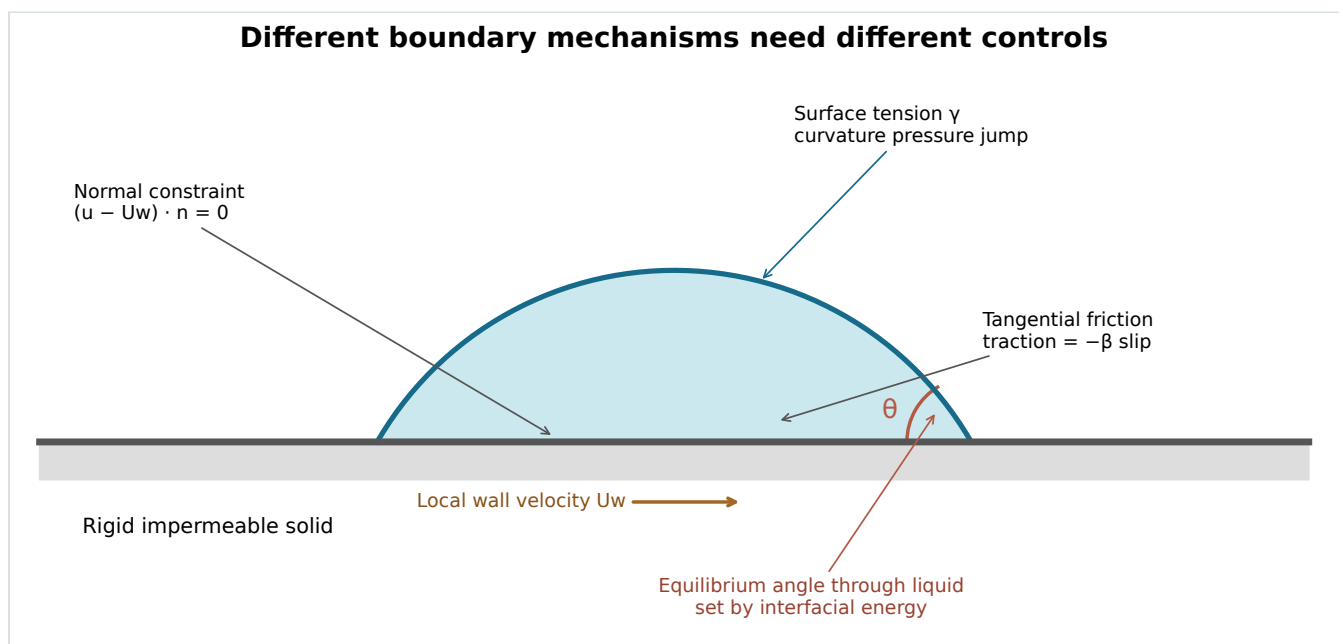
Qian, Wang, and Sheng combine Cahn–Hilliard interfacial free energy with a generalized Navier boundary condition containing uncompensated Young stress. They compare the continuum model with molecular-dynamics profiles, including Couette and Poiseuille configurations. That is stronger evidence than an attractive drop image, but it does not license transplanting their fitted molecular parameters to an arbitrary macroscopic liquid or treating diffuse-interface mobility as a harmless numerical knob. [R06](#)

For contact-angle hysteresis, an engineering pinned state may allow $\theta_R \leq \theta_{app} \leq \theta_A$ with $U_{cl} = 0$. An advancing or receding branch applies outside that interval. Store θ_R , θ_A , the observation scale, and the dynamic law separately. Static friction, capillary pinning, and high bulk viscosity can all delay motion while representing different physics.

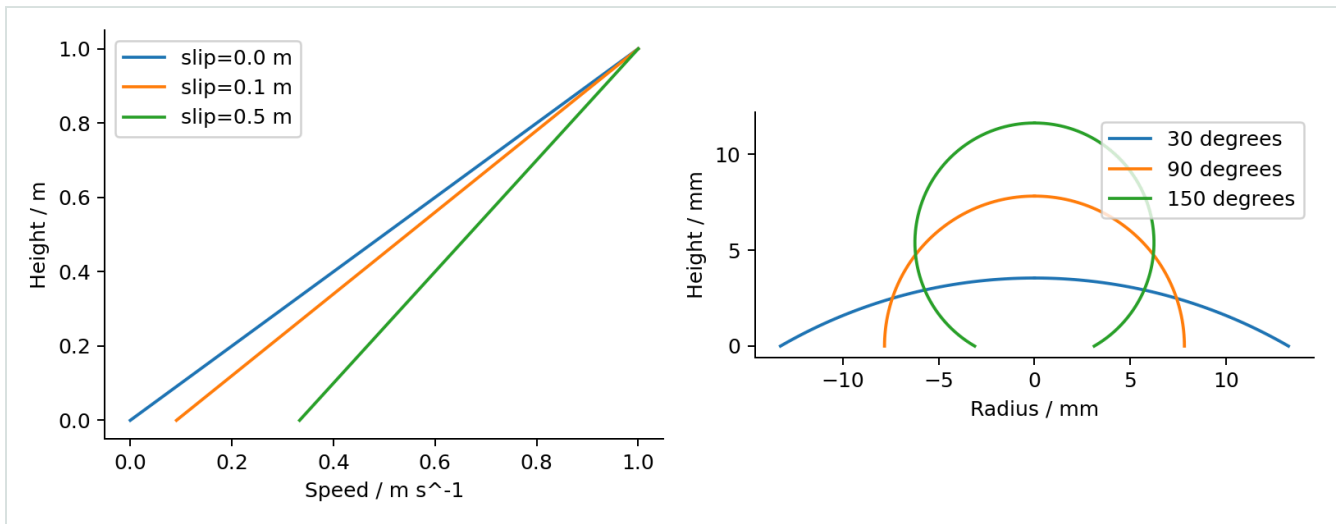
Partial wetting has an admissible range

Young balance requires $\sigma > 0$ and $|(\gamma_{sv} - \gamma_{sl})/\sigma| \leq 1$. Outside that interval, clamping the inferred cosine hides a changed model. The spreading coefficient $S = \gamma_{sv} - \gamma_{sl} - \sigma$ distinguishes the ideal complete-wetting tendency when $S \geq 0$; a precursor film or disjoining pressure then needs extra length scales. The fixed-volume cap reference deliberately stays within partial wetting and does not provide a film model.

With fixed total solid area, surface energy can be written $E_s = \sigma(A_{lv} - \cos\theta_e A_{sl})$ up to a constant. Changing μ changes the relaxation path, while changing θ_e changes this ideal energy landscape. A nearly stationary highly viscous drop may be far from equilibrium. An empirical SPH cohesion/adhesion force is a possible visual model [R11](#), but its coefficient is not automatically the gradient of this area energy or a calibrated interfacial tension.



Boundary geometry, tangential dissipation and interfacial energy supply different conditions; arrows are schematic.



Original reference illustration for 23 Wetting, adhesion and the contact line

24 Slip and wall friction

Impermeability sets relative normal velocity. A tangential boundary law must be selected separately. No-slip sets tangential fluid velocity equal to the wall. Free-slip sets tangential traction to zero. Navier slip relates traction and relative tangential speed. None of these selects a wetting angle.

Dissipative Navier law

With a normal/sign convention chosen so traction on the fluid opposes slip, let $t_t = -\beta(u - u_s)_t$, where $\beta \geq 0$ has units Pa s/m. Relative power is

$$t_t \cdot (u - u_s)_t = -\beta |(u - u_s)_t|^2 \leq 0.$$

`slip_power_nonpositive` proves the scalar component version, and component sums extend it to a tangent plane. The inequality concerns relative dissipation. A moving wall can still increase fluid kinetic energy through external work. It does not prove a spatially assembled slip discretization has this sign.

For positive viscosity, slip length $\ell_s = \mu/\beta$. The limiting cases must be represented explicitly: $\ell_s = 0$ is no-slip, while zero β is free-slip and corresponds to unbounded slip length. Avoid dividing by zero or approximating both limits with the same finite parameter. When $\mu = 0$, this relation cannot define the same viscous boundary model.

Couette reference

A lower stationary wall at $y=0$ and upper wall moving at U at $y=H$ give steady affine speed

$$u(y) = U \frac{y + \ell_s}{H + \ell_s}$$

when only the lower wall has Navier slip and the upper wall has no-slip. Then $u(0) = \ell_s u'(0)$, $u(H) = U$, and shear stress is $\mu U / (H + \ell_s)$. Increasing slip raises lower-wall speed and reduces shear at fixed U . It does not change equilibrium contact angle.

The reference generator verifies both boundary equations and exports profiles for slip lengths 0, 0.1 and 0.5 m in a unit-height slab. The browser uses actual profile samples for its arrows and shows the shear and dissipated power. These large illustrative slip lengths clarify the mathematics; they are not fitted values for water on a named material.

Friction is a model choice

Dry Coulomb friction bounds a tangential force using normal contact force. Applying it to a liquid-solid wall is a separate constitutive assumption, not the default Newtonian no-slip boundary law. A penalty damping term likewise has its own units and finite-step dissipation. Label it as a numerical/constitutive choice if used. A single “stickiness” parameter cannot consistently identify all of these mechanisms.

Cox's primary contact-line analysis explains why a microscopic mobility mechanism matters [E9]. A macroscopic slip length selected for a resolved Couette profile need not serve as the microscopic regularization of a contact-line model. Using it for both requires an explicit scale argument and validation.

Proposed state and tests

`TangentialLaw` distinguishes no-slip, free-slip and finite Navier coefficient. `WettingLaw` owns angle/energies separately. Boundary elimination must include known moving-wall terms; diagnostics retain relative dissipation and external wall power independently.

Tests include both signs of U , rotated wall frames, uniform wall/fluid translation, curved-wall interpolation and zero/coefficient limits. A stationary relative state must dissipate zero. Refinement should approach the analytic Couette profile and traction. The finite proof does not establish those geometric limits, and this edition supplies only the affine oracle rather than a production moving-wall implementation.

Both walls slipping: a distinct oracle

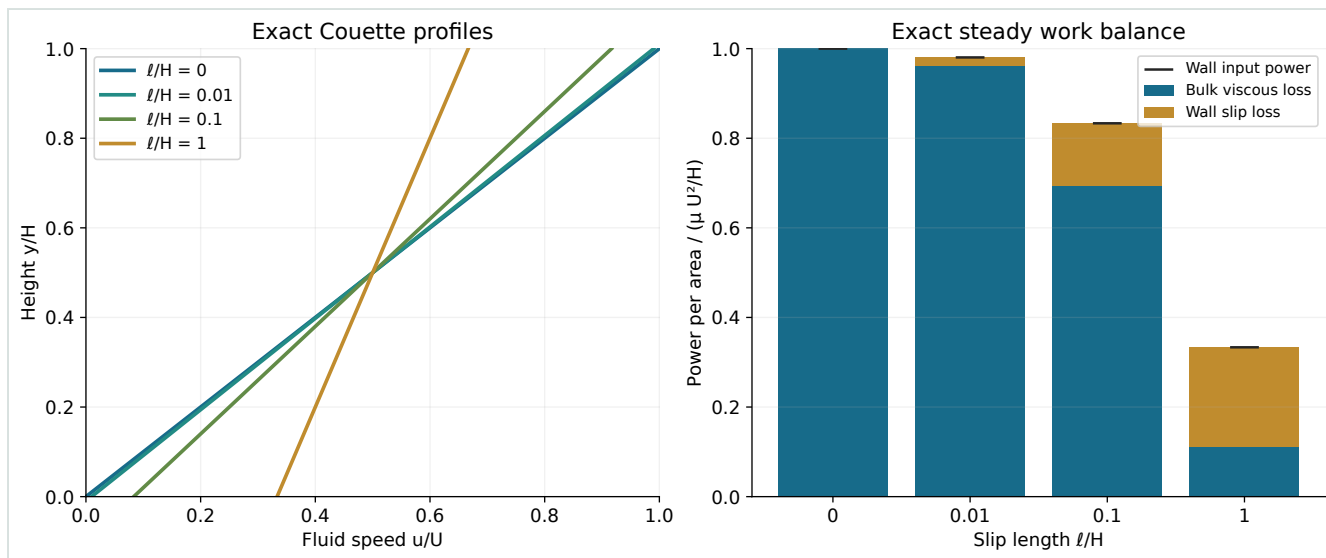
The browser lab uses lower-wall slip only. With equal finite slip lengths ℓ at both stationary-lower/moving-upper walls, the separate exact profile is

$$u(y) = U \frac{y + \ell}{H + 2\ell}, \quad \tau = \mu \frac{U}{H + 2\ell}.$$

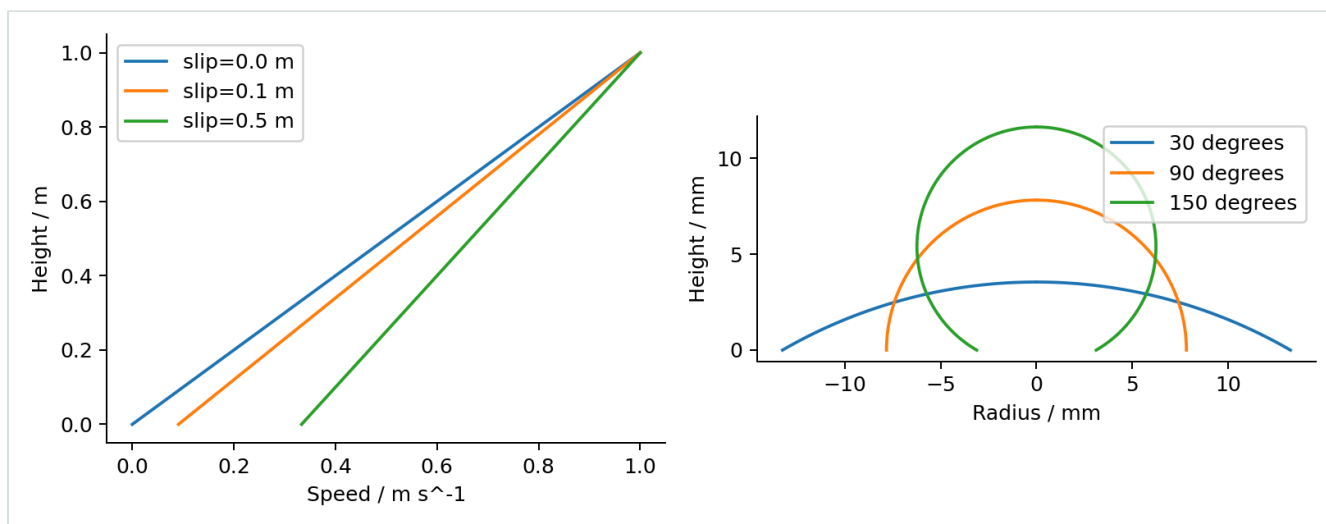
The two relative wall speeds have magnitude $U\ell/(H+2\ell)$, and actuator power partitions as

$$\tau U = \mu \left(\frac{U}{H + 2\ell} \right)^2 H + 2 \frac{\mu}{\ell} \left(\frac{U\ell}{H + 2\ell} \right)^2.$$

Handle $\ell=0$ as a no-slip constraint instead of dividing by zero. This formula is an independent analytical fixture; it must not replace the one-wall denominator in the existing browser data. The original comparison figure below uses symmetric two-wall slip and displays bulk/wall power separately.



Exact fully developed Couette profiles with symmetric Navier slip; wall input partitions into bulk and wall dissipation.



Original reference illustration for 24 Slip and wall friction

25 Surface tension and integrated qualification

Surface tension supplies an interface stress. It is not viscosity or a wall adhesion force. With constant σ , outward liquid normal and curvature sum $\kappa = \nabla \cdot n$, a spherical interface has $\kappa = 2/R$. Neglecting normal viscous stresses gives $p_l - p_a = \sigma\kappa$. With viscous liquid stress and negligible air viscous stress, the normal jump instead includes $n^T \tau_l n$. Chapter 9’s sign convention remains authoritative.

Pressure-force balance

A discrete surface force and pressure gradient should use compatible locations, masks and differences. Otherwise even an accurately solved stationary drop can develop parasitic velocity. Basilisk’s original surface-tension implementation expresses an interfacial potential using curvature and applies an explicit capillary stability bound [E10]. Its code is a reference for examining balanced construction, not a borrowed Rheon backend.

The capillary lab holds the same cap volume and angle while varying tension through 0.03, 0.072 and 0.12 N/m. Shape remains the analytic zero-gravity equilibrium cap; pressure jump changes linearly with tension. For $V = 10^{-6}$ m³ and angle 90 degrees, R is approximately 0.007815926 m and the jump at 0.072 N/m is approximately 18.42392 Pa. The reference records use full precision, with these prose numbers rounded.

Time scales and resolution

The inertial capillary scale is $t_\sigma = \sqrt{\rho h^3 / \sigma}$. The viscous diffusion scale is $t_\nu = h^2 / \nu$. The gravity-capillary length is $\ell_c = \sqrt{\sigma / (\Delta \rho g)}$. These are dimensional scales, not universal accepted step sizes. Actual stability constants depend on discretization and time integration; implicit tension can change stability while retaining geometric error.

Dimensionless groups separate regimes: $Re = \rho UL / \mu$, $Ca = \mu U / \sigma$, $We = \rho U^2 L / \sigma$, and $Bo = \Delta \rho g L^2 / \sigma$. A spherical-cap reference assumes negligible Bo; a shear decay reference excludes an interface. Combining their pictures does not validate a coupled solver.

Progressively richer teaching examples

The Pages edition presents six linked 3D laboratories: finite MAC projection; earliest triangle-mesh collision with prescribed motion; layered hydrostatic density; implicit viscous mode; Navier-slip Couette flow; and constant-volume wetting/capillarity. Each points to its chapter, original reference implementation, stored data and applicable proof. Parameters select or interpolate declared reference states, rather than pretending to run a general liquid simulator.

The original smoke render remains a separate accepted Rust demonstration with its original source identity. This edition does not relabel that smoke as a liquid calculation, or claim that the educational analytic caps establish production realism. Readers can compare the evidence levels in Appendix F.

Production integration sequence

After research review, implement one-way collision geometry and gravity first. Add free-surface classification/transport with volume and spatial metrics, then constant and variable density using a consistent pressure correction. Add tensor viscosity and wall traction, then a declared contact-angle reconstruction and balanced capillarity. Dynamic contact lines and two-way solids come after their assumptions are selected. This is a research implementation sequence, not a merge gate for the existing bounded smoke milestone.

An accepted liquid step must publish interface, material fields, velocity, time and geometry version together. Failure in geometry, transport, nonlinear stress or pressure keeps the last coherent state. Capillary, viscous and Courant restrictions must be reported with the actual accepted interval; no hidden parameter changes should imitate a successful solve.

Evidence that earns a capability claim

Use hydrostatic rest, variable-density layers, viscous decay/Couette flow, translating volume, a stationary droplet and a sessile cap before combined pouring scenes. Report pressure jump, parasitic speed, volume drift, contact angle, wall flux/work and boundary versions. Refine h and dt independently. Then add moving meshes, pinning/depinning and separation, with independent geometry references. Tests must name a physical regime and error budget before advertising calibrated material behavior.

The new Lean contracts remove ambiguity in finite identities and sign conditions. The reference programs qualify local mechanisms. Neither proves arbitrary mesh assembly, full three-dimensional liquid transport, contact-line convergence, floating-point production behavior or performance. Those are explicit research gaps, not reasons to reject the existing smoke demonstration.

A future liquid qualification suite

The following fixtures are planned for a quantitative liquid implementation. Proposed error targets are starting engineering choices, not universal tolerances or merge gates for the accepted smoke milestone. The local references qualify only the finite/analytical mechanisms described earlier.

Analytical fixtures

F1 Planar Couette flow

Use a periodic streamwise channel with $0 \leq y \leq H$, lower wall speed 0 and upper wall speed U . For stationary fully developed incompressible Newtonian flow with no slip,

$$u_x(y) = Uy/H, \quad \tau = \mu U/H, \quad P_{wall}/A = \mu U^2/H.$$

Use synthetic SI values $H = 0.01$ m, $U = 0.1$ m/s, $\mu = 0.1$ Pa·s, $\rho = 1000$ kg/m³. Then $\nu = 10^{-4}$ m²/s, $Re = 10$ using gap width, $t_v = 1$ s, $\tau = 1$ Pa, and $P/A = 0.1$ W/m². Initialize the analytic steady profile for a discrete equilibrium test; start from rest separately to test transient relaxation. Do not claim steady state merely after an arbitrary number of frames.

Proposed gates: relative velocity L2 error below 1%, wall shear error below 2%, and steady wall-work minus dissipation residual below 1% of wall power on the finest test grid. Also verify that changing ρ while holding μ fixed changes transient timescale but not the steady profile.

F2 Symmetric Navier slip in Couette flow

For equal slip lengths ℓ at both walls,

$$u_x(y) = U \frac{y + \ell}{H + 2\ell}, \quad \tau = \mu \frac{U}{H + 2\ell}.$$

Sweep $\ell/H = 0, 0.01, 0.1, 1$. Use a constraint for the zero-slip limit, not division by zero in $\beta = \mu/\ell$. Check both fluid slip velocities and both traction signs. For finite ℓ , test

$$\tau U = \mu \left(\frac{U}{H + 2\ell} \right)^2 H + 2 \frac{\mu}{\ell} \left(\frac{U\ell}{H + 2\ell} \right)^2.$$

This partitions actuator power into bulk and wall dissipation. A solver matching the profile but omitting wall dissipation is still missing an energy term.

F3 Pressure driven Poiseuille flow

For a planar channel, no slip, and $G = -dp/dx > 0$,

$$u_x(y) = \frac{G}{2\mu} y(H - y), \quad Q' = \frac{GH^3}{12\mu}.$$

Q' has units m^2/s and is flow rate per out-of-plane width. With equal Navier slip length ℓ , add $G\ell H/(2\mu)$ to the velocity and multiply Q' by $1 + 6\ell/H$. A uniform body acceleration reproducing the pressure gradient is $a = G/\rho$, not G itself. Check profile, flow rate, wall shear, and input pressure power per unit out-of-plane width $GQ'Lx$ (W/m). Its time integral is work per unit width.

Use the Couette material values and $G = 120$ Pa/m for a no-slip mean speed 0.01 m/s. Proposed gates: below 1% relative flow-rate/profile error on the finest grid and decreasing errors under refinement. Repeat with a rotated or curved embedded channel only when an appropriate analytic or independently converged reference is supplied.

F4 Viscous mode decay and rigid motion

For a periodic mode $u = (U \sin(ky), 0)$, the convective term vanishes and

$$u_x(t) = U \sin(ky) e^{-\nu k^2 t}, \quad E(t)/E(0) = e^{-2\nu k^2 t}.$$

This distinguishes μ from ν and reveals excess numerical damping. With backward Euler and constant operator eigenvalue λ , the discrete amplitude ratio after N steps is $U_N/U_0 = (1 + \nu\lambda\Delta t)^{-N}$, where $\lambda \geq 0$ is the eigenvalue of the positive discrete negative-Laplacian operator before multiplication by ν , with units m^{-2} . Compare against the discrete formula for the integrator test and the exponential for the PDE convergence test.

Independently apply only the viscous step to uniform translation and a rigid rotation sampled on a free domain. Check zero strain, zero viscous force, and unchanged velocity. Rotation tests boundary traction and symmetric-gradient consistency; decay of a nonrigid sine mode tests actual dissipation.

F5 Static pressure jump and parasitic currents

Turn gravity off, place a circular drop away from walls, set zero velocity, and use constant γ . In a strictly two-dimensional planar cross-section with translational invariance, $\Delta p = \gamma/R$. For a three-dimensional sphere, $\Delta p = 2\gamma/R$. Do not use the sphere formula for a planar 2D disk.

Start with $R = 0.001$ m and synthetic $\gamma = 0.072$ N/m, giving 72 Pa in planar 2D and 144 Pa in 3D. These are declared synthetic parameters, not a measured fluid specification. Use $\rho_l = 1000$ kg/m³ and $\tau_\gamma = \sqrt{(\rho_l R^3/\gamma)}$; select and report gas density/viscosity or the one-phase approximation. Sweep radii of 8, 16, 32, and 64 grid cells, drop offsets, and at least one density ratio beyond unity.

Measure bulk-phase mean pressure away from the interface band, maximum and RMS velocity, curvature bias/variance, volume drift, and interface error. Normalize parasitic velocity by $U_\gamma = \sqrt{(\gamma/(\rho_l R))}$; additionally report $\mu U_{\max}/\gamma$ when $\mu > 0$. A proposed quantitative gate is pressure-jump error below 2%, relative volume drift below 0.1% over ten τ_γ , and $U_{\max}/U_\gamma$ below 10^{-3} on the finest fixture. These targets may need stricter tolerances for long capillary relaxation.

Basilisk's original spurious-current test supplies a reproducible scientific precedent with explicit diameter, Laplace number, viscous damping time, and resolution sweep. It is a reference design to reproduce, not a result already obtained here. [R24](#)

F6 Sessile drops at several contact angles

Set gravity to zero, keep fixed liquid volume, and use a smooth flat rigid substrate. For a 2D circular segment with angle θ through the liquid, using radians in the formula,

$$A = R^2(\theta - \sin \theta \cos \theta), \quad a = R \sin \theta, \quad h = R(1 - \cos \theta).$$

For a 3D spherical cap,

$$V = \frac{\pi R^3}{3}(2 - 3 \cos \theta + \cos^3 \theta), \quad a = R \sin \theta, \quad h = R(1 - \cos \theta).$$

At obtuse angles the spherical cap can overhang its footprint, so a height-field-only visualization cannot represent every shape. Compare $\theta = 30^\circ$, 60° , 90° , 120° , and 150° first. Extreme angles are a separate capability claim and require appropriate curvature stencils and film resolution. Report angle measured at a stated distance from the wall, fitted radius, footprint, height, and remaining kinetic energy. Proposed gates are angle error below 2° and dimensions within 2% on the finest baseline grid, with converging errors and stable volume.

The original Basilisk 2D and 3D sessile-drop tests provide implementation-level references. Preserve the distinction between a circular segment and a spherical cap; the published 2D page uses an area-like volume convention. [R22](#), [R23](#)

F7 Dynamic capillary oscillation

Perturb a drop slightly and measure frequency and damping. Use the correct dimensionality and phase model. For an inviscid 3D spherical drop in a dynamically negligible exterior, the degree- l mode has $\omega l^2 = l(l-1)(l+2)\gamma/(\rho R^3)$, $l \geq 2$. With a dynamically important exterior or significant viscosity, use the corresponding two-fluid/damped theory rather than that formula. [R27](#)

Keep perturbation amplitude small and verify amplitude independence over a decreasing-amplitude sequence. Report oscillation frequency separately from amplitude decay. A method can produce the correct equilibrium radius while being badly overdamped. The original Basilisk oscillation fixture explicitly analyzes frequency error and equivalent numerical viscosity; use it as an additional reproducible baseline for its own dimensional setup. [R25](#)

F8 Moving contact line and capillary penetration

For a selected slip/diffuse-interface closure, compare spreading radius and apparent angle versus time while refining the grid at fixed physical regularization length. If ℓ_s or ϵ is tied to Δx , report that as a changing-model sequence. Do not label it ordinary convergence to a fixed physical problem.

For a long horizontal circular capillary of radius r , with negligible inertia/gravity/gas resistance, quasi-steady Poiseuille flow, no slip outside the regularized contact region, and constant θ , the Lucas–Washburn reference is

$$x^2(t) - x^2(0) = \frac{r\gamma \cos \theta}{2\mu} t.$$

Use $\theta < 90^\circ$ and a finite initial filled length to avoid the unphysical infinite startup speed of the inertia-free approximation. Fit only a time window where its hypotheses hold; entrance losses, inertia, gravity, dynamic angle, and trapped gas can all invalidate it. The original Washburn publication supplies the square-root penetration law and its assumptions. [R26](#)

F9 Moving walls and two way solids

Translate a wall through initially stationary fluid and rotate a paddle with prescribed angular speed. Plot accumulated actuator work against changes in kinetic/potential/surface energy and dissipation. Then move both fluid and obstacle at the same constant velocity; a Galilean-consistent relative-velocity treatment should not generate drag from their common translation. Repeat as the obstacle crosses grid cells.

For a freely moving solid in a closed fluid system without external forces, check combined linear momentum. Check total angular momentum only with compatible rotational geometry and transfer. During a fixed-geometry implicit projection, T4 predicts a particular nonnegative numerical energy loss, not exact kinetic-energy conservation.

Geometry stress fixtures

Fixture	Sweep	Future capability observation
Thin closed plate	Thickness/ Δx from 4 to 0.1; sub-cell offsets and orientations	No unreported disappearance or leakage; displaced volume error disclosed

Fixture	Sweep	Future capability observation
Zero-thickness two-sided shell	Crossings of every interpolation/advection stencil	Barrier respected or explicit unsupported status
Narrow slot	Width/ $\Delta x = 0.25, 0.5, 1, 2, 4, 8$	Conductance convergence when resolved; no invented sub-grid accuracy
Sliver cut cell	Fluid fraction approaching zero	Stability, flux conservation, redistribution locality, no negative mass
Moving cut surface	Swept cell fractions and common translation	Uniform-field preservation and mass budget under covering/uncovering
Rotating paddle	Angular speed and interaction-point offsets	Correct local wall speed, torque, and work signs
Dirty mesh	Open seam, self-intersection, reversed patch, degenerate triangles	Deterministic rejection or documented repair; never silently plausible output
Stale deformation map	Bending or scaling obstacle after preprocessing	Update/rebuild or explicit unsupported status

Collision leakage should be measured as fluid mass crossing a labeled impermeable barrier, not merely particles visibly inside a solid at the end of a frame. For a closed static tank, declare a finite reference observation time $T = L_{\text{ref}}/U_{\text{ref}}$ with $U_{\text{ref}} > 0$, or select t_y or t_v when there is no reference flow. Proposed regression gates are relative mass drift below 0.1% over ten such T and zero topological barrier crossings in exact-geometry tests; engineering fixtures should additionally report the numerical tolerance used to classify a crossing.

A Algorithm reference

These algorithms specify future implementation behavior. The numerical companion executes a smaller independent subset. The pseudocode is not a tested Rust solver. Each routine names ownership and failure boundaries.

Validated grid construction

Inputs are positive counts, finite positive spacing, finite origin and caller-supplied memory limit. Compute all capacities and bytes with checked arithmetic. Reject overflow, unsupported dimensions or budget excess before allocation. Allocate all required buffers before publishing a state.

```
make_grid(nx, ny, nz, spacing, origin, memory_limit):
    require finite(spacing) and spacing > 0
    require finite(origin) and positive counts
    cells = checked_product(nx, ny, nz)
    xfaces = checked_product(checked_add(nx, 1), ny, nz)
    yfaces = checked_product(nx, checked_add(ny, 1), nz)
    zfaces = checked_product(nx, ny, checked_add(nz, 1))
    bytes = checked_layout_bytes(cells, xfaces, yfaces, zfaces)
    if bytes > memory_limit: return ResourceLimit
    allocate state and workspace or return AllocationFailure
    return ValidatedGrid
```

The indexing theorem applies after finite-range checks. Each view retains its shape so face arrays cannot accidentally use cell strides.

Matrix free pressure

For each pressure cell gather incident faces with nonnegative coefficients. Closed walls contribute known flux to the right-hand side. Prescribed pressure is eliminated consistently.

```
apply_A(p, out, topology):
    for each pressure cell i:
        total = 0
        for neighbor j across internal face e:
            total += weight[e] * (p[i] - p[j])
        for each prescribed-pressure face e:
            total += boundary_diagonal[e] * p[i]
        out[i] = total
```

This gives nonnegative diagonals, positive only with at least one positive retained coefficient, and nonpositive off-diagonals. Known boundary pressure adds a matching right-hand-side term. A pinned pressure needs matching row and column elimination.

Pressure solve lifecycle

The lifecycle operates on the gauge-fixed SPD system or a separately justified admissible subspace. The preconditioner is assumed SPD by a separately justified construction. Check initial residual before any quotient. Final acceptance uses a freshly computed residual.

```
solve(A, M, b, p, tolerances, iteration_limit):
    r = b - A(p)
    if not finite(r): return NonfiniteState
    if accepted(r, b): return Converged
    z = M_inverse(r)
    d = z
    rz = dot(r, z)
    if not finite(rz) or rz <= 0: return Breakdown
    for k in 0 .. iteration_limit:
        Ad = A(d)
        curvature = dot(d, Ad)
        if not finite(curvature) or curvature <= 0:
            return Breakdown
        alpha = rz / curvature
        p += alpha * d
        r -= alpha * Ad
        if candidate_tolerance_met(r):
            true_r = b - A(p)
            if accepted(true_r, b): return Converged
            r = true_r
            z = M_inverse(r)
            d = z
            rz = dot(r, z)
            if not finite(rz) or rz <= 0: return Breakdown
            continue
        z = M_inverse(r)
        next_rz = dot(r, z)
        if not finite(next_rz) or next_rz <= 0:
            return Breakdown
        beta = next_rz / rz
        d = z + beta * d
        rz = next_rz
    return IterationLimit(p, b - A(p))
```

Production also validates updated vectors and supports cancellation. Tiny negative curvature still indicates breakdown of expected SPD arithmetic; precision escalation or retry must be explicit. The stopping function combines absolute and relative scales, then verifies physical divergence.

A fixed boundary substep

This first-order proposal holds geometry fixed over a substep. Reordering requires validation.

```
step(accepted, boundary_snapshot, requested_dt):
    validate configuration and state version
    choose dt and report its limiting reason
```

```

initialize candidate buffers
advect velocity into separate face arrays
apply declared body accelerations
apply viscosity if enabled
impose prescribed normal wall velocities
build pressure topology and component gauges
build rhs through the same divergence operator
solve pressure or return a declared failure
subtract the matching pressure gradient
verify boundary consistency
recompute divergence and finite-state diagnostics
transport scalars at the declared velocity time level
integrate sources with explicit amounts
validate field bounds and diagnostics
publish candidate atomically at time + dt

```

Liquids need separately derived interface transport, classification, velocity extension and free-surface pressure stages. Adding a surface renderer does not turn this into a liquid algorithm.

Conservative amounts

Compute each shared face flux once from immutable old state. A gather update reads a flux array without races:

```

for each face e:
    flux[e] = selected_flux(old, velocity, geometry, e)
for each cell i:
    amount_next[i] = amount_old[i]
    for incident face e:
        amount_next[i] += dt * signed_inflow(i, e) * flux[e]
    amount_next[i] += integrated_source[i]

```

The conserved quantity is amount, not concentration. Divide by cell volume only when constructing concentration. Moving cells require geometric transport terms absent from this static-cell argument.

Diagnostics

A packet should contain state ID, boundary version, start/end time, accepted substeps, limiting reason, maximum and RMS divergence, true residual, iterations, total tracer or liquid amount, kinetic-energy estimate, allocated bytes and termination reason. Define units and normalization. Missing measurements are unavailable, not zero.

B Reproduction and evidence

Source includes chapter Markdown, appendices, document and figure builders, experiments, covers and Lean. Word equations are native and editable; PDF is the rendered reading copy. Cover fluid imagery is illustrative, not simulated evidence.

Numerical environment

Executed fixtures use Python 3.12.14, NumPy 2.3.5 and SciPy 1.17.0 on Linux x86_64. Figures use Matplotlib 3.10.8. Requirements pin these packages. The deterministic seed is 20261003. The script writes results.json including full residual histories.

From this book directory, preserve the committed observations before generating new outputs. The qualified Linux x86_64 replay uses Haswell dispatch on hardware that supports that kernel, with both wheel BLAS libraries at OpenBLAS 0.3.30. These process-local settings reproduce the separately recorded CG profile, not the historical CG trajectory byte-for-byte; historical JSON stays unchanged.

```
python3 -m pip install -r companion/requirements.txt
baseline=$(mktemp -d)
cp companion/results.json companion/depth-results.json "$baseline/"
OPENBLAS_CORETYPE=Haswell OPENBLAS_NUM_THREADS=1 python3 companion/experiments.py
OPENBLAS_CORETYPE=Haswell OPENBLAS_NUM_THREADS=1 python3 companion/depth_experiments.py
OPENBLAS_CORETYPE=Haswell OPENBLAS_NUM_THREADS=1 python3 companion/verify_results.py \
  --baseline-dir "$baseline" --cg-profile haswell-openblas-0.3.30
python3 companion/make_figures.py
```

Run these commands in a disposable copy if the committed working-tree evidence must remain intact. Verification checks historical/source hashes and the loaded runtime, substitutes only the explicitly qualified CG history, and retains the original comparison tolerances for all fields. See the companion README for profile limits. Assertions cover graph projection, rational conservation, periodic transport bounds, diffusion decay and memory arithmetic. They do not qualify production runtime or a GPU backend. Run experiment assertions with normal Python, since -O disables those scientific experiment assertions.

Lean environment

The toolchain is leanprover/lean4:v4.19.0. Mathlib is pinned to c44e0c8ee63ca166450922a373c7409c5d26b00b and transitive revisions are locked. The workflow uses a pinned checkout action and official versioned elan, read-only permissions and no retained checkout credentials.

```
cd proofs
python3 scripts/check_sources.py
lake exe cache get
lake build
```

```
lake env lean AxiomAudit.lean
python3 scripts/test_audit.py
```

Inventory digests are reviewed, not regenerated in CI. Changed statements require mathematical review, inventory updates and full qualification. A digest establishes source identity, not truth.

The first successful qualification is [workflow run 37141752646](#), associated with source head ecf97d3a943ebe30d20acc1cf01b94dc08ff96a2. The runner checked the pull-request merge ref against then-current main. The retained log records actual checkout and all proof/audit stages. Later proof changes require new qualification.

Boundaries and rendering

Proofs establish properties of Lean definitions. Fixtures establish small reference calculations. Rust architecture, liquids and performance remain implementation work. No production throughput, latency or GPU-memory benchmark was executed.

Rebuild and render after layout edits, inspect every page, and check native equation structure separately. Export success alone does not establish readable notation or clean pagination.

Extended mechanism fixtures

Run `python companion/depth_experiments.py` for twelve restricted fixtures. They cover staggered affine sampling, rotating-field backtracing, a two-level Dirichlet multigrid mechanism, axis-aligned interface flux, sphere curvature, weighted viscous energy, frozen-position affine particle transfer and SPH constraint gradients.

The affine sampler errors are at most 2.23×10^{-16} , while the intentionally wrong x-face offset creates a -0.07 bias. Over one unit of rotation time, midpoint tracing error decreases from 0.00166645 at 10 steps to 0.0000260416 at 80; Euler decreases from 0.0511230 to 0.00626930.

The 31-unknown Dirichlet multigrid fixture reduces residual from 1.13620 to 0.00873160 in one cycle and near floating precision in eight. Its exact coarse-space correction discrepancy is 3.17×10^{-15} . This nodal one-dimensional mechanism fixture is not the proposed cell-centered three-dimensional fluid hierarchy.

For a radius-0.5 sphere, curvature errors decrease from 0.0392195 at $h=0.1$ to 0.000624805 at $h=0.0125$. The weighted viscous fixture decreases energy from 7 to 0.941037. Its tiny computed negative minimum eigenvalue, about -2.49×10^{-16} , is numerical roundoff around the exact constant nullspace, not a proof of negative dissipation.

The 27-node quadratic particle support reproduces its affine matrix to 3.85×10^{-16} and represented angular momentum to 1.16×10^{-16} . Removing one support node and renormalizing leaves a first-moment defect of 0.205778. The SPH analytical constraint gradient differs from finite differences by at most 1.12×10^{-10} . These results validate the specified local algebra and derivatives, not a full APIC, SPH or PBF simulation.

The boundary review fixtures distinguish linear from affine divergence: omitting pressure boundary work creates a 0.21 identity defect. A skew pressure connection gives normal gradient 3.2 instead of 2 despite positive geometry. An offset fraction step confirms opposite fraction-gradient and level-set-gradient signs. A compatible chain has retained

residual maximum 0.001 but full maximum 0.003, so a 0.0015 threshold must reject it despite the reduced solve criterion.

C Primary sources

The works below provide historical and technical context. Graph definitions, worked arithmetic, recommendations, experiments and Lean contracts are independently specified for Rheon. No third-party figures or long quotations are reproduced. Publisher-only citations identify original works without implying full-paper reproduction.

[S1] Jos Stam. Stable Fluids. SIGGRAPH 1999, 121-128. DOI 10.1145/311535.311548. [Author paper](#). Semi-Lagrangian graphics simulation.

[S2] Francis H. Harlow and J. Eddie Welch. Numerical Calculation of Time-Dependent Viscous Incompressible Flow of Fluid with Free Surface. Physics of Fluids 8, 2182-2189, 1965. DOI 10.1063/1.1761178. [Paper](#). Marker-and-cell foundation.

[S3] Alexandre J. Chorin. Numerical Solution of the Navier-Stokes Equations. Mathematics of Computation 22, 745-762, 1968. [Author paper](#). Projection methods.

[S4] Robert Bridson and Matthias Müller-Fischer. Fluid Simulation. SIGGRAPH 2007 Course Notes. [Author notes](#). Broad numerical implementation context.

[S5] Jonathan Richard Shewchuk. An Introduction to the Conjugate Gradient Method Without the Agonizing Pain. 1994. [Author paper](#). Krylov methods and conditioning.

[S6] Andrew Selle, Ronald Fedkiw, ByungMoon Kim, Yingjie Liu and Jarek Rossignac. An Unconditionally Stable MacCormack Method. Journal of Scientific Computing 35, 350-371, 2008. [Publisher](#). Corrected advection.

[S7] Christopher Batty, Florence Bertails and Robert Bridson. A Fast Variational Framework for Accurate Solid-Fluid Coupling. ACM Transactions on Graphics, 2007. [Author paper](#). Boundary coupling.

[S8] Stanley Osher and James A. Sethian. Fronts Propagating with Curvature-Dependent Speed: Algorithms Based on Hamilton-Jacobi Formulations. Journal of Computational Physics 79, 12-49, 1988. [Author paper](#). Level sets.

[S9] C. W. Hirt and B. D. Nichols. Volume of Fluid Method for the Dynamics of Free Boundaries. Journal of Computational Physics 39, 201-225, 1981. [Publisher](#). Volume fractions.

[S10] Douglas Enright, Ronald Fedkiw, Joel Ferziger and Ian Mitchell. A Hybrid Particle Level Set Method for Improved Interface Capturing. Journal of Computational Physics 183, 83-116, 2002. [Publisher](#). Particle interface correction.

[S11] Christopher Batty and Robert Bridson. A Simple Finite Difference Method for Time-Dependent, Variable Coefficient Stokes Flow on Irregular Domains. 2010 preprint. [Paper](#). Coupled viscosity and pressure.

[S12] Ronald Fedkiw, Jos Stam and Henrik Wann Jensen. Visual Simulation of Smoke. SIGGRAPH 2001. [Author paper](#). Smoke and confinement.

[S13] Matthias Müller, David Charypar and Markus Gross. Particle-Based Fluid Simulation for Interactive Applications. SCA 2003. [Author paper](#). SPH.

- [S14] Lvmin Zhang, Anyi Rao and Maneesh Agrawala. Adding Conditional Control to Text-to-Image Diffusion Models. 2023. [Paper](#). Spatial conditioning; not a Rheon-specific error guarantee.
- [S15] Miles Macklin and Matthias Müller. Position Based Fluids. ACM Transactions on Graphics 32, 2013. [Author preprint](#). Density constraints.
- [S16] Chenfanfu Jiang, Craig Schroeder, Andrew Selle, Joseph Teran and Alexey Stomakhin. The Affine Particle-In-Cell Method. ACM Transactions on Graphics 34, 2015. [Publisher](#). APIC.
- [S17] Chenfanfu Jiang, Craig Schroeder and Joseph Teran. An Angular Momentum Conserving Affine-Particle-In-Cell Method. 2016 preprint; Journal of Computational Physics, 2017. [Paper](#). Specified transfer conservation.
- [S18] David Goldberg. What Every Computer Scientist Should Know About Floating-Point Arithmetic. ACM Computing Surveys, 1991. [Authorized reprint](#). Rounding and exceptional arithmetic.
- [S19] Rheon repository README, revision 0cb48429789699578b640d30d9d5dc99c6921ce4. [Pinned source](#). Project scope; no production solver at this revision.
- [S20] MrScripty. Coding Standards, revision dcc56f26e884ade260770beceba2501d3746200d. [Pinned source](#). Ownership, evidence and performance guidance.
- [S21] Lean community. [Lean 4.19.0](#) and [pinned mathlib](#). Formal environment.
- [S22] Yousef Saad. Iterative Methods for Sparse Linear Systems, second edition. SIAM, 2003. [Author hosted book](#). Multigrid and iterative-method context. The companion hierarchy and its measured results are independently constructed.

D Notation and provenance

Velocity, pressure, density, dynamic viscosity, kinematic viscosity, spacing and physical step are $\mathbf{u}, p, \rho, \mu, \nu, h, \Delta t$. Advected scalar, level set and liquid fraction are q, ϕ, f_i . Units follow the model, not the letter alone.

In the graph model, B is cells by oriented internal faces, negative at tail and positive at head. Physical divergence is $-B\mathbf{u}$ at unit spacing; pressure difference is $B^T p$; $A = BWB^T$. Euclidean graph norms are unweighted. Physical integrals require volume or face metrics. MiB is 1,048,576 bytes; GiB is 1,073,741,824 bytes.

This research edition is credited to Puma at the user's request. Covers were generated specifically for the book and visually reviewed. They are illustrative, not numerical evidence. Original prompts and hashes accompany the artwork. Explanatory diagrams are independently generated from definitions and experiment data; no cited-paper figures are reproduced.

The native-equation publishing approach follows the project family's research-book style. Scientific content and theorem claims are Rheon-specific. The proof environment reuses a previously qualified pinned Lean/mathlib combination with independently authored fluid contracts.

The dense baseline, typed outcomes, component gauges and immutable snapshots are recommendations for Rust implementation review. Sparse grids, APIC, non-Newtonian materials and integrated diffusion evaluation remain extensions. No application GUI, calibrated presets or measured production performance is claimed.

The repository's original license is Apache 2.0. Cited papers retain their own rights and are linked rather than redistributed. Generated illustration provenance is documented without asserting identical copyright treatment in every jurisdiction. No publisher affiliation, ISBN, peer-review status or physical validation is claimed.

E A worked matrix laboratory

Small exact systems make useful implementation fixtures because every entry has a meaning. They reveal sign errors, gauge mistakes and false conservation claims without a large visual scene.

Three cells on a line

Orient two faces from cell 0 to 1 and from 1 to 2. With unit weights,

$$B = \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}, \quad A = BB^T = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}.$$

Take face velocity $(2, -1)$ and time step $1/2$. Incidence is $(-2, 3, -1)$, so right-hand side is $(-4, 6, -2)$. It sums to zero, as required. Fixing $p_0 = 0$ leaves

$$\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \end{bmatrix}.$$

The second equation gives $p_2 = p_1 - 2$, hence $p_1 = 4$, $p_2 = 2$. Gradient is $(4, -2)$. Subtracting half of it gives zero velocity. A closed one-dimensional chain has no nonzero divergence-free internal flux; the result follows from topology, not excessive numerical damping.

Now stop at $p = (0, 3.9, 2)$. Gradient is $(3.9, -1.9)$, corrected velocity is $(0.05, -0.05)$, and incidence is $(-0.05, 0.10, -0.05)$. Residual is $(-0.10, 0.20, -0.10)$, exactly twice the corrected incidence. This fixture checks the time-step factor and the sign of physical divergence.

A cycle retains circulation

A three-cell cycle has faces 0 to 1, 1 to 2 and 2 to 0:

$$B = \begin{bmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}.$$

The constant face vector (c, c, c) has zero incidence. With unit metric, projecting velocity $(3, 0, 0)$ onto this nullspace yields $(1, 1, 1)$. Its circulation survives while compressive content is removed. Initial squared norm is 9; final squared norm is 3. A test expecting zero velocity in every closed domain would be wrong because it confuses a chain with a cycle.

This graph is an algebraic teaching fixture, not a spatially resolved three-cell physical vortex. Its role is to make the nullspace distinction visible. A production mesh must separately establish which cycles and coefficients correspond to its geometry.

Deriving energy reduction

Let W be positive definite, $u^+ = u - \Delta t W B^T p$, and $Bu^+ = 0$. Then the removed vector $v = \Delta t W B^T p$ is orthogonal to u^+ in the weighted inner product:

$$(u^+)^T W^{-1} v = \Delta t (Bu^+)^T p = 0.$$

Since $u = u^+ + v$,

$$\|u\|_{W^{-1}}^2 = \|u^+\|_{W^{-1}}^2 + \|v\|_{W^{-1}}^2.$$

This establishes exact norm nonincrease for the stated closed compatible projection. If some weights vanish, W^{-1} is undefined and the argument needs restriction to active faces or a different metric. Moving boundaries and prescribed flux add work; approximate solves leave a nonzero cross term. The result is derived here but is not one of the current checked Lean theorems.

For an approximate pressure, the cross term is tied to residual. In the unweighted case,

$$(u^+)^T (\Delta t B^T p) = \Delta t (Bu^+)^T p = \Delta t^2 r^T p.$$

Thus solver error can affect energy even when the algebraic operator is correct. A small residual in one norm does not bound this term without also controlling pressure in the dual norm.

Residual and solution error

On a gauge-fixed SPD system, exact solution p_* and approximate p satisfy $A(p_* - p) = r$. Therefore

$$\|p_* - p\|_2 \leq \lambda_{\min}(A)^{-1} \|r\|_2.$$

The smallest eigenvalue depends on geometry, gauge and scaling. A universal pressure-error claim from a residual threshold is invalid without this spectral factor. Velocity error can be better related through the energy norm:

$$\|B^T(p_* - p)\|_2^2 = (p_* - p)^T A(p_* - p).$$

These are exact linear-system relations. Floating arithmetic, variable weights and physical metrics need corresponding versions. Their practical lesson is to choose diagnostics from the consumer: divergence tolerance, velocity error and pressure accuracy are related but not identical acceptance criteria.

A conservation fixture with fractions

Take scalar amounts $(1/3, 1/2, 1/6)$, cyclic face fluxes $(2/7, -1/5, 1/9)$, and time step $1/10$. Each cell adds incoming minus outgoing amount. The result is $(199/630, 96/175, 61/450)$, whose sum is exactly one. The companion executes this with rational arithmetic.

This fixture isolates cancellation. It does not prove the selected flux is physically accurate or positivity-preserving for every input. A complete transport validation adds the required flux law, time restriction, geometry and boundary terms. Conservation is a structural property; accuracy depends on how the flux approximates the intended equation.

F Expansion contracts, sources and evidence

This edition adds original finite contracts and local numerical references. The historical 20 theorems retain their source-bound evidence. The previous expansion receipt expansion/proof-qualification.json retains its 33-theorem/45-declaration source binding. Current integration qualification is recorded separately in expansion/bounded-proof-qualification.json: all 42 public theorems compile, 60 declarations pass the transitive axiom audit, and negative audit/source-integrity fixtures pass. The latest bounded module contributes nine theorems and six definitions. All new derivations and implementations are Rheon research work. No cited researcher reviewed or endorsed this edition.

A map from requirements to evidence

Requirement	New exact statement	Executed local reference	Remaining production gap
External forces	force_work_identity	signed finite-step work	spatial force interpolation and IEEE execution
Moving boundary	sealed_flux_compatibility	balanced/unbalanced component flux	swept mesh and geometric conservation
Planar collision; numerical mesh query	wall_first_hit, wall_hit_on_surface, clipped_segment_in_halfspace	exact rational crossing/clipping plus independent cube queries	finite-facet containment, earliest mesh query, robust predicates and cut cells
Positive density	positive_transmissibility, weighted_pressure_energy	layered hydrostatic increments	phase-interface coefficient assembly
Liquid volume	volume_source_balance	unequal-volume rational ledger	bounded multidimensional interface transport
Viscosity	viscous_work, dissipation_nonnegative, backward_euler_work, backward_euler_energy_nonincrease	exact rational operator step; dense solve versus periodic eigenmode	symmetric-gradient assembly, free-surface traction, forcing and approximate solves
Slip	slip_power_nonpositive	affine Couette boundary equations	curved wall frame and assembly
Wetting/adhesion	young_adhesion_identity, adhesion_bounds	fixed-volume spherical caps	dynamic angle, hysteresis and microscopic law
Surface tension	no full capillary theorem claimed	pressure jump and independent volume integral	balanced moving-interface force and parasitic currents

Assumptions of the new theorems

The domain is a finite set and values are exact reals. Internal_amount_balance and volume_source_balance assume balanced incidence columns. Sealed_flux_compatibility additionally assumes every cell constraint is solved, proving a necessary global compatibility condition rather than existence. Weighted_pressure_energy assumes positive areas, densities and distances and bounds a finite weighted-square energy. It does not establish that geometric assembly yields these quantities.

Hydrostatic_increment assumes nonzero density/distance and a prescribed exact pressure increment. The implicit energy results assume an exact discrete work equation; nonincrease additionally assumes nonnegative masses, time and dissipation. BoundedPhysics now derives that work identity from defined finite strain and exact coordinate update equations with fixed nonnegative weights; it does not identify a production stencil or prove solver convergence. Young_adhesion_identity assumes Young balance; adhesion_bounds treats the cosine as a bounded scalar, without proving trigonometry or interfacial thermodynamics. Slip dissipation assumes a nonnegative coefficient. Force_work_identity is a finite-step algebraic identity without a stability assertion.

No theorem proves mesh orientation, robust arbitrary-triangle intersection, wetting existence, contact-line convergence, liquid volume bounds, production Rust correspondence, floating-point error or performance. The educational reference has no general liquid solver.

Bounded assumptions and witnesses

The planar lemmas use $g(x)=n \cdot x - c$, permitted side $g \geq 0$, a strictly positive start and strictly negative proposed endpoint. The normal need not be unit length. They prove an interior first hit and clipping to it; grazing, wall starts, finite-facet containment and moving-wall queries are excluded. The viscosity lemmas use arbitrary fixed finite E and μ , $K=E^T \text{diag}(\mu)E$, nonnegative masses/time/weights and the exact coordinate backward-Euler equations. Their conclusion is a weighted squared-norm inequality, not solvability, stencil validity or a full timestep result.

evidence/bounded-physics-contracts/witness.py checks rational first contact at $1/2$, the forbidden unclipped continuation, out-of-range hits without endpoint signs, an implicit squared-norm decrease $2 \rightarrow 1/2$, and counterexamples with negative viscosity or an omitted update equation. Normal and optimized Python outputs agree. These witnesses are separate from the nine kernel-checked theorems and the browser's triangle-query/shear fixtures.

Shared data and reproduction

Run `python3 docs/research-book/expansion/reference.py` from the repository. It writes `reference-data.json`, original figures and a source/data-bound receipt. The JSON schema identifies every lab as an analytic or local numerical oracle. The site consumes it directly; the figures use the same values. Numerical acceptance uses independent dense/modal, geometric and rational checks, not an aesthetic comparison.

Build the teaching edition with `python3 docs/education/build.py`; its README records math, browser and PDF steps. The generated full Markdown preserves all chapters and appendices, including this source map. Historical evidence files are not rewritten.

Primary sources added

[E1] Batty, Bertails and Bridson (2007). A Fast Variational Framework for Accurate Solid-Fluid Coupling. [Primary source](#). Prescribed/dynamic coupling, adjoint reaction, wall separation; Sections 2 and 4.

[E2] Robert Bridson. Author-hosted mesh_query and Tunicate implementations. [Primary source](#). Original geometry implementations linked under Downloads; independently written Rheon segment fixture uses no copied source.

[E3] Basilisk contributors. Embedded boundaries, original source. [Primary source](#). Geometry metrics, interpolation and small-cell policies; current online code, accessed 2026-10-04.

[E4] Batty and Bridson (2008). Accurate Viscous Free Surfaces for Buckling, Coiling, and Rotating Liquids. [Primary source](#). Section 3: normal/tangential traction, viscous variational construction.

[E5] Basilisk contributors. Implicit viscous diffusion, original source. [Primary source](#). Coupled stress, residual and relaxation implementation; current code, accessed 2026-10-04.

[E6] Basilisk contributors. 3D sessile drop, original test. [Primary source](#). Constant-volume spherical-cap comparison; no code/figures copied.

[E7] Afkhami and Bussmann (2009; author preprint dated 2008). Height functions for applying contact angles to 3D VOF simulations. IJNMF 61, 827–847. DOI 10.1002/flid.1974. [Primary source](#). Section 2.5: wall frame and 3D height-function geometry.

[E8] Basilisk contributors. Contact angles, original source. [Primary source](#). 3D orientation and shallow-angle support restrictions; current code, accessed 2026-10-04.

[E9] R. G. Cox (1986). The dynamics of the spreading of liquids on a solid surface. Part 1. Viscous flow. JFM 168, 169–194. DOI 10.1017/S0022112086000332. [Primary source](#). Publisher abstract: contact-line stress singularity, microscopic regularization and small-capillary-number asymptotics. No full-paper reproduction.

[E10] Basilisk contributors. Surface tension, original source. [Primary source](#). Interfacial potential and capillary restriction; current code, accessed 2026-10-04.

Verified expertise, without endorsement

[Christopher Batty](#). Author university page lists liquid/gas simulation, viscous flows, surface tension and embedded boundaries. This verifies the relevance of the source author, not an external review of Rheon.

[Shahriar Afkhami](#). Author NJIT page identifies computational interfacial flows, dynamic contact lines and complex liquids. This verifies the relevance of the source author, not an external review of Rheon.

[Robert Bridson](#). Author UBC page lists fluid simulation research, original implementations and primary publications. This verifies the relevance of the source author, not an external review of Rheon.

Additional primary-source ledger

The focused contribution inspected these original works and implementations on 2026-10-04. R identifiers avoid collisions with the historical S bibliography. Access depth is retained; moving code URLs are references, not vendored qualified dependencies.

[R01] Batty and Bridson, Accurate Viscous Free Surfaces for Buckling Coiling and Rotating Liquids, 2008. [Primary source](#). Full original text inspected, especially sections 3–5. Coupled symmetric-gradient viscosity, free-surface traction, variational dissipation, rigid-rotation counterexample.

[R02] Brackbill, Kothe, and Zemach, A Continuum Method for Modeling Surface Tension, JCP 100, 335–354, 1992. [Primary source](#). Original-paper text surfaced through academic-hosted PDF. Continuum surface force and wall adhesion boundary treatment. DOI 10.1016/0021-9991(92)90240-Y.

- [R03] Young, An Essay on the Cohesion of Fluids, 1805. [Primary source](#). Original archival metadata verified; the journal DOI endpoint returned 403. Used only for historical attribution of material-dependent contact angle, not to attribute modern notation verbatim. DOI 10.1098/rstl.1805.0005.
- [R04] Huh and Scriven, Hydrodynamic Model of Steady Movement of a Solid Liquid Fluid Contact Line, 1971. [Primary source](#). Original abstract inspected. Creeping-flow contact-line stress/dissipation singularity under adherence. DOI 10.1016/0021-9797(71)90188-3.
- [R05] Cox, The Dynamics of the Spreading of Liquids on a Solid Surface Part 1 Viscous Flow, JFM 168, 169–194, 1986. [Primary source](#). Original abstract and bibliographic details inspected. Small-Ca matched-asymptotic/slip assumptions verified. Full publisher text was not retrieved; do not present the cubic shorthand as its general finite-angle equation. DOI 10.1017/S0022112086000332.
- [R06] Qian, Wang, and Sheng, Molecular Scale Contact Line Hydrodynamics of Immiscible Flows, PRE 68, 016306, 2003. [Primary source](#). Original full text inspected. Generalized Navier boundary condition, Cahn–Hilliard framework, molecular comparison and fitted dynamic parameters. DOI 10.1103/PhysRevE.68.016306.
- [R07] Batty, Bertails, and Bridson, A Fast Variational Framework for Accurate Solid Fluid Coupling, 2007. [Primary source](#). Original project, paper text, and sample-code descriptions inspected. Kinetic projection, static irregular-boundary samples, and author hindsight on improved weights. Sample archive license not verified.
- [R08] Popinet, An Accurate Adaptive Solver for Surface Tension Driven Interfacial Flows, JCP 228, 5838–5866, 2009. [Primary source](#). Original publisher abstract/conclusion inspected. Balanced-force VOF with height-function curvature and stationary-drop equilibrium. Author PDF endpoint timed out. DOI 10.1016/j.jcp.2009.04.042.
- [R09] Basilisk original contact-angle implementation. [Primary source](#). Source and explicit limitations inspected. Height-function normal fallback and shallow-angle limitation are visible in the implementation.
- [R10] Basilisk original surface-tension implementation. [Primary source](#). Source inspected. Explicit capillary timestep restriction and curvature potential construction.
- [R11] Akinci, Akinci, and Teschner, Versatile Surface Tension and Adhesion for SPH Fluids, TOG 32(6), 2013. [Primary source](#). Original text inspected. Cohesion and boundary attraction as practical SPH models. DOI 10.1145/2508363.2508395.
- [R12] Weiler, Koschier, Brand, and Bender, A Physically Consistent Implicit Viscosity Solver for SPH Fluids, CGF 37(2), 2018. [Primary source](#). Author abstract and accepted-paper text inspected. Conservation, refinement, and spurious-viscosity evaluation criteria.
- [R13] Bender et al., Implicit Frictional Boundary Handling for SPH. [Primary source](#). Original manuscript passages inspected. Boundary viscosity/friction extension and separate boundary coefficient. Filename uses 2020; final issue metadata was not verified.
- [R14] SPlisHSPlasH Weiler2018 implementation. [Primary source](#). Source inspection confirms μ = configured viscosity $\times$ rest density and separate boundary coefficient. No build/execution.
- [R15] Macklin and Müller, Position Based Fluids, 2013. [Primary source](#). Original algorithm inspected. Positional density constraints, artificial pressure, velocity postprocessing, and performance/accuracy tradeoff.

- [R16] Koschier and Bender, Density Maps for Improved SPH Boundary Handling, SCA 2017. [Primary source](#). Original author abstract inspected. Precomputed continuous boundary density contribution.
- [R17] Bender, Kugelschadt, Weiler, and Koschier, Volume Maps An Implicit Boundary Representation for SPH, MIG 2019. [Primary source](#). Original text inspected. Kernel-independent geometric-volume contribution, rigid boundaries, comparison with density maps.
- [R18] Guendelman, Selle, Losasso, and Fedkiw, Coupling Water and Smoke to Thin Deformable and Rigid Shells, SIGGRAPH 2005. [Primary source](#). Original abstract and paper passages inspected. Infinitesimally thin surfaces, ray-based visibility, and incompressibility treatment.
- [R19] Basilisk original embedded-boundary implementation. [Primary source](#). Source inspected. Volume/face geometry, small-cell CFL restriction, tracer redistribution. Static implementation details are not asserted to establish arbitrary moving-boundary correctness.
- [R20] Basilisk core license text. [Primary source](#). GPL version 3 text verified. Per-file scope and downstream compatibility still require checking.
- [R21] SPLisHSPlasH repository license. [Primary source](#). MIT notice verified. Repository README separately warns that dependencies have their own terms.
- [R22] Basilisk original 2D sessile-drop test. [Primary source](#). Source and analytic geometry comparison inspected. The implementation tests 15° – 165° and notes shallower limitations.
- [R23] Basilisk original 3D sessile-drop test. [Primary source](#). Original source inspected as the 3D comparison. No local reproduction performed.
- [R24] Basilisk original parasitic-current test. [Primary source](#). Source inspected. Constant density, diameter 0.8, Laplace-number-controlled viscosity, viscous-time observation, resolution study.
- [R25] Basilisk original drop-oscillation test. [Primary source](#). Source/diagnostics inspected. Frequency error and equivalent numerical viscosity are separate quantities. Its dimensional setup must be retained when reproducing it.
- [R26] Washburn, The Dynamics of Capillary Flow, Physical Review 17, 273–283, 1921. [Primary source](#). Original abstract inspected. Cylindrical-capillary square-root penetration formula. Full text is access controlled; no access bypass attempted.
- [R27] Gong et al., Contactless Surface Tension Measurement of Molten Oxides Using Oscillating Drop Method in an Aerodynamic Levitator. [Primary source](#). Original measurement equations and experimental cautions inspected. Supports Rayleigh frequency convention and the need to control external forcing, mode splitting, and rotation. No material values imported as Rheon presets.
- [R28] François, Cummins, Dendy, Kothe, Sicilian, and Williams, A Balanced Force Algorithm for Continuous and Sharp Interfacial Surface Tension Models within a Volume Tracking Framework, JCP 213, 141–173, 2006. [Primary source](#). Original author-uploaded text inspected; publisher endpoint failed. Exact-curvature balanced-force experiment is an appropriate control before testing estimated curvature. DOI 10.1016/j.jcp.2005.08.004.

[R29] SPlisHSPlasH Akinci2013 implementation. [Primary source](#). Original source inspected. Separate fluid and boundary force coefficients, with boundary-method branches. No assumption that every branch reproduces every original-paper conservation claim.

Next operator proofs, not compiled claims

The nine focused targets are stationary implicit strain/wall dissipation; moving-wall work with residual norm; affine pressure constraint work; two-way impulse cancellation; rigid strain null modes; conservative redistribution/geometric conservation; balanced capillary rest; work-compatible surface energy; and contact-line friction/pinning. The current checked Physics module covers finite energy identities, conservation, positive coefficients, relative slip power and Young algebra. The new BoundedPhysics module supplies the unforced finite-strain work implication within the first target; wall terms, operator solvability, assembled strain validity and the larger surface-evolution targets remain future work.

In particular, a nonnegative stiffness is insufficient for a rigid-rotation null mode. A conservative redistribution matrix needs column sums one, but positivity and barrier connectivity need additional conditions. Balanced rest assumes M is symmetric positive definite, an exact solve, and a capillary force actually in the chosen gradient range. Surface-energy conservation requires a discrete gradient satisfying its exact work identity; triangle area is not globally convex in arbitrary vertex coordinates. A scalar contact-coordinate dissipation proof cannot establish a 3D contact-line PDE.

The contributed `expansion/contact/sanity_checks.py` executes 28 algebra/formula checks, including 100 synthetic SPD systems for each of the stationary, moving-wall and affine-projection identities. These remain numerical sanity checks, separate from Lean compilation and from a Rheon liquid run. No third-party research code or paper figure is bundled.